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Compositions and methods for production of salidroside, icariside D2, and precursors of salidroside and icariside D2

Abstract

Transgenic host cells, vectors useful for making transgenic host cells, and kits useful for making transgenic host cells are described. Also described are transgenic plants. In some embodiments, transgenic host cells express a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, transgenic host cells express a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). The transgenic host cells are useful for biosynthesis of one or more of salidroside, icariside D2, tyrosol, and 4-hydroxyphenylacetaldehyde.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/224,257 (now U.S. Pat. No. 11,408,009), filed on Dec. 18, 2018, which claims the benefit of U.S. Provisional Application No. 62/607,271, filed on Dec. 18, 2017. The entire teachings of the above applications are incorporated herein by reference.

INCORPORATION BY REFERENCE OF MATERIAL IN ASCII TEXT FILE

(1) This application incorporates by reference the Sequence Listing contained in the following ASCII text file being submitted concurrently herewith: File name: 03992060002_SL.txt; created May 2, 2025, 506,369 Bytes in size.

BACKGROUND

(2) Salidroside, also known as tyrosol 8-O-glucoside, is naturally produced by plants within the *Rhodiola* genus. Salidroside is of particular interest and value because of its unique reported biological activities (Cifani et al., 2010; Guan et al., 2012; Panossian et al., 2014). However, commercially available salidroside in its pure form is currently obtained through a lengthy purification process from its native plant host, which poses a significant bottleneck hindering further clinical development of salidroside as a potential therapeutic agent. Accordingly, improved methods of making salidroside are needed.

SUMMARY

(3) Salidroside is a bioactive tyrosine-derived phenolic natural product found in medicinal plants under the *Rhodiola* genus. In addition to their anti-fatigue and anti-anoxia roles in traditional medicine, *Rhodiola* total extract and salidroside have also displayed medicinal properties as anti-cardiovascular disease, and anti-cancer, agents. The resulting surge in global demand of *Rhodiola* plants and salidroside has driven some species close to extinction.

(4) Described herein is a *Rhodiola* salidroside biosynthetic pathway that was elucidated utilizing comprehensive transcriptomics and metabolomics datasets for *Rhodiola rosea*. This pathway includes a pyridoxal phosphate (PLP)-dependent 4-hydroxyphenylacetaldehyde synthase (4HPAAS) that directly converts tyrosine to 4-HPAA. Genes encoding the subsequent 4-HPAA reductase (4HPAR) and tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT), respectively, were further identified to complete salidroside biosynthesis in *Rhodiola*. As described herein, heterologous production of salidroside can be achieved in yeast *Saccharomyces cerevisiae* as well as in plant *Nicotiana benthamiana* through transgenic expression of *Rhodiola* salidroside biosynthetic genes. Accordingly, the methods and compositions described herein provide useful tools for engineering sustainable production of salidroside in heterologous hosts.

(5) Described herein are vectors and kits that include vectors. Those vectors include a nucleic acid

encoding one or more of a 4-hydroxyphenylacetaldehyde synthase (4HPAAS), a 4-hydroxyphenylacetaldehyde reductase (4HPAR), a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT), and a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). Described herein are methods of using the vectors and kits to make a transgenic host cell having a transgene encoding one or more of a 4HPAAS, a 4HPAR, a T8GT, and a T4GT. Described herein are methods of making one or more of 4-hydroxyphenylacetaldehyde (4-HPAA), tyrosol, tyrosol 8-O-glucoside (salidroside), and icariside D2 in a transgenic host cell. The tyrosol, salidroside, and/or icariside D2 can subsequently be obtained, e.g., by separation and purification processes. A variety of transgenic host cells can be used, such as yeast cells, plant cells, and bacterial cells. In some embodiments, the tyrosol, tyrosol 8-O-glucoside (salidroside), or icariside D2 can be obtained in greater quantities than by purification from the native plant host. In some embodiments, the tyrosol, tyrosol 8-O-glucoside (salidroside), or icariside D2 can be obtained more cost-effectively than by purification from the native plant host.

(6) Certain embodiments provide a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS), wherein the 4HPAAS has at least 70% sequence identity to either SEQ ID NO: 2 (*Rhodiola rosea* 4HPAAS), or a biologically active fragment thereof. The 4HPAAS includes: a) an amino acid residue selected from the group consisting of F, L, I, M and V at a position corresponding to the F residue at position 343 in SEQ ID NO: 2; b) an amino acid residue selected from the group consisting of N and D at a position corresponding to the H residue at position 198 in SEQ ID NO: 2; or c) a combination thereof.

(7) Certain embodiments provide a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR), wherein the 4HPAR includes at least 70% amino acid sequence identity to SEQ ID NO: 4, or a biologically active fragment thereof.

(8) Certain embodiments provide a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the T8GT comprises a plant secondary product glycosyltransferase (PSPG) motif. In some embodiments, the T8GT comprises at least 70% amino acid sequence identity to one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20, or a biologically active fragment thereof.

(9) Certain embodiments provide a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). In some embodiments, the T4GT comprises a plant secondary product glycosyltransferase (PSPG) motif. In some embodiments, the T4GT comprises at least 70% amino acid sequence identity to one or more of SEQ ID NO: 6, SEQ ID NO: 8, SEQ ID NO: 10, SEQ ID NO: 12, and SEQ ID NO: 14, or a biologically active fragment thereof.

(10) Certain embodiments provide a kit that includes: a) a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS); b) a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR); and c) one or more of i) a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) and ii) a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). In some embodiments, the kit includes both a T8GT and a T4GT.

(11) Some embodiments provide a host cell that includes a transgene encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the host cell further includes a transgene encoding 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some embodiments, the host cell further includes a transgene encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, the host cell further includes a transgene encoding both a 4HPAR and a 4HPAAS. In some embodiments, a single transgene encodes multiple genes, such as one or more of the T8GT, the 4HPAR, and the 4HPAAS. In some embodiments, separate transgenes encode one or more of T8GT, 4HPAR, and 4HPAAS.

(12) Some embodiments provide a host cell that includes a transgene encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, the host cell further includes a transgene encoding 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some

embodiments, the host cell further includes a transgene encoding tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the host cell further includes a transgene encoding tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). In some embodiments, the host cell further includes a transgene encoding both a 4HPAR and a T8GT. In some embodiments, the host cell further includes a transgene encoding both a 4HPAR and a T4GT. In some embodiments, a single transgene encodes multiple genes, such as one or more of the 4HPAAS, the 4HPAR, the T8GT, and the T4GT. In some embodiments, separate transgenes encode one or more of the 4HPAAS, the 4HPAR, the T8GT, and the T4GT.

(13) Some embodiments provide a method of making a transgenic host cell. The method can include introducing a vector into the host cell, wherein the vector includes a nucleic acid encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). The method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR). The method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS).

(14) Some embodiments provide a method of making a transgenic host cell. The method can include introducing a vector into the host cell, wherein the vector includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). The method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR). The method can further include introducing into the host cell a vector that includes a nucleic acid encoding tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) or a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). In some embodiments, the method can further include introducing into the host cell a vector that includes a nucleic acid encoding a T8GT and a nucleic acid encoding a T4GT.

(15) Certain embodiments provide a method of making tyrosol 8-O-glucoside (salidroside). In some embodiments, the salidroside is made in a host cell. In certain embodiments, the salidroside is made in a cell-free system or cell lysate. The method can include expressing in a host cell a transgene that encodes a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the host cell includes tyrosol, either produced endogenously or provided to the cell exogenously.

(16) In some embodiments, the host cell further expresses a transgene that encodes a 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some embodiments, the host cell further expresses a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, tyrosol is secreted by the host cell into the cell culture media, from which it can be obtained.

(17) Certain embodiments provide a method of making 4-hydroxyphenylacetaldehyde (4-HPAA). In some embodiments, the 4-HPAA is made in a host cell. In some embodiments, the 4-HPAA is made in a cell-free system or lysate. The method can include expressing in the host cell a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, the host cell includes L-tyrosine, produced endogenously or provided to the cell exogenously. In some embodiments, the method further includes making tyrosol in the host cell, and the host cell further expresses a transgene encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some embodiments, tyrosol is secreted by the host cell into the cell culture media, from which it can be obtained.

(18) In some embodiments, the host cell is a yeast cell, such as *Saccharomyces cerevisiae*. In some embodiments, the host cell is a plant cell, such as a cell from a *Nicotiana benthamiana* plant. In some embodiments, the host cell is a bacterial cell, such as *Escherichia coli* or *Agrobacterium tumefaciens*.

(19) In some embodiments, nucleic acids encoding two or more of 4HPAAS, 4HPAR, T8GT, and T4GT are included in a single vector. In some embodiments, the transgene encoding an enzyme

(e.g., 4HPAAS, 4HPAR, T8GT, and T4GT) can be integrated into the genome of the host transgenic cell.

(20) Certain embodiments provide a transgenic plant, such as a *Nicotiana benthamiana* plant, that includes a transgene encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT).

(21) Some embodiments provide a transgenic plant, such as a *Nicotiana benthamiana* plant, that includes a transgene encoding a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT).

(22) Some embodiments provide a transgenic plant, such as a *Nicotiana benthamiana* plant, that includes a transgene encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS).

(23) Certain embodiments provide a transgenic plant, such as a *Nicotiana benthamiana* plant, that includes a transgene encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR).

(24) Certain embodiments provide an isolated deoxyribonucleic acid (DNA) coding sequence encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the nucleic acid includes SEQ ID NO: 13. In some embodiments, the nucleic acid includes SEQ ID NO: 15. In some embodiments, the nucleic acid includes SEQ ID NO: 17. In some embodiments, the nucleic acid includes SEQ ID NO: 19.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

(2) The foregoing will be apparent from the following more particular description of example embodiments, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating embodiments.

(3) FIGS. 1A-C show salidroside biosynthesis in *R. rosea*. FIG. 1A is a photograph of greenhouse-grown *R. rosea*. FIG. 1B is a chromatogram showing metabolic profiling of *R. rosea* root and crown tissues by LC-HRAM-MS. Enrichment of tyrosol and salidroside is observed in the root. Extracted ion chromatogram (XIC) is shown with mass windows set to display the [M-H].sup.- ion for tyrosol and the [M+NH.sub.4].sup.+ ion for salidroside. The identity of the metabolites was verified in comparison to authentic standards. FIG. 1C is a schematic showing alternative salidroside biosynthetic pathways in *Rhodiola*.

(4) FIGS. 2A-C show identification and characterization of the Rr4HPAAS. FIG. 2A is a simplified maximum likelihood (ML) phylogenetic tree of land plant AAADs. A fully annotated version of this tree is shown in FIG. 11. The three major groups of the tree have been annotated as the basal (green), TyDC (blue) and TDC (red) clades based on taxonomic distribution, cladding, and conservation of the substrate-specifying active site residue. Representative characterized enzymes are labeled at the tree branches, while the *R. rosea* TDC, AAS and 4HPAAS are displayed in bold. The scale measures evolutionary distances in substitutions per amino acid. FIG. 2B is LC-UV chromatograms of the reaction product of L-tyrosine and Rr4HPAAS enzyme (with and without NaBH.sub.4 reduction) in comparison to enzyme assay conducted using PsTyDC as a control. The identity of the products was verified by comparison with authentic standards. FIG. 2C is a graph showing kinetic characterization of Rr4HPAAS against various aromatic amino acid substrates.

(5) FIGS. 3A-C show identification and characterization of two *R. rosea* 4HPARs. FIG. 3A is a simplified ML phylogenetic tree of angiosperm ADHs. A fully annotated version of this tree is shown in FIG. 14. Major clades are annotated based on representative characterized enzymes when possible. The two *R. rosea* 4HPARs and the previously characterized SIPARs are labeled at the tree branches. The scale measures evolutionary distances in substitutions per amino acid. FIG. 3B is

LC-MS chromatograms of the reaction product of 4-HPAA and 0.2 μ g recombinant Rr4HPAR1 after incubation for various time points. FIG. 3C is LC-MS chromatograms of the reaction product of 4-HPAA and 15 μ g recombinant Rr4HPAR2 after incubation for various time points. The identity of the tyrosol product was verified by comparison with an authentic standard.

(6) FIGS. 4A-C show identification and characterization of *R. rosea* tyrosol-modifying UGTs. FIG. 4A is a maximum likelihood (ML) phylogenetic tree of 34 *R. rosea* UGTs together with 88 full-length UGTs encoded by the *A. thaliana* genome. UGTs that show T4GT and T8GT activities are denoted by black circles and stars, respectively. Bootstrap values (based on 500 replicates) are indicated at the major nodes. The scale measures evolutionary distances in substitutions per amino acid. FIG. 4B a chart showing relative in vivo T4GT and T8GT activities of *R. rosea* UGTs as examined in engineered yeast. FIG. 4C is a graph showing Michaelis-Menten kinetic characterization of four *R. rosea* tyrosol-modifying UGTs.

(7) FIGS. 5A-C show heterologous production of tyrosine-derived metabolites in transgenic *N. benthamiana* as detected by LC-HRAM-MS. FIG. 5A is a chromatogram showing that *N. benthamiana* transiently expressing Rr4HPAAS or Pc4HPAAS produces both salidroside and icaricide D2. FIG. 5B is a chromatogram showing that *N. benthamiana* transiently expressing PsTyDC produces tyramine. FIG. 5C is a chromatogram showing that *N. benthamiana* leaves transiently co-expressing Rr4HPAAS and RrT4GT or RrT8GT produce predominantly icaricide D2 or salidroside, respectively. XICs are shown with mass windows set to display the [M+NH₄]⁺ ion for salidroside and icaricide D2, and the [M+H]⁺ ion for tyramine. The identity of the metabolites was verified by comparison with authentic standards.

(8) FIG. 6 shows the chemical structures of a number of *Rhodiola* glycosylated natural products.

(9) FIG. 7 is a multiple sequence alignment highlighting the sequence regions that can influence enzyme substrate selectivity in select plant AAAD family members. Sequences represent various enzymes from the TyDC and TDC clades. The residue framed in black (identified as Gly 370 for *C. roseus*) can impact substrate selectivity (indolic vs. phenolic). Columns framed in blue indicate greater than 70% conservation of residue physico-chemical properties. Identical amino acids are in white font boxed in red, while similar residues are displayed in red font. FIG. 7 includes SEQ ID Nos. 207 through 219, in order from top-to-bottom.

(10) FIGS. 8A-B show total ion count of the root and crown *R. rosea* extractions. FIG. 8A is a chromatogram of positive ion mode metabolites. FIG. 8B is a chromatogram of negative ion mode metabolites.

(11) FIG. 9 is a chromatogram showing relative abundance of rosavin between *R. rosea* tissue types. The differential location of the natural product rosavin [M+NH₄]⁺ ion between *R. rosea* root and crown. The identity of rosavin was verified by comparison to an authentic standard.

(12) FIG. 10 is a multiple sequence alignment highlighting the residue that dictate decarboxylation and aldehyde synthase chemistry in plant AAADs family members. Sequences represent plant AAADs performing either decarboxylation chemistry or aldehyde synthase chemistry (highlighted in green). The three identified *R. rosea* AAAD sequences are also displayed. Investigation of the activity influencing residue (boxed in black) suggests that the *R. rosea* sequence from the TDC clade likely catalyzes decarboxylation chemistry while the basal and TyDC clade *R. rosea* AAS catalyze aldehyde synthase chemistry. Columns framed in blue indicate greater than 70% conservation of residue physico-chemical properties. Identical amino acids are in white font boxed in red, while similar residues are displayed in red font. FIG. 10 includes SEQ ID Nos. 220 through 240, in order from top-to-bottom.

(13) FIG. 11 is a phylogenetic tree of embryophyte AAADs. This tree is populated with sequences from all Phytozome V12 embryophyte species, the three AAAD like sequences from the *R. rosea* transcriptomes (shown in bold) and all attainable characterized NCBI AAAD sequences (also shown in bold). Green, red and blue branches correspond to the basal, TDC and TyDC clades, respectively. These clades were determined through the application of the indolic substrate

selective active site glycine (red clade), the phenolic substrate selective serine (blue clade), their taxonomic distribution (green clade exists in all sampled species and is most closely related to chlorophytes species) and representative characterized sequences.

(14) FIG. 12 is a graph showing relative hydrogen peroxide production for the Rr4HPAAS, the RrAAS and the PsTyDC. 100 μ L reaction mixtures containing 50 μ g of recombinant enzyme, 2 mM L-tyrosine, 50 mM Tris pH 8.0 and 200 μ M PLP were incubated at 30° C. for various time points prior to quenching with 100 μ L of 0.8 M formic acid. Hydrogen peroxide levels of quenched reaction mixtures were subsequently analyzed using Pierce Quantitative Peroxide Assay Kit against a standard curve of hydrogen peroxide.

(15) FIG. 13 is a chart showing relative TPM for the Rr4HPAAS transcript between the root and crown transcriptomes.

(16) FIG. 14 is a phylogenetic tree of angiosperm ADHs. This tree is populated with sequences from various Phytozome V12 angiosperm species, ADLs like sequences from the *R. rosea* transcriptomes and two characterized SLPAR sequences. Characterized *A. thaliana* enzymes, the two SLPARs and the two Rr4HPARs are show in bold. The different colors have been applied to distinguish between various clades.

(17) FIG. 15 is a chromatogram showing the enzymatic reduction of phenylacetaldehyde to phenylethyl alcohol by Rr4HPAR1 and Rr4HPAR2. Chromatogram of positive ion mode metabolites between 50 and 300 m/z show the depletion of phenylacetaldehyde and production of phenylethyl alcohol when exposed to NADPH and Rr4HPAR1 or RrPAR2. Reactions were carried out in 200 μ L 50 mM Tris, pH 8.0 in the presence of 2 mM phenylacetaldehyde, 5 mM NADPH and 50 μ g of recombinant enzyme. The reactions were incubated at 30° C. for 25 minutes prior to quenching with 200 μ L of 0.8 M formic acid, extracted with 100 μ L of ethyl acetate and analyzed by gas chromatography-mass spectrometry. Phenylethyl alcohol was verified by comparison to an authentic standard.

(18) FIG. 16 is a chromatogram showing the enzymatic reduction of 4-HPAA to tyrosol by Rr4HPAR1 and Rr4HPAR2. LC-UV chromatograms of products generated from coupled enzyme assays conducted using Rr4HPAAS in combination with Rr4HPAR1 or Rr4HPAR2, respectively. Enzyme assay conducted using Rr4HPAAS alone is included as a control. 100 μ L reaction mixtures containing 50 mM Tris pH 8.0, 4 mM tyrosine, 2 μ g cataylase and 100 μ g of Rr4HPAAS were incubated at 30 degrees C. for 1 hour. 10 mM NADPH and 10 μ g of Rr4HPAR1 or Rr4HPAR2 was then added and incubated for an additional 15 minutes at which point the reactions were quenched with an equal volume of 0.8 M formic acid and analyzed by LC-UV. The identity of the product was verified by comparing the elution profile and UV spectrum to that of an authentic tyrosol standard.

(19) FIG. 17 is a phylogenetic tree of the 113 curated full-length non redundant *R. rosea* transcriptome UGTs. Sequences profiled for tyrosol glycosylation activity in yeast have been annotated RrUGT1-34.

(20) FIG. 18 is a chromatogram of the tyrosol [M-H]⁻ ion generated in transgenic yeast expressing the Rr4HPAAS, the Rr4HPAAS+Rr4HPAR1 or Rr4HPAAS+Rr4HPAR2. The identity was verified by comparison to commercially purchased tyrosol.

(21) FIGS. 19A-B show *R. rosea* tyrosol UGTs as compared to previously characterized *Rhodiola* tyrosol UGTs. FIG. 19A is a chromatogram of the tyrosol glycoside [M+NH₄]^{sup.+} production using the newly described RrUGT3, RrUGT33 and the previously described *R. sachalinensis* UGTs (GenBank: AAS55083 and EU567325). FIG. 19B is a graph showing relative icariside D2 and salidroside production from RrT8HGT, RrT4GHT RsAAS55083 and RsEU567325. The identity of the ions was confirmed by comparison to NMR verified standards.

(22) FIG. 20 is the structure of salidroside.

(23) FIG. 21 is the structure of icariside D2.

(24) FIG. 22 is a .sup.1H NMR spectrum (400 MHz, CDCl₃) of salidroside isolated from *N*.

benthamiana leaves overexpressing *R. rosea* salidroside biosynthetic genes. δ : 9.16 (1H, s, OH), 7.03 (2H, d, $J=8.4$, 4-H, 8-H), 6.65 (2H, d, $J=8.4$, 5-H, 7-H), 4.92 (3H, m, Glu-OH), 4.47 (1H, s, Glu-OH), 4.16 (1H, d, $J=7.6$, 1'-H), 3.87 (1H, m, 1-H), 3.65 (1H, m, 6'-H), 3.56 (1H, m, 1-H), 3.42 (1H, m, 6'-H), 3.12 (1H, m, 3'-H), 3.07 (1H, m, 5'-H), 3.04 (1H, m, 4'-H), 2.95 (1H, m, 2'-H), 2.73 (2H, m, 2-H).

(25) FIG. 23 is a ^{13}C NMR spectrum (100 MHz, CDCl_3) of salidroside isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes. δ : 155.6 (6-C), 129.7 (4, 8-C), 128.6 (3-C), 115.0 (5, 7-C), 102.8 (1'-C), 76.9 (3'-C), 76.8 (5'-C), 73.4 (2'-C), 70.1 (1-C), 69.9 (4'-C), 61.1 (6'-C), 34.8 (2-C).

(26) FIG. 24 is a heteronuclear multiple bond correlation (HMBC) spectrum of salidroside isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes.

(27) FIG. 25 is a ^1H NMR spectrum (400 MHz, CDCl_3) of icariside D2 isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes. δ : 7.11 (2H, d, $J=8.8$, 4-H, 8-H), 6.92 (2H, d, $J=8.8$, 5-H, 7-H), 5.28 (1H, s, Glu-OH), 5.09 (1H, s, Glu-OH), 5.02 (1H, s, Glu-OH), 4.78 (1H, d, $J=7.2$, 1'-H), 4.61 (1H, m, 1-H), 4.56 (1H, m, 1-H), 3.68 (1H, s, 6'-H), 3.55 (1H, m, OH), 3.45 (1H, s, 6'-H), 3.14-3.32 (4H, m, 2', 3', 4', 5'-H), 2.66 (2H, m, 2-H).

(28) FIG. 26 is a ^{13}C NMR spectrum (100 MHz, CDCl_3) of icariside D2 isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes. δ : 155.8 (6-C), 132.7 (3-C), 129.7 (4, 8-C), 116.1 (5, 7-C), 100.6 (1'-C), 77.0 (3'-C), 76.6 (5'-C), 73.3 (2'-C), 69.7 (4'-C), 62.4 (1-C), 60.7 (6'-C), 38.2 (2-C).

(29) FIG. 27 is a chromatogram of the tyrosol $[\text{M}-\text{H}]^+$ ion generated in transgenic *N. benthamiana* expressing the Rr4HPAA. The addition of either the RrT4HGT or the RrT8HGT depletes the tyrosol substrate in the production of icariside D2 or salidroside. The identity of the ions was verified against authentic standards.

(30) FIG. 28 is a chart showing titer of salidroside producing *S. cerevisiae* strains with and without substrate feeding. The first bar of the bar graph represents salidroside titer in wild type (WT) BY4743 yeast while the second bar demonstrates salidroside titer from the yeast strains expressing the native Rr4HPAAS and RrT8GT genes in separate pTEF 2 μ plasmids. The third bar of the graph illustrates the salidroside titer from the *S. cerevisiae* strain transformed with a pTDH3 promoter 2 μ multi gene plasmid containing coRr4HPAAS and coRrT8GT genes. The fourth and fifth bars show salidroside production from the aforementioned codon optimized multi gene plasmid with the addition of either 4 mM L-tyrosine or 4 mM tyrosol. The final bar shows the salidroside production from a strain containing the multi gene coRr4HPAAS and coRrT8GT plasmid additionally transformed with a second 2 μ pTDH3 ARO4 K229L and ARO7 G141S multi gene plasmid.

(31) FIG. 29A-B are chromatograms of the salidroside $[\text{M}+\text{NH}_4]^+$ ion generated in transgenic *S. cerevisiae*. FIG. 29A is a chromatogram of salidroside production in wild type (WT), native Rr4HPAAS and RrT8GT expressing, coRr4HPAAS and coRrT8GT expressing or ARO4 K229L, ARO7 G141S, coRr4HPAAS and coRrT8GT expressing *S. cerevisiae* strains. FIG. 29B is a chromatogram of salidroside production in *S. cerevisiae* expressing coRr4HPAAS and coRrT8GT with and without the addition of L-tyrosine and tyrosol.

(32) FIG. 30 is a graph showing total ion count of salidroside producing transgenic *S. cerevisiae*. Salidroside, labeled in the chromatogram, appears as one of the principle metabolites.

(33) FIG. 31 is a multiple sequence alignment of key residues within biochemically characterized plant AAADs. The multiple sequence alignment of FIG. 31 shows portions of the full alignment of FIG. 32. FIG. 31 includes SEQ ID Nos. 241 through 257 corresponding to sequences beginning at position 190, in order from top-to-bottom; and SEQ ID Nos. 275-291 corresponding to sequences beginning at position 332, in order from top-to-bottom.

(34) FIG. 32 is a multiple sequence alignment of key residues within biochemically characterized plant AAADs. FIG. 32 includes SEQ ID Nos. 258 through 274, in order from top-to-bottom.

(35) FIG. 33 is a chart showing sequence conservation for plant AAAD activity dictating residues.

Multiple sequence alignments of the queried AAAD sequences evaluated for active site conservation using WebLogo. Polar amino acids are green, basic amino acids are blue, acidic amino acids are red and hydrophobic amino acids are black. The y-axis units (bits) display the maximum entropy for the given residue. The representative residues from the Rr4HPAAS MF674522 sequence are listed below with residue numbers.

(36) FIGS. 34A-C show product formation of PsTyDC and mutants. FIG. 34A is a chromatogram showing the reduced enzyme product of tyrosine incubated with wildtype PsTyDC. FIG. 34B is a chromatogram showing the reduced enzyme product of tyrosine incubated with PsTyDC Y350F. FIG. 34C is a chromatogram showing the reduced enzyme product of tyrosine incubated with PsTyDC H204N.

(37) FIG. 35 is a depiction of active site conformations of *Catharanthus roseus* tryptophan decarboxylase. In this homodimer, the A chain is shown in green while the B chain is shown in blue. The active site ligand (tryptophan) is shown in yellow. The active site lysine bound pyridoxal phosphate (LLP) cofactor is visible in the B chain.

DETAILED DESCRIPTION

(38) A description of example embodiments follows.

(39) *Rhodiola* and Salidroside Biosynthesis

(40) The *Rhodiola* genus consists of approximately 90 species of high-altitude and cold tolerant perennial plants of the Crassulaceae family native to the arctic regions of Eurasia and North America (FIG. 1A). Select species from this genus have a long history in traditional medicine with purported roles in bolstering immunity, memory and learning, while ameliorating depression, altitude sickness and fatigue (Fu, 2009; Lei et al., 2006). Recent studies of *Rhodiola* extract have also demonstrated antioxidant and anti-inflammatory properties with potential applications in the prevention of cardiovascular diseases and cancer (Gauger et al., 2010; Khanum et al., 2005; Skopinska-Rozewska et al., 2008; Tu et al., 2008; Zhang et al., 2007). Extensive phytochemical analysis of *Rhodiola* has identified a number of specialized glycosides, including rosiridin, rhodionin, rosarin, rosin, rosavin and salidroside (FIG. 6) (Du and Xie, 1995; Rohloff, 2002; Yang et al., 2012; Yousef et al., 2006). Salidroside, or tyrosol 8-O-glucoside, is of particular interest and value because of its unique reported biological activities (Cifani et al., 2010; Guan et al., 2012; Panossian et al., 2014). However, commercially available salidroside in its pure form is currently obtained through a lengthy purification process from its native plant host, which poses a significant bottleneck hindering further clinical development of salidroside as a potential therapeutic agent. Moreover, surging global demand of wild *Rhodiola* plants as a herbal supplement has led to overharvesting of these ecologically vulnerable plants from their native habitats with some species now threatened by extinction (Booker et al., 2016; Dorji, 2016).

(41) Metabolic engineering is a promising approach to gain access to high-value plant natural products as an alternative to direct compound isolation from plant hosts (O'Connor, 2015). Previous attempts to engineer salidroside biosynthesis in heterologous hosts have utilized a selection of plant and yeast enzymes to assemble artificial salidroside biosynthetic pathways (Bai et al., 2014; Chung et al., 2017). Although these studies demonstrated the feasibility of engineering salidroside production in bacterial hosts (Bai et al., 2014), an unresolved native salidroside biosynthetic pathway in planta hinders further development and improvement of salidroside biosynthetic strategies in bacteria and other alternative chassis organisms. In postulated salidroside biosynthetic pathway, the salidroside aglycone tyrosol is generated from tyrosine through sequential decarboxylation, oxidative deamination, and aldehyde reduction reactions, catalyzed by three discrete enzymes, tyrosine decarboxylase (TyDC), monoamine oxidase (MAO) and 4HPAR, respectively (FIG. 1C) (Lan et al., 2013). Tyrosol is then glycosylated at its 8-OH group by a regio-specific uridine 5'-diphospho-glucosyltransferase (UGT) to yield salidroside. Although the proposed salidroside pathway seems plausible, to date, only one enzyme of this proposed pathway, *Rhodiola crenulata* TyDC (RcTyDC) (GenBank AFN89854.1), has been previously recombinantly

expressed and experimentally examined (Lan et al., 2013). Overexpression of this TyDC-like gene in *R. crenulata* hairy roots culture led to increased accumulation of salidroside (Lan et al., 2013). (42) TyDCs, together with tryptophan decarboxylases (TDCs) and aromatic acetaldehyde synthases (AASs), encompass a large family of PLP-dependent enzymes broadly referred to as the plant AAAD family (Facchini et al., 2000; Kaminaga et al., 2006). As their respective names imply, TyDCs, TDCs and AAS catalyze discrete decarboxylation or decarboxylation-deamination reactions using specific aromatic amino acids as substrates.

(43) To resolve *Rhodiola* salidroside biosynthesis, tissue-specific transcriptomics and metabolomics datasets were generated for *R. rosea*. Using a combination of differential expression analysis, phylogenetic analysis, biochemical characterization, and heterologous expression, a set of *Rhodiola* genes encoding 4HPAAS, 4HPAR, and T8GT to complete salidroside biosynthesis from tyrosine were identified. In addition, a number of regio-specific T4GTs capable of producing icaricide D2 were identified. The newly acquired knowledge about phenolic glycoside biosynthesis in *Rhodiola* allowed reconstitution of salidroside or icaricide D2 biosynthesis in yeast *S. cerevisiae* as well as in the plant *N. benthamiana*.

(44) Aromatic Amino Acid Decarboxylases (AAAD) Family of Enzymes

(45) TyDCs, together with tryptophan decarboxylases (TDCs) and aromatic acetaldehyde synthases (AASs), encompass a large family of PLP-dependent enzymes broadly referred to as the plant AAAD family (Facchini et al., 2000; Kaminaga et al., 2006). Thus, the AAAD family encompasses enzymes with aromatic amino acid decarboxylase activity and enzymes with aromatic acetaldehyde synthase activity. (Torrens-Spence et al., 2012; Torrens-Spence et al., 2013). Without wishing to be bound by theory, the catalytic mechanism of the AAAD family of enzymes is contingent on the conformational change of two active site loops, which is illustrated in FIG. 35 with respect to a tryptophan decarboxylase from *Catharanthus roseus*. The large loop from the A chain (342-359) undergoes a dramatic conformational change from a solvent exposed active site “open” conformation to an active site obscured “closed” conformation. Concurrently, a small loop from chain B (201-205) undergoes a crank shaft conformational change to move from a solvent exposed “open” conformation to a pyridoxal phosphate (LLP) associated “closed” conformation. Key residues in these dynamic loops play important roles in the catalytic mechanism of AAAD enzymes. In the tryptophan decarboxylase from *Catharanthus roseus*, tyrosine 348 (Chain A) functions as a catalytic acid to donate a proton to the carbanion intermediate in the decarboxylation reaction mechanism while histidine 203 (Chain B) functions as a molecular chaperon responsible for coordinating and enabling proton donation of the acid tyrosine 348. Substitution of either residue abolishes the protonation and enables a peroxy-alimine intermediate through the attack of molecular oxygen which spontaneously decomposes to yield the corresponding aromatic acetylaldehyde, peroxide and ammonia aldehyde synthase products. Consequently, substitutions at either location function as a primary sequence means for biochemical functional prediction. One of skill in the art will understand that the precise location within the sequence (here, tyrosine at 348 and histidine at 203) varies among related enzymes within the AAAD family.

(46) Nucleic Acids

(47) As used herein, the term “nucleic acid” refers to a polymer comprising multiple nucleotide monomers (e.g., ribonucleotide monomers or deoxyribonucleotide monomers). “Nucleic acid” includes, for example, DNA (e.g., genomic DNA and cDNA), RNA, and DNA-RNA hybrid molecules. Nucleic acid molecules can be naturally occurring, recombinant, or synthetic. In addition, nucleic acid molecules can be single-stranded, double-stranded or triple-stranded. In certain embodiments, nucleic acid molecules can be modified. In the case of a double-stranded polymer, “nucleic acid” can refer to either or both strands of the molecule.

(48) The terms “nucleotide” and “nucleotide monomer” refer to naturally occurring ribonucleotide or deoxyribonucleotide monomers, as well as non-naturally occurring derivatives and analogs thereof. Accordingly, nucleotides can include, for example, nucleotides comprising naturally

occurring bases (e.g., adenosine, thymidine, guanosine, cytidine, uridine, inosine, deoxyadenosine, deoxythymidine, deoxyguanosine, or deoxycytidine) and nucleotides comprising modified bases known in the art.

(49) As used herein, “wildtype” refers to the canonical amino acid sequence as found in nature. As those of skill in the art would appreciate, a nucleic acid sequence can be modified, e.g., for codon optimization in a host cell (e.g., bacteria, yeast, and plant host cells).

(50) As used herein, the term “sequence identity,” refers to the extent to which two nucleotide sequences, or two amino acid sequences, have the same residues at the same positions when the sequences are aligned to achieve a maximal level of identity, expressed as a percentage. For sequence alignment and comparison, typically one sequence is designated as a reference sequence, to which a test sequences are compared. The sequence identity between reference and test sequences is expressed as the percentage of positions across the entire length of the reference sequence where the reference and test sequences share the same nucleotide or amino acid upon alignment of the reference and test sequences to achieve a maximal level of identity. As an example, two sequences are considered to have 70% sequence identity when, upon alignment to achieve a maximal level of identity, the test sequence has the same nucleotide or amino acid residue at 70% of the same positions over the entire length of the reference sequence.

(51) Alignment of sequences for comparison to achieve maximal levels of identity can be readily performed by a person of ordinary skill in the art using an appropriate alignment method or algorithm. In some instances, the alignment can include introduced gaps to provide for the maximal level of identity. Examples include the local homology algorithm of Smith & Waterman, *Adv. Appl. Math.* 2:482 (1981), the homology alignment algorithm of Needleman & Wunsch, *J. Mol. Biol.* 48:443 (1970), the search for similarity method of Pearson & Lipman, *Proc. Nat'l. Acad. Sci. USA* 85:2444 (1988), computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, Wis.), and visual inspection (see generally Ausubel et al., *Current Protocols in Molecular Biology*).

(52) When using a sequence comparison algorithm, test and reference sequences are input into a computer, subsequent coordinates are designated, if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters. A commonly used tool for determining percent sequence identity is Protein Basic Local Alignment Search Tool (BLASTP) available through National Center for Biotechnology Information, National Library of Medicine, of the United States National Institutes of Health. (Altschul et al., 1990).

(53) In various embodiments, two nucleotide sequences, or two amino acid sequences, can have at least, e.g., 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity. When ascertaining percent sequence identity to one or more sequences described herein, the sequences described herein are the reference sequences.

(54) Some embodiments of the invention relate to a nucleic acid coding sequence (e.g., dsDNA, cDNA) encoding one or more of the enzymes described herein, including those nucleic acid sequences provided in SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 11, SEQ ID NO: 13, SEQ ID NO: 15, SEQ ID NO: 17, and SEQ ID NO: 19.

(55) Enzymes

(56) As used herein, the term 4-hydroxyphenylacetaldehyde synthase (4HPAAS) refers to an enzyme that catalyzes conversion of L-tyrosine to 4-hydroxyphenylacetaldehyde. Methods and assays for determining whether an enzyme catalyzes conversion of L-tyrosine to 4-hydroxyphenylacetaldehyde are known in the art, and include enzyme activity assays and liquid chromatography to assess retention time of metabolites, as described herein. Chemical structure can also be assessed by nuclear magnetic resonance (NMR) or liquid chromatography-mass

spectrometry. An example of a 4HPAAS is SEQ ID NO: 2, which is the amino acid sequence of a 4HPAAS identified in *Rhodiola rosea* (Rr4HPAAS). In some embodiments, a 4HPAAS has at least about 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 2, or a biologically active fragment thereof. In some embodiments, a 4HPAAS has: a) an amino acid residue selected from the group consisting of F, L, I, M and V at a position corresponding to the F residue at position 343 in SEQ ID NO: 2; b) an amino acid residue selected from the group consisting of N and D at a position corresponding to the H residue at position 198 in SEQ ID NO: 2; or c) a combination thereof. Typically, a 4HPAAS has at least 70% sequence identity to SEQ ID NO: 2, or a biologically active fragment thereof, and also: a) an amino acid residue selected from the group consisting of F, L, I, M and V at a position corresponding to the F residue at position 343 in SEQ ID NO: 2; b) an amino acid residue selected from the group consisting of N and D at a position corresponding to the H residue at position 198 in SEQ ID NO: 2; or c) a combination thereof. An example of a nucleic acid coding sequence that encodes a 4HPAAS is SEQ ID NO: 1, which encodes an amino acid having SEQ ID NO: 2. Many different nucleic acids can encode the 4HPAAS of SEQ ID NO: 2 due to the degeneracy of the genetic code. Nucleic acids can also differ from SEQ ID NO: 1, for example, as a result of one or more substitutions (e.g., silent substitutions).

(57) In some embodiments, modified enzymes can be used in the methods and host cells described herein to provide 4HPAAS activity in those host cells and methods. Typically, those modified enzymes have a) an amino acid residue selected from the group consisting of F, L, I, M and V at a position corresponding to the F residue at position 343 in SEQ ID NO: 2; b) an amino acid residue selected from the group consisting of N and D at a position corresponding to the H residue at position 198 in SEQ ID NO: 2; or c) a combination thereof. In certain embodiments, modified *Papaver somniferum* tyrosine decarboxylase (PsTyDC) enzymes comprising a substitution of the active site histidine (e.g., with N or D) at the position corresponding to the H residue at position 198 in SEQ ID NO: 2, and/or the active site tyrosine (e.g., with F, L, I, M or V) corresponding to the F residue at position 343 in SEQ ID NO: 2, can be used in the methods and host cells described herein to provide 4HPAAS activity in those host cells and methods. In some embodiments, modified nucleic acids encoding the modified enzymes can be used in the vectors, kits, and methods described herein. In some embodiments, those nucleic acids may be codon optimized for expression in a host cell.

(58) As used herein, the term 4-hydroxyphenylacetaldehyde reductase (4HPAR) refers to an enzyme that catalyzes conversion of 4-hydroxyphenylacetaldehyde to tyrosol. Methods and assays for determining whether an enzyme catalyzes conversion of 4-hydroxyphenylacetaldehyde to tyrosol are known in the art, and include enzyme activity assays and liquid chromatography to assess retention time of metabolites, as described herein. Chemical structure can also be assessed by nuclear magnetic resonance (NMR) or liquid chromatography-mass spectrometry. An example of a 4HPAR is SEQ ID NO: 4, which is the amino acid sequence of a 4HPAR identified in *Rhodiola rosea* (Rr4HPAR). In some embodiments, a 4HPAR has at least about 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 4, or a biologically active fragment thereof. An example of a nucleic acid that encodes a 4HPAR is SEQ ID NO: 3, which encodes an amino acid having SEQ ID NO: 4. Many different nucleic acids can encode the 4HPAR of SEQ ID NO: 4 due to the degeneracy of the genetic code. Nucleic acids can also differ from SEQ ID NO: 3, for example, as a result of one or more substitutions (e.g., conservative substitutions, non-conservative substitutions), deletions, or insertions, or a combination thereof, with respect to the wild-type Rr4HPAR sequence (SEQ ID NO: 3).

(59) As used herein, the term tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) refers to an enzyme that catalyzes conversion of tyrosol to tyrosol 8-O-glucoside (salidroside). Methods and assays for determining whether an enzyme catalyzes conversion of tyrosol to tyrosol 8-O-glucoside

(salidroside) are known in the art, and include enzyme activity assays and liquid chromatography to assess retention time of metabolites, as described herein. Chemical structure can also be assessed by nuclear magnetic resonance (NMR) or liquid chromatography-mass spectrometry. Examples of T8GTs are SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20, which are the amino acid sequences of T8GTs identified in *Rhodiola rosea* (RrT8GTs). In some embodiments, a T8GT has at least about 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20, or biologically active fragments thereof. Examples of nucleic acids that encode T8GTs are SEQ ID NO: 13, SEQ ID NO: 15, SEQ ID NO: 17, and SEQ ID NO: 19, which encode amino acids having SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20, respectively. Many different nucleic acids can encode the T8GTs due to the degeneracy of the genetic code. Nucleic acids can also differ, for example, as a result of one or more substitutions (e.g., silent substitutions), with respect to any of the wild-type RrT8GT nucleic acid sequences.

(60) As used herein, the term tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT) refers to an enzyme that catalyzes conversion of tyrosol to tyrosol 4-O-glucoside (icaraside D2). Methods and assays for determining whether an enzyme catalyzes conversion of tyrosol to tyrosol 4-O-glucoside (icaraside D2) are known in the art, and include enzyme activity assays and liquid chromatography to assess retention time of metabolites, as described herein. Chemical structure can also be assessed by nuclear magnetic resonance (NMR) or liquid chromatography-mass spectrometry. Examples of T4GTs are SEQ ID NO: 6, SEQ ID NO: 8, SEQ ID NO: 10, SEQ ID NO: 12, and SEQ ID NO: 14, which are the amino acid sequences of T4GTs identified in *Rhodiola rosea* (RrT4GTs). In some embodiments, a T4GT has at least about 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to one or more of SEQ ID NO: 6, SEQ ID NO: 8, SEQ ID NO: 10, SEQ ID NO: 12, and SEQ ID NO: 14, or biologically active fragments thereof. Examples of nucleic acids that encode T8GTs are SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 11, and SEQ ID NO: 13, which encode amino acids having SEQ ID NO: 6, SEQ ID NO: 8, SEQ ID NO: 10, SEQ ID NO: 12, and SEQ ID NO: 14, respectively. Many different nucleic acids can encode the T4GTs due to the degeneracy of the genetic code. Nucleic acids can also differ, for example, as a result of one or more substitutions (e.g., silent substitutions) with respect to any of the wild-type RrT4GT nucleic acid sequences.

(61) Vectors

(62) The terms “vector”, “vector construct” and “expression vector” mean the vehicle by which a DNA or RNA sequence (e.g. a foreign gene) can be introduced into a host cell, so as to transform the host and promote expression (e.g. transcription and translation) of the introduced sequence. Vectors typically comprise the DNA of a transmissible agent, into which foreign DNA encoding a protein is inserted by restriction enzyme technology. A common type of vector is a “plasmid”, which generally is a self-contained molecule of double-stranded DNA that can readily accept additional (foreign) DNA and which can readily introduced into a suitable host cell. A large number of vectors, including plasmid and fungal vectors, have been described for replication and/or expression in a variety of eukaryotic and prokaryotic hosts.

(63) The terms “express” and “expression” mean allowing or causing the information in a gene or DNA sequence to become manifest, for example producing a protein by activating the cellular functions involved in transcription and translation of a corresponding gene or DNA sequence. A DNA sequence is expressed in or by a cell to form an “expression product” such as a protein. The expression product itself, e.g. the resulting protein, may also be said to be “expressed” by the cell. A polynucleotide or polypeptide is expressed recombinantly, for example, when it is expressed or produced in a foreign host cell under the control of a foreign or native promoter, or in a native host cell under the control of a foreign promoter. Gene delivery vectors generally include a transgene (e.g., nucleic acid encoding an enzyme) operably linked to a promoter and other nucleic acid

elements required for expression of the transgene in the host cells into which the vector is introduced. Suitable promoters for gene expression and delivery constructs are known in the art. For bacterial host cells, suitable promoters, include, but are not limited to promoters obtained from the *E. coli* lac operon, *Streptomyces coelicolor* agarase gene (dagA), *Bacillus subtilis* levansucrase gene (sacB), *Bacillus licheniformis* alpha-amylase gene (amyL), *Bacillus stearothermophilus* maltogenic amylase gene (amyM), *Bacillus amyloliquefaciens* alpha-amylase gene (amyQ), *Bacillus licheniformis* penicillinase gene (penP), *Bacillus subtilis* xy1A and xy1B genes, and prokaryotic beta-lactamase gene (See e.g., Villa-Kamaroff et al., *Proc. Natl. Acad. Sci. USA* 75: 3727-3731, 1978), as well as the tac promoter (See e.g., DeBoer et al., *Proc. Natl. Acad. Sci. USA* 80: 21-25, 1983). Examples of promoters for filamentous fungal host cells, include, but are not limited to promoters obtained from the genes for *Aspergillus oryzae* TAKA amylase, *Rhizomucor miehei* aspartic proteinase, *Aspergillus niger* neutral alpha-amylase, *Aspergillus niger* acid stable alpha-amylase, *Aspergillus niger* or *Aspergillus awamori* glucoamylase (glaA), *Rhizomucor miehei* lipase, *Aspergillus oryzae* alkaline protease, *Aspergillus oryzae* triose phosphate isomerase, *Aspergillus nidulans* acetamidase, and *Fusarium oxysporum* trypsin-like protease (See e.g., WO 96/00787), as well as the NA2-tpi promoter (a hybrid of the promoters from the genes for *Aspergillus niger* neutral alpha-amylase and *Aspergillus oryzae* triose phosphate isomerase), and mutant, truncated, and hybrid promoters thereof. Examples of yeast cell promoters can be from the genes for *Saccharomyces cerevisiae* enolase (ENO-1), *Saccharomyces cerevisiae* galactokinase (GAL1), *Saccharomyces cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP), and *Saccharomyces cerevisiae* 3-phosphoglycerate kinase. Other useful promoters for yeast host cells are known in the art (See e.g., Romanos et al., *Yeast* 8:423-488, 1992). The selection of a suitable promoter is within the skill in the art. The recombinant plasmids can also comprise inducible, or regulatable, promoters for expression of an enzyme in cells.

(64) Various gene delivery vehicles are known in the art and include both viral and non-viral (e.g., naked DNA, plasmid) vectors. Viral vectors suitable for gene delivery are known to those skilled in the art. Such viral vectors include, e.g., vector derived from the herpes virus, baculovirus vector, lentiviral vector, retroviral vector, adenoviral vector and adeno-associated viral vector (AAV). Vectors derived from plant viruses can also be used, such as the viral backbones of the RNA viruses Tobacco mosaic virus (TMV), Potato virus X (PVX) and Cowpea mosaic virus (CPMV), and the DNA geminivirus Bean yellow dwarf virus. The viral vector can be replicating or non-replicating.

(65) Non-viral vectors include naked DNA and plasmids, among others. Non-limiting examples include pKK plasmids (Clonetech), pUC plasmids, pET plasmids (Novagen, Inc., Madison, Wis.), pRSET or pREP plasmids (Invitrogen, San Diego, Calif.), or pMAL plasmids (New England Biolabs, Beverly, Mass.), and such vectors may be introduced into many appropriate host cells, using methods disclosed or cited herein or otherwise known to those skilled in the relevant art.

(66) In certain embodiments, the vector comprises a transgene operably linked to a promoter. The transgene encodes a biologically active molecule, such as an enzyme described herein.

(67) To facilitate the introduction of the gene delivery vector into host cells, the vector can be combined with different chemical means such as colloidal dispersion systems (macromolecular complex, nanocapsules, microspheres, beads) or lipid-based systems (oil-in-water emulsions, micelles, liposomes).

(68) Some embodiments relate to a vector comprising a nucleic acid encoding any enzyme described herein. In certain embodiments, the vector is a plasmid, and includes any one or more plasmid sequences such as, e.g., a promoter sequence, a selection marker sequence, or a locus-targeting sequence. Suitable plasmid vectors include p423TEF 2 μ , p425TEF 2 μ , and p426TEF 2 μ . Another suitable vector is pHis8-4 (Whitehead Institute, Cambridge, Massachusetts, United States of America), which is identified as SEQ ID NO: 94. Another suitable vector is pEAQ-HT, which is identified as SEQ ID NO: 95. Another suitable vector is pJKW 1410, which is identified as SEQ ID

NO: 96. pJKW 1410 is a backbone vector used to construct the multi gene yeast expression vector used for solidoside production in the work described in the Examples.

(69) Although the genetic code is degenerate in that most amino acids are represented by multiple codons (called “synonyms” or “synonymous” codons), it is understood in the art that codon usage by particular organisms is nonrandom and biased towards particular codon triplets. Accordingly, in some embodiments, the vector includes a nucleotide sequence that has been optimized for expression in a particular type of host cell (e.g., through codon optimization). Codon optimization refers to a process in which a polynucleotide encoding a protein of interest is modified to replace particular codons in that polynucleotide with codons that encode the same amino acid(s), but are more commonly used/recognized in the host cell in which the nucleic acid is being expressed. In some aspects, the polynucleotides described herein are codon optimized for expression in a bacterial cell, e.g., *E. coli*. In some aspects, the polynucleotides described herein are codon optimized for expression in a yeast cell, e.g., *S. cerevisiae*.

(70) Host Cells

(71) A wide variety of host cells can be used, including fungal cells, bacterial cells, plant cells, insect cells, and mammalian cells.

(72) In some embodiments, the host cell is a fungal cell, such as a yeast cell and an *Aspergillus* spp cell. A wide variety of yeast cells are suitable, such as cells of the genus *Pichia*, including *Pichia pastoris* and *Pichia stipitis*; cells of the genus *Saccharomyces*, including *Saccharomyces cerevisiae*; cells of the genus *Schizosaccharomyces*, including *Schizosaccharomyces pombe*; and cells of the genus *Candida*, including *Candida albicans*.

(73) In some embodiments, the host cell is a bacterial cell. A wide variety of bacterial cells are suitable, such as cells of the genus *Escherichia*, including *Escherichia coli*; cells of the genus *Bacillus*, including *Bacillus subtilis*; cells of the genus *Pseudomonas*, including *Pseudomonas aeruginosa*; and cells of the genus *Streptomyces*, including *Streptomyces griseus*.

(74) In some embodiments, the host cell is a plant cell. A wide variety of cells from a plant are suitable, including cells from a *Nicotiana benthamiana* plant. In other embodiments, the plant belongs to a genus selected from the group consisting of *Arabidopsis*, *Beta*, *Glycine*, *Helianthus*, *Solanum*, *Triticum*, *Oryza*, *Brassica*, *Medicago*, *Prunus*, *Malus*, *Hordeum*, *Musa*, *Phaseolus*, *Citrus*, *Piper*, *Sorghum*, *Daucus*, *Manihot*, *Capsicum*, and *Zea*.

(75) In some embodiments, the host cell is an insect cell, such as a *Spodoptera frugiperda* cell, such as *Spodoptera frugiperda* Sf9 cell line and *Spodoptera frugiperda* Sf21

(76) In some embodiments, the host cell is a mammalian cell.

(77) In some embodiments, the host cell is an *Escherichia coli* cell, and the vector is pHs8-4. In some embodiments, the host cell is a *Nicotiana benthamiana* cell, and the vector is pEAQ-HT. In some embodiments, the cell is a *Saccharomyces cerevisiae* cell, and the vector is a p423TEF 2μ plasmid, a p425TEF 2μ plasmid, or a p426TEF 2μ plasmid.

(78) As used herein, the term “host cell” encompasses cells in cell culture and also cells within an organism (e.g., a plant).

(79) Some embodiments relate to a host cell comprising a vector as described herein. In certain embodiments, the host cell is an *Escherichia coli* cell, a *Nicotiana benthamiana* cell, or a *Saccharomyces cerevisiae* cell.

(80) In some embodiments, the host cells are cultured in a cell culture medium, such as a standard cell culture medium known in the art to be suitable for the particular host cell. In some embodiments, the culture medium is supplemented with one or more of L-tyrosine, 4-hydroxyphenylacetaldehyde (4-HPAA), and tyrosol. In some embodiments, the culture medium is supplemented with tyrosine, for example, between 0.1 mM and 100 mM L-tyrosine. In some embodiments, the culture medium is supplemented with 4-HPAA, for example, between 0.1 mM and 100 mM of 4-HPAA. In some embodiments, the culture medium is supplemented with tyrosol, for example, between 0.1 mM and 100 mM of tyrosol.

(81) Methods of Making Transgenic Host Cells

(82) Described herein are methods of making a transgenic host cell. The transgenic host cells can be made, for example, by introducing one or more of the vector embodiments described herein into the host cell.

(83) In one embodiment, the method comprises introducing into a host cell a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In another embodiment, the method can also include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR) in addition to introducing a nucleic acid encoding a T8GT. In another embodiment, the method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS) in addition to introducing one or more nucleic acids encoding one or more of T8GT and 4HPAR.

(84) In another embodiment, the method comprises introducing a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In another embodiment, the method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR) in addition to introducing a nucleic acid encoding a 4HPAAS. In another embodiment, the method can further include introducing into the host cell a vector that includes one or more of a nucleic acid encoding tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) and a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT), in addition to introducing a nucleic acid encoding one or more of a 4HPAAS and a 4HPAR. In some embodiments, the method can further include introducing into the host cell a vector that includes a nucleic acid encoding a T8GT and a nucleic acid encoding a T4GT, in addition to introducing a nucleic acid encoding one or more of a 4HPAAS and a 4HPAR.

(85) In some embodiments, nucleic acids encoding two or more of 4HPAAS, 4HPAR, T8GT, and T4GT are included in a single vector, such that a single vector encoding one or more enzymes is introduced into a host cell.

(86) In some embodiments, one or more of the nucleic acids are integrated into the genome of the host cell. In some embodiments, the nucleic acids to be integrated into a host genome can be introduced into the host cell using any of a variety of suitable methodologies known in the art, including, for example, CRISPR-based systems (e.g., CRISPR/Cas9; CRISPR/Cpf1), TALEN systems and *Agrobacterium*-mediated transformation. However, as those skilled in the art would recognize, transient transformation techniques can be used that do not require integration into the genome of the host cell. In some embodiments, nucleic acid (e.g., plasmids) can be introduced that are maintained as episomes, which need not be integrated into the host cell genome.

(87) In certain embodiments, the nucleic acid is introduced into a tissue, cell, or seed of a plant cell. Various methods of introducing nucleic acid into the tissue, cell, or seed of plants are known to one of ordinary skill in the art, such as protoplast transformation. The particular method can be selected based on several considerations, such as, e.g., the type of plant used. For example, the floral dip method, as described herein, is a suitable method for introducing genetic material into a plant. In certain embodiments, the nucleic acid can be delivered into the plant by an *Agrobacterium*.

(88) In some embodiments, a host cell is selected or engineered to have increased activity of the synthesis pathway for one or more of L-tyrosine, 4-hydroxyphenylacetaldehyde (4-HPAA) and tyrosol. In some embodiments, a host cell is selected or engineered to have increased activity of the synthesis pathway for L-tyrosine. In some embodiments a host cell may be selected or engineered to have reduced feedback inhibition of one or more enzymes in the L-tyrosine synthesis pathway. In some embodiments, the host cell is engineered to increase uptake of a precursor, such as L-tyrosine, 4-HPAA, or tyrosol, from the medium.

(89) Methods of Making Salidroside, Icariside D2, and Salidroside Precursors

(90) Described herein are methods of making salidroside, icariside D2, and salidroside precursors. Salidroside, icariside D2, and salidroside precursors can be produced by expressing one or more of

the enzymes described herein in a host cell.

(91) Some embodiments provide a method of making tyrosol 8-O-glucoside (salidroside) in a host cell. The method can include expressing in a host cell a transgene that encodes a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the host cell synthesizes tyrosol and includes, either endogenously or transgenically, enzymes to synthesize tyrosol. In some embodiments, tyrosol is provided in the culture media. In some embodiments, the host cell further expresses a transgene that encodes a 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some embodiments, the host cell further expresses a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS).

(92) Some embodiments provide a method of making tyrosol 8-O-glucoside (salidroside) in a host cell. The method can include expressing in a host cell a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS) and a transgene that encodes a 4-hydroxyphenylacetaldehyde reductase (4HPAR). The host cell expresses, either endogenously or transgenically, one or more enzymes that catalyze conversion of tyrosol to tyrosol 8-O-glucoside (salidroside).

(93) Certain embodiments provide a method of making 4-hydroxyphenylacetaldehyde (4-HPAA) in a host cell. The method can include expressing in the host cell a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, the host cell includes L-tyrosine, produced endogenously or provided to the cell exogenously. In some embodiments, L-tyrosine is provided in the cell culture medium. In some embodiments, the method further includes making tyrosol in the host cell, and the host cell further expresses a transgene encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR).

(94) In some embodiments, particularly those optimized for producing salidroside, the host cell can have low or absent T4GT activity in order to reduce competition from T4GT for the substrate tyrosol. In some embodiments, the host cell is engineered to reduce or eliminate expression of T4GT.

(95) In some embodiments, particularly those optimized for producing icaraside D2, the host cell can have low or absent T8GT activity in order to reduce competition from T8GT for the substrate tyrosol. In some embodiments, the host cell is engineered to reduce or eliminate expression of T8GT.

(96) In some embodiments, a host cell (e.g., a bacterial host cell) endogenously expresses enzymes that catalyze the production of salidroside or icaraside D2 from tyrosol. For example, some bacteria express UGTs that exhibit T8GT and/or T4GT activity (Fan et al., 2017). In some embodiments, nucleic acids encoding the bacterial-derived T8GTs can be used in vectors and methods described herein. In some embodiments, host cells and methods can express a T8GT that is a bacterial-derived T8GT.

(97) In some embodiments, one or more copies of one or more of the nucleic acids are integrated into the genome of the host cell. However, as those skilled in the art would recognize, transient transformation techniques can be used that do not require integration into the genome of the host cell.

(98) Methods of obtaining, or extracting, salidroside, icaraside D2, and precursors of salidroside and icaraside D2 are described herein and are well known to one of ordinary skill in the art. For example, as described herein, salidroside, icaraside D2, and/or precursors of salidroside and icaraside D2 can be separated by liquid chromatography. Larger scale separation can be obtained by, e.g., simulated moving bed (SMB) chromatography and/or ion exchange chromatography. Any of the methods described herein can further include isolating salidroside, icaraside D2, and/or a salidroside precursor from a host cell. Any of the methods described herein can include harvesting tissue (e.g., leaves, roots) of a transgenic plant described herein and processing the harvested tissue to obtain salidroside, icaraside D2, and/or a precursor of salidroside and icaraside D2 therefrom.

(99) Values and Ranges

(100) Unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or subrange within the stated ranges in various embodiments, unless the context clearly dictates otherwise. “About” in reference to a numerical value generally refers to a range of values that fall within $\pm 8\%$, in some embodiments $\pm 6\%$, in some embodiments $\pm 4\%$, in some embodiments $\pm 2\%$, in some embodiments $\pm 1\%$, in some embodiments $\pm 0.5\%$ of the value unless otherwise stated or otherwise evident from the context.

EXEMPLIFICATION

Example #1: Results

(101) Generating Metabolomics and Transcriptomics Resources for *R. rosea*

(102) To survey the metabolic profile of *Rhodiola* cultivated under lab conditions, crown tissue (aerial tissue including leaves and stems) and root tissue were collected separately from a three-month old greenhouse-grown *R. rosea* plant (FIG. 1A). The fresh tissues were extracted by 50% methanol, and analyzed by untargeted liquid chromatography high-resolution accurate-mass mass spectrometry (LC-HRAM-MS). This analysis confirmed the presence of tyrosol, rosavin and salidroside in greenhouse-grown *R. rosea*, all of which accumulate at much higher levels in the root compared to the crown (FIGS. 1, 8, and 9).

(103) The higher accumulation of these metabolites in the root suggests that the requisite biosynthetic genes may also obey a similar tissue-specific expression pattern. An RNA-Seq experiment was then performed using total RNAs prepared from the two tissues. This experiment yielded about 30 million paired-end sequencing reads (100×100 bp) per sample. While 84,645 and 105,132 unique transcripts were assembled de novo from the crown and root tissues separately, a total of 128,623 unique transcripts were assembled combining all raw sequencing reads from both tissues. The combined transcriptome was evaluated as 90.3% complete by the metric of Benchmarking Universal Single-Copy Orthologs (BUSCO) (Simao et al., 2015). The Transcripts Per Million (TPM) value of unique transcripts in each tissue type was calculated to infer the relative expression level of the corresponding genes (Li et al., 2010). The identification and prioritization of candidate salidroside biosynthetic genes from the *R. rosea* transcriptome were based upon our hypothetical salidroside biosynthetic model, subsequent large-scale phylogenetic analyses, and the relative expression level of plausible candidate genes in the two examined tissue types. The biochemical function of selected candidate genes was further investigated both in vitro and in vivo.

(104) *R. rosea* Contains a Neofunctionalized 4HPAAS

(105) A BLAST search using PsTyDC as the query against the *R. rosea* transcriptome identified three AAAD homologs. Using the sequence motifs correlating to AAAD substrate specificity and catalytic mechanism (Torrens-Spence et al., 2014; Torrens-Spence et al., 2013), two of the three *R. rosea* AAAD homologs were predicted to possibly function as AASs, and the other is likely to catalyze decarboxylation chemistry (FIG. 10). A phylogenetic analysis including the three *R. rosea* AAAD homologs together with other AAAD sequences from taxonomically diverse plant species was conducted (FIGS. 2B and 11). Whereas the predicted *R. rosea* decarboxylase candidate clusters within the TDC clade (red) containing largely previously known TDCs, the two *R. rosea* AAS candidates fall into two distinct clades, designated as the basal clade (green) and the TyDC clade (blue), respectively (FIGS. 2B and 11). It is noted that the TyDC-type AAS candidate isolated in this study is likely orthologous to the RcTyDC previously reported by Bai et al. (Bai et al., 2014), sharing 96% sequence identity at the protein level.

(106) To experimentally assess the biochemical activities of the two *R. rosea* AAS candidates, full-length open reading frame corresponding to both the basal and TyDC-type AAS candidate genes from *R. rosea* cDNA were cloned. Their encoded proteins were recombinantly expressed in *E. coli*, purified to homogeneity, and tested for enzymatic activity using L-tyrosine as the substrate. Both enzymes readily yield hydrogen peroxide, a co-product of AAS as opposed to canonical TyDC

(Kaminaga et al., 2006), while the TyDC-type AAS candidate exhibits much higher activity than the basal AAS candidate (FIG. 12). To confirm the chemical identity of the AAS reaction products, the enzyme assays were analyzed by LC coupled with a UV detector (FIG. 2B). Incubation of L-tyrosine with both AAS candidate enzymes led to the production of 4-HPAA, which is distinct from the tyramine product yielded by PsTyDC as a control (FIG. 2B). The identity of the 4-HPAA product was further confirmed by sodium borohydride reduction of 4-HPAA to yield tyrosol (FIG. 2B). Notably, the transcript corresponding to the TyDC-type AAS candidate is highly enriched in the root versus the crown (FIG. 13), whereas such pattern was not observed for the basal AAS candidate. In light of these results, the TyDC-type AAS candidate is likely the primary AAS involved in salidroside biosynthesis in *R. rosea* root. The TyDC-type AAS candidate is referred to as Rr4HPAAS hereafter.

(107) The Michaelis-Menten kinetics of Rr4HPAAS was measured against four aromatic amino acids, namely L-tyrosine, L-3,4-dihydroxyphenylalanine (L-DOPA), L-phenylalanine, and L-tryptophan (FIG. 2C and Table 1). Rr4HPAAS demonstrates the highest catalytic efficiency toward L-tyrosine ($k_{\text{sub.cat}}/K_{\text{sub.m}}=11.7 \text{ s.sup.-1 mM.sup.-1}$) followed by L-DOPA ($k_{\text{sub.cat}}/K_{\text{sub.m}}=9.1 \text{ s.sup.-1 mM.sup.-1}$), whereas L-phenylalanine and L-tryptophan are much less preferred substrates (FIG. 2C and Table 1). L-DOPA and any potential phenolic compound derived from it were not detected in the *R. rosea* metabolomics datasets, and thus the kinetic characteristics of Rr4HPAAS is consistent with its role in salidroside biosynthesis. These results also suggest that the previously reported RcTyDC was likely functionally mischaracterized (Bai et al., 2014).

(108) Identification and Biochemical Characterization of *Rhodiola* Phenolic Aldehyde Reductases

(109) To identify *R. rosea* enzymes involved in the next step of salidroside biosynthesis, a BLAST search was conducted using the previously characterized *Solanum lycopersicum* PAR (SlPAR, GenBank: ABR15768.1) as a query (Tieman et al., 2007) against our *R. rosea* transcriptome. A phylogenetic analysis was performed using the returned *R. rosea* hits together with other homologous ADHs from select plant species (FIGS. 3A and 14). This analysis revealed two *R. rosea* ADH homologs, referred to as RrPAR-like1 and RrPAR-like2, that cluster phylogenetically with SlPAR (Tieman et al., 2007), and share 76% and 58% protein sequence identity to SlPAR, respectively. Both genes were cloned from *R. rosea* cDNA as candidate genes encoding 4HPAR.

(110) To examine the biochemical activity of the two 4HPAR candidates, recombinant enzymes were expressed in *E. coli*, purified to homogeneity, and assayed against 4-HPAA or phenylacetaldehyde in the presence of NADPH as the co-substrate. Both enzymes are capable of reducing phenylacetaldehyde to phenylethanol with RrPAR-like1 displaying higher activity (FIG. 15). Likewise, RrPAR-like1 exhibited orders of magnitude higher specific activity towards 4-HPAA ($6.9 \mu\text{mol min.sup.-1 mg.sup.-1}$) than RrPAR-like2 ($8.4 \text{ nmol min.sup.-1 mg.sup.-1}$) (FIGS. 3B, 3C, and 16). Nonetheless, RrPAR-like1 and RrPAR-like2 were renamed as Rr4HPAR1 and Rr4HPAR2, respectively, as both enzymes displayed 4-HPAA reductase activity.

(111) Identification of Regio-Specific Tyrosol-Modifying UGTs from *R. rosea*

(112) To complete the salidroside biosynthetic pathway, candidate UGT genes encoding T8GT were identified. The UGT superfamily is one of the largest enzyme families in the plant kingdom (Li et al., 2001). The plant secondary product glycosyltransferase (PSPG) motif is described in Gachon et al., 2005, particularly at FIG. 2A and associated text. The transcriptome was queried using a UGT superfamily signature motif (Li et al., 2001), and conducted an unbiased phylogenetic analysis using 113 curated non-redundant full-length UGT homologs retrieved from the *R. rosea* transcriptome (FIG. 17). Thirty-four candidate UGT genes were then prioritized for further functional analysis according to a combination of criteria including phylogenetic distribution pattern and transcript levels in the root and crown transcriptome datasets.

(113) To facilitate functional assessment of a large number of UGT candidates, an in vivo tyrosol glycosylation assay in the yeast *S. cerevisiae* was devised. Initial iterations of the tyrosol-producing

yeast strains were generated by transforming wild type *S. cerevisiae* BY4743 with separate 2-micron TEF-promoter expression plasmids containing Rr4HPAAS and Rr4HPAR1, respectively. It was later observed that yeast contains endogenous ADH activity sufficient to reduce 4-hydroxyphenylacetaldehyde produced by Rr4HPAAS to tyrosol. Therefore, the Rr4HPAR1-containing plasmid was omitted in the final tyrosol-producing strain (FIG. 18). Each of the 34 *R. rosea* UGT candidate genes, carried on the yeast 2-micron TEF-promoter expression plasmids, was transformed into the background strain expressing Rr4HPAAS. After auxotrophic selection, colonies were cultured, harvested and subjected to metabolic profiling by LC-HRAM-MS. From this screen, we identified three UGTs (RrUGT 29, 32, and 33) with regio-specific T8GT activity, four UGTs (RrUGT 2, 3, 7, and 13) with regio-specific T4GT activity, and RrUGT17 with both T8GT and T4GT activities (FIG. 4B). Further phylogenetic analysis of the 34 cloned *R. rosea* UGTs against the 88 unique and complete *A. thaliana* UGTs suggests a correlation between the cladding of the UGTs and their respective biochemical activities (FIG. 4A) (Li et al., 2001). The UGTs that contain T4GT activity appear to be phylogenetically diverse with representative enzymes falling into the D, G, E and K groups, while all of identified T8GTs cluster within the G group (FIG. 4A). These results also show that RrUGT 3 and RrUGT33, the most active T4GT and T8GT, respectively, display significantly higher regio-specific tyrosol glycoside-producing activities than the two previously reported UGTs from *R. sachalinensis* (GenBank: AAS55083 and EU567325) (FIG. 19) (Ma et al., 2007; Yu et al., 2011).

(114) Using recombinant enzymes produced and purified from *E. coli*, the kinetic parameters for the salidroside-producing RrUGT29 and RrUGT33 and the icariside D2-producing RrUGT2 and RrUGT3 were measured (FIG. 4C and Table 1). RrUGT33 exhibits the highest T8GT catalytic efficiency with a $k_{\text{sub.cat}}/K_{\text{sub.m}}$ value of $420.6 \text{ s.sup.-1 mM.sup.-1}$ and was subsequently referred to as RrT8GT (Table 1). In contrast, RrUGT3 exhibits the greatest T4GT catalytic efficiency with a $k_{\text{sub.cat}}/K_{\text{sub.m}}$ value of $117.2 \text{ s.sup.-1 mM.sup.-1}$ and was subsequently referred to as RrT4GT (Table 1).

(115) Heterologous Production of Salidroside and Icariside D2 in *N. benthamiana*

(116) To further evaluate the biochemical function of *R. rosea* tyrosol glycoside biosynthetic genes in planta, these genes were expressed in *N. benthamiana* leaves using the *Agrobacterium tumefaciens*-mediated transient protein production technique (Sainsbury et al., 2009) followed by LC-HRAM-MS-based metabolic profiling. To first demonstrate the biochemical function of Rr4HPAAS in planta, Rr4HPAAS alone was transiently expressed in *N. benthamiana* leaves. PsTyDC and the previously reported *Petroselinum crispum* 4HPAAS (Pc4HPAAS, GenBank: AAA33861) (Torrens-Spence et al., 2012) were also tested in parallel as controls. Interestingly, expression of Rr4HPAAS or Pc4HPAAS in *N. benthamiana* led to significant accumulation of both salidroside and icariside D2 in *N. benthamiana* leaves (FIG. 5A). The chemical identity of these compounds was confirmed by both LC-HRAM-MS and nuclear magnetic resonance (NMR) analyses (FIGS. 20-26). This result suggests that 4-HPAA produced by transgenic 4HPAAS can be readily metabolized by endogenous *N. benthamiana* reductase and glycosyltransferase enzymes to yield both salidroside and icariside D2. In contrast, the expression of PsTyDC yielded tyramine in high abundance in *N. benthamiana* leaves with no measurable production of tyrosol glycosides (FIG. 5B). Next, the in planta regio-specificity of RrT8GT and RrT4GT in tyrosol glycosylation was evaluated. Co-expression of either RrT8GT or RrT4GT with Rr4HPAAS led to regio-specific glycosylation of tyrosol and accumulation of salidroside or icariside D2, respectively (FIG. 5C). Meanwhile, the accumulation of free tyrosol was reduced in these plants compared to those with Rr4HPAAS expression alone (FIG. 27). Notably, the paired expression of Rr4HPAAS and one of the two regio-specific *R. rosea* tyrosol glycosyltransferases yielded up to 2% dry weight for salidroside or icariside D2 production in *N. benthamiana* leaves. This set of in planta experiments demonstrate that Rr4HPAAS and regio-specific RrT8GT are specialized metabolic enzymes underpinning salidroside biosynthesis in *Rhodiola*. Although icariside D2 does not naturally

accumulate in *Rhodiola*, the identification of the regio-specific RrT4GT adds to the tool box for metabolic engineering of valuable tyrosol-derived glycosides.

(117) Optimization of Salidroside Production in *S. cerevisiae*

(118) The complete elucidation of salidroside biosynthesis in *Rhodiola* provides new opportunities for bioengineering of sustainable salidroside production in heterologous hosts. Although *N. benthamiana* has been used for the commercial production of high value natural products and recombinant proteins, its scalability currently does not match to industrial yeast fermentation. To increase the salidroside titer in yeast, the Rr4HPAAS and RrT8GT genes were optimized according to *S. cerevisiae* codons, and assembled in a custom 2 μ plasmid for constitutive expression driven by pTDH3 promoter in yeast (FIGS. 28 and 29A). The increased promoter strength and codon optimization of Rr4HPAAS (coRr4HPAAS) and RrT8GT (coRrT8GT) resulted in a 2.5-fold increase in salidroside titer as compared to the initial strain. To probe the potential bottlenecks in salidroside biosynthesis in yeast, we next fed the culture containing the codon optimized construct with either L-tyrosine or tyrosol. Both feeding experiments demonstrated significant increase in salidroside titer, suggesting that improved tyrosine flux may further improve salidroside titer (FIGS. 28 and 29B). Thus, a yeast strain was engineered to include the previously described feedback-insensitive mutants of the yeast L-tyrosine pathway enzymes ARO4 and ARO7 (Gold et al., 2015). Incorporation of both ARO4 K229L and ARO7 G141S into the prior best engineered yeast strain produced salidroside as one of the most abundant metabolites with a titer of 1.5 mg L⁻¹, when grown for 48 h in 4% glucose 2 $\times$ yeast nitrogen base in shake flasks (FIGS. 28, 29A, and 30). In summary, this preliminary metabolic engineering exercise in yeast yielded a prototype salidroside-producing strain, which can be improved through additional rounds of targeted and untargeted genetic modifications to further increase titer.

Example #1: Discussion

(119) As described herein, the *R. rosea* ortholog of the previously reported RcTyDC is a 4HPAAS, which catalyzes the direct conversion of tyrosine to 4-HPAA. This discovery therefore corrected a major long-standing misconception about the biosynthetic route towards tyrosol, an important precursor for many important phenolic natural products in plants (Chapple et al., 1986; Wyk, 2010).

(120) UGTs play important roles in plant specialized metabolism as they alter the solubility, reactivity, bioactivity, intercellular and subcellular transport of a wide array of plant metabolites by glycosylation (Jones and Vogt, 2001). The resulting glycosides also have profound impact on human health with diverse pharmacological and nutraceutical indications (Jones and Vogt, 2001). Since natural product glycosides often contain distinct pharmacokinetic properties as compared to their aglycones, chemical derivatization via glycosylation has received considerable attention in pharmaceutical research (Gantt et al., 2011). Several *R. rosea* UGTs capable of producing salidroside and icariside D2 from the aglycone tyrosol in a regio-specific manner were identified. In this gene-mining process, a library was established containing phylogenetically diverse UGTs from *R. rosea*, which likely contain enzymes responsible for the biosynthesis of other phenolic glycosides from *Rhodiola*, such as rosiridin, rhodionin, rosarin, rosin, and rosavin (FIG. 6).

(121) Unlike bacterial natural product biosynthetic pathways, which are encoded by operons ubiquitously present in bacterial genomes, enzyme-encoding genes of a given plant specialized metabolic pathway often scatter randomly across the plant genome, making metabolic pathway elucidation unattainable simply by genome mining. Plants, like many other multicellular eukaryotes, contain rich tissue types where specific natural products accumulate under developmental and environmental regulations. In recent years, this feature of plant specialized metabolism has been exploited for pathway and enzyme discovery in medicinal plants that lack classical genetic tools (Torrens-Spence et al., 2016). Through mining transcriptomics and metabolomics datasets generated separately from the root and crown tissues of *R. rosea*, candidate salidroside biosynthetic genes were prioritized based on correlation between transcript and

metabolite abundances in these two tissues. Extended phylogenomics analyses of the involved enzyme families further provided additional information that facilitates salidroside biosynthetic gene discovery. The biochemical functions of the identified candidate enzymes were then examined in vitro using recombinantly expressed proteins, and in vivo through expression of the candidate enzymes in heterologous hosts, e.g. yeast and *N. benthamiana* in this case. Collectively, this work describes a rare de novo elucidation of the complete biosynthetic pathway of a given plant natural product. The workflow adopted in this study is generally applicable for future investigation of other largely unexplored specialized metabolic pathways in non-model plants, and will ultimately contribute to a capability of synthesizing structurally diverse plant natural products through the means of metabolic engineering.

Materials and Methods

(122) Reagents

(123) Salidroside, tyrosine, tyramine, tyrosol, phenylacetaldehyde, phenylethyl alcohol, sodium borohydride, NADPH, UDP-glucose, and PLP were purchased from Sigma-Aldrich. 4-HPAA was purchased from Santa Cruz Biotechnology, Inc.

(124) Plant Materials

(125) *R. rosea* seeds were purchased from Horizon Herbs. Seeds were stratified at 4° C. for three days, and germinated in potting soil. *R. rosea*, *P. crispum*, and *N. benthamiana* plants were grown under a 16-h-light/8-h-dark photoperiod at 23° C. in a local greenhouse.

(126) RNA Isolation, Library Preparation, Transcriptome Assembly, cDNA Production and Molecular Cloning

(127) Tissue of seventy-day-old *R. rosea* plants were harvested for total RNA extraction using the Qiagen's RNeasy Mini Kit (Qiagen). RNA quality was assessed by Bioanalyzer (Agilent Technologies). For the RNAseq experiment, strand-specific mRNA libraries were prepared using total RNA prepared separately from the root and crown tissue using the TruSeq Stranded mRNA Library Prep Kit (Illumina), and sequenced on a HiSeq2000 sequencer (Illumina) in paired-end mode (PE100). Sequence FASTQ files were trimmed for sequencing adaptors using Trimmomatic (Bolger et al., 2014) and assembled into de novo transcriptomes using Trinity in strand-specific mode (Grabherr et al., 2011). Gene expression statistics (TPM values) were determined by RSEM (Li and Dewey, 2011). Completeness of the combined *R. rosea* root and crown transcriptome was evaluated using the BUSCO tool, with 'embryophyta_odb9' set as lineage and 'Arabidopsis' set as model species (Simao et al., 2015). Putative coding regions were predicted using Transdecoder (Haas et al., 2013). Transcripts and predicted protein sequences were annotated with TPM values and closest BLAST hits using in-house scripts. Transcriptome mining was performed on a local BLAST server (Anurag Priyam, 2015). First-strand cDNAs were synthesized by RT-PCR using total RNA sample as template and the Invitrogen SuperScript™ III kit (Invitrogen) with the oligo(dT)20 primer. The coding sequences (CDS) of candidate genes were amplified from cDNAs by PCR using gene-specific primers (Table 2). Select *R. rosea* and *R. sachalinensis* genes were also synthesized as gBlocks (IDT) with yeast codon optimization. Gibson assembly was used to ligate PCR amplicons or gBlocks into several base vectors. These include pHis8-4, a bacterial expression vector containing an N-terminal 8×His tag followed by a tobacco etch virus (TEV) cleavage site for recombinant protein production in *E. coli*; pEAQ-HT, a binary vector designed for transient expression of heterologous proteins in *N. benthamiana* (Peyret and Lomonossoff, 2013); p423TEF, p425TEF and p426TEF 2μ plasmids (Mumberg et al., 1995) with various auxotrophic growth markers for constitutive expression in *S. cerevisiae*; and a custom plasmid containing 2μ, pTDH3, tTDH1, HIS3 for constitutive multi gene expression in *S. cerevisiae* (Lee et al., 2015).

(128) Sequence Alignment and Phylogenetic Analysis

(129) The protein multiple sequence alignments were generated using ClustalW2 with default settings (Thompson et al., 2002). ESPript 3.0 (Gouet et al., 2003) was used to display the multiple sequence alignments. The phylogeny was inferred using the Maximum Likelihood method based

on the Poisson correction model (Li, 1965). The bootstrap consensus unrooted trees were inferred from 500 replicates to represent the phylogeny of the analyzed enzyme families (Sanderson and Wojciechowski, 2000). The phylogenetic analysis of the AAAD family includes 242 sequences from the Phytozome V12 embryophyte species with fully sequenced genome (*A. thaliana*, *G. raimondii*, *P. trichocarpa*, *M. domestica*, *M. truncatula*, *E. grandis*, *K. laxiflora*, *S. lycopersicum*, *A. coerulea*, *Z. mays*, *B. distachyon*, *O. sativa*, *Z. marina*, and *A. trichopoda*), the *R. rosea* transcriptome, and previously characterized AAAD proteins. The phylogenetic analysis of ADHs includes 346 PAR homologs from the Phytozome V12 embryophyte species and *R. rosea* transcriptome. The phylogenetic analysis of UGTs contains 113 non-redundant full-length UGT homologs from the *R. rosea* transcriptome. A second UGT tree was also generated using the 34 cloned *R. rosea* UGTs in addition to the 88 full length and unique UGTs from *Arabidopsis thaliana* (Li et al., 2001). Initial trees for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using a JTT model, and then selecting the topology with superior log likelihood value. All phylogenetic analyses were conducted in MEGA7 (Kumar et al., 2016).

(130) *Agrobacterium*-Mediated Transient Expression of Heterologous Proteins in *N. benthamiana*

(131) *A. tumefaciens* (LBA4404) containing the transgene construct was grown to optical density (OD) 600 of 1.6 in 50 mL of YM medium (0.4 g/L yeast extract, 10 g/L mannitol, 0.1 g/L NaCl, 0.2 g/L MgSO₄·7H₂O, 0.5 g/L K₂HPO₄·3H₂O), washed with washing buffer (10 mM MES (2-(N-morpholino)ethanesulfonic acid), pH 5.6), and resuspended in MMA buffer (10 mM MES, pH 5.6, 10 mM MgCl₂, 100 μM acetosyringone) to OD 600 of 0.8. For co-expressing multiple genes, individual *A. tumefaciens* cultures containing the unique transgene constructs were grown, pelleted, and washed separately. The cultures were then resuspended together at a higher optical density so that each individual culture was present at a concentration equivalent to OD 600 of 0.8. 1 mL of culture was used to infiltrate the underside of six-week-old *N. benthamiana* leaves.

(132) Metabolomic Profiling by LC-HRAM-MS

(133) Crown tissue and root tissue of a three-month-old *R. rosea* plant was harvested and stored at -80° C. before subsequent metabolomic analysis. Various transgene-carrying *S. cerevisiae* BY4743 strains and transiently transformed *N. benthamiana* plants were generated to test the activity of candidate genes involved in the tyrosol glycoside biosynthesis. 3 mL of saturated *S. cerevisiae* culture was used to inoculate 50 mL of synthetic minimal media (SD) in a shake flask. After 24 hours of shaking at 30° C., the culture was pelleted by centrifugation, washed with water, and stored at -80° C. before further processing. *N. benthamiana* leaf tissue was harvested 5 days after *Agrobacterium* infiltration and was stored at -80° C. before further processing. Frozen yeast or plant tissue was disrupted with a TissueLyser (Qiagen) using acid-washed metal beads in 50% methanol (500 μL per 100 mg fresh weight). The extracts were then analyzed by LC-HRAM-MS. Metabolite profiling was conducted on a QExactive benchtop orbitrap mass spectrometer equipped with an Ion Max source and a HESI II probe, which was coupled to a Dionex UltiMate 3000 UPLC system (Thermo Fisher Scientific). 2 μL of each sample was injected onto a 150×2.1 mm ZIC-PHILIC column (5 μm particle size, EMD Millipore). Solvent A was 20 mM ammonium carbonate, 0.1% ammonium hydroxide; solvent B was acetonitrile. The column oven and autosampler tray were held at 25° C. and 4° C., respectively. The chromatographic gradient was run at a flow rate of 0.15 mL/min as follows: 0-20 min, linear gradient from 80% to 20% solvent B; 20-20.5 min, linear gradient from 20% to 80% solvent B; 20.5-28 min, hold at 80% solvent B. The mass spectrometer was operated in full-scan, polarity-switching mode with the spray voltage set to 3.0 kV, the heated capillary held at 275° C., and the HESI probe held at 350° C. The sheath gas flow was set to 40 units, the auxiliary gas flow was set to 15 units, and the sweep gas flow was set to 1 unit. The MS data acquisition was performed in a range of 70-1000 m/z, with the resolution set at 70,000, the AGC target at 10 e6, and the maximum injection time at 20 msec. The raw data was converted to

mzML format using MSConvert (Chambers et al., 2012), and analyzed using MetaboAnalyst (Xia and Wishart, 2016) and MZmine2 (Pluskal et al., 2010).

(134) Small Molecule Isolation and NMR

(135) For large-scale compound isolation from *Agrobacterium*-transformed *N. benthamiana* leaves, 15 g (dry weight) of *N. benthamiana* leaves (harvested 5 days post infection) were extracted with 70% EtOH. The solvent was evaporated from the extracts under reduced pressure using a rotary evaporator (Buchi). The residue was suspended in 100 mL of water, and extracted successively with hexane, chloroform and butanol. The water-soluble portion was separated by Sephadex LH20 using a H.sub.2O/MeOH gradient of 0-100% MeOH. Fractions 26-32 and 36-44 were combined separately for further purification by a preparative HPLC (Shimadzu) equipped with a SPD-20A UV-VIS detector and a 150×21.2 mm 100 Å Kinetex 5 µC.sub.18 column (Phenomenex). 7 mg of salidroside and 13 mg of icariside D2 were purified using water (solvent A) and a 60-minute gradient of 5-80% acetonitrile (solvent B) at a flow rate of 10 mL/min. The samples were dried by lyophilization and subjected to NMR analysis in DMSO-d.sub.6. The solution NMR spectra were recorded on a Bruker AVANCE-400 NMR spectrometer with a Spectro Spin superconducting magnet.

(136) Recombinant Protein Production and Purification

(137) BL21(DE3) *E. coli* containing appropriate constructs were grown at 37° C. in terrific broth (TB) to OD 600 of 0.9, induced with 0.15 mM isopropyl-β-D-thiogalactoside (IPTG), and allowed to grow for an additional 20 h at 18° C. Cells were harvested by centrifugation, washed with phosphate buffered saline (PBS) (137 mM NaCl, 2.7 mM KCl, 10 mM Na.sub.2HPO.sub.4 and 1.8 mM KH.sub.2PO.sub.4), resuspended in 150 mL of lysis buffer (50 mM Tris pH 8.0, 0.5 M NaCl, 20 mM imidazole, and 0.5 mM dithiothreitol (DTT)), and lysed with five passes through a M-110L microfluidizer (Microfluidics). The resulting crude protein lysate was clarified by centrifugation prior to Qiagen Ni-NTA gravity flow chromatographic purification. After loading the clarified lysate, His-tagged recombinant protein-bound Ni-NTA resin was washed with 20 column volumes of lysis buffer, and eluted with 1 column volume of elution buffer (50 mM Tris pH 8.0, 0.5 M NaCl, 250 mM imidazole and 0.5 mM DTT). 1 mg of His-tagged TEV protease was added to the eluted protein, followed by dialysis at 4° C. for 16 h in dialysis buffer (50 mM Tris pH 8.0, 0.1 M NaCl, 20 mM imidazole and 2 mM DTT). After dialysis, protein solution was then passed through Ni-NTA resin to remove uncleaved protein and His-tagged TEV. The recombinant protein was further purified by gel filtration on a fast protein liquid chromatography (FPLC) system (GE Healthcare Life Sciences). The principle peaks were collected, verified for molecular weight by SDS-PAGE, and stored in storage buffer (20 mM Tris pH 8.0, 25 mM NaCl, and 0.5 mM DTT) at a protein concentration of 10 mg/mL. The purity of the recombinant protein was evaluated by ImageJ densitometric analysis using bovine serum albumin as the standard (Schneider et al., 2012). 200 µM PLP was added to all buffers during the purification of all AAAD family enzymes.

(138) Enzyme Assays

(139) The AAS enzyme assays were performed in 100 µL of reaction buffer (50 mM Tris, pH 8.0) containing 50 µg of recombinant enzyme, 200 µM PLP. Kinetic reactions were incubated with a range of amino acid substrate concentrations (1 µM-4 mM) at 30° C. for 30 minutes prior to quenching with 100 µL of 0.8 M formic acid. The reaction mixture was centrifuged, and the supernatant was analyzed by Pierce Quantitative Peroxide Assay Kit (Pierce) against a standard curve of hydrogen peroxide to demonstrate AAS activity or determine AAS kinetic parameters. Rr4HPAAS reactions were also analyzed by LC-MS-UV. 50 µL of reaction mixture was analyzed by an Ultimate 3000 liquid chromatography system (Dionex), equipped with a 150 mm C18 Column (Kinetex 2.6 µm silica core shell C18 100 Å pore, Phenomenex) and coupled to an UltiMate 3000 diode-array detector (DAD) in-line UV-Vis spectrophotometer (Dionex) and a TSQ Quantum Access MAX triple-quadrupole mass spectrometer (Thermo-Scientific). To resolve chromatographically L-tyrosine, tyrosol and 4-HPAAA, compounds were separated through the use

of an isocratic mobile phase containing 50 mM monopotassium phosphate pH 4.6, 15% (v/v) acetonitrile and 0.5 mM octyl sulfate. Rr4HPAAS product formation was quantified using the UV absorbance at 280 nm and compared to analytical standards using the diode array detector wavelength at a wavelength range of 200-500 nm and chromatographic retention time. The reduction of aldehyde products was achieved by addition of saturated sodium borohydride in ethanol or by addition of 10 mM NADPH and 10 g of Rr4HPAR1 or Rr4HPAR2.

(140) The phenylacetaldehyde reductase activity assays using Rr4HPAR1 and Rr4HPAR2 were carried out in 200 μ L reaction buffer (50 mM Tris, pH 8.0) at the presence of 2 mM phenylacetaldehyde, 5 mM NADPH and 5 μ g of recombinant enzyme. The reactions were incubated at 30° C. for various time points, quenched with an equal volume of 0.8 M formic acid, and extracted by 100 μ L of ethyl acetate. The organic phase was then analyzed by gas chromatography-mass spectrometry (GC-MS) using an 5% Phenyl Methyl Silox column (30 m $\times$ 250 m $\times$ 0.25 m, Agilent) with a temperature gradient as follows: 0-1 min 45° C., 4-13.33 min 45-185° C. The quadrupole MS was set to EI mode, electron energy at 70 eV, MS-source temperature at 230° C., MS-quad temperature at 150° C., scan mass range at 50-300 m/z and SIM for 120 m/z and 122 m/z. EI-MS spectra were compared against analytical standards. The 4-HPAA reductase activity assays were carried out in 100 μ L of 50 mM Tris pH 8.0 with the addition of 5 mM 4-HPAA, 10 mM NADPH and 0.2 μ g of Rr4HPAR1 or 15 ag of Rr4HPAR2. The reactions were incubated at 30° C. and then quenched at various time points with the addition of 100 μ L methanol. The reaction mixture was then centrifuged and analyzed by LC-MS. Compounds were separated by reversed-phase chromatography with a ramp gradient of solvent A (0.1% formic acid in H.sub.2O) and solvent B (0.1% formic acid in acetonitrile): 10% solvent B for 0.5 min, 5-40% solvent B over 8.5 min, 95% solvent B for 1.8 min followed by a final equilibration of 10% solvent B for 1 min with a flow rate at 0.7 mL/min. Product formation was measured using select ion monitoring in positive mode for a centroid center mass of 121.065 with a scan width of 0.002. The specific activity was determined at a five-minute reaction time point, and quantified against a standard curve of tyrosol.

(141) Kinetic characterization of UGTs was conducted in 200 μ L reaction buffer (50 mM Tris, pH 8.0) containing 10 mM UDP-glucose and various concentrations of tyrosol (0.01-5.0 mM). Reactions were started with addition of recombinant enzyme, incubated at 30° C. for 10 minutes, and quenched by addition of 200 μ L of methanol. The reaction mixed was then analyzed by LC-HRMS as described above. Compounds were separated by reversed-phase chromatography with a ramp gradient of solvent A (0.1% formic acid in H.sub.2O) and solvent B (0.1% formic acid in acetonitrile): 5% solvent B for 0.5 min, 5-55% solvent B over 6 min, 55-5% solvent B over 1.0 min and a final equilibration of 5% solvent B for 1 min with a flow rate at 0.6 mL/min. Product formation was measured using select ion monitoring in positive mode for a centroid center mass of 318.15. Product mass was calculated by comparison to a standard curve of the NMR verified plant purified salidroside and icarisperone D2 samples.

(142) Kinetic constants such as K_m and V_{max} were determined by fitting raw data to the Michaelis-Menten equation using the nonlinear regression function in Prism (version 7.0).

(143) Accession Codes

(144) The sequences of *R. rosea* genes reported in this article are deposited into NCBI GenBank under the following accession numbers: Rr4HPAAS (MF674522), RrAAS (MF674523), Rr4HPAR1-2 (MF674524-MF674525) and RrUDP1-34 (MF674526-MF674558, MG385659). Raw RNA-Seq reads have been submitted to NCBI SRA (SRR5936536 and SRR5936537). The de novo transcriptomes assembled from the raw reads have been submitted to NCBI TSA (GFVD000000000 for merged transcriptome, GFVE000000000 for crown transcriptome, and GFVF000000000 for root transcriptome). Raw and mzTab format feature called metabolomic data from the *R. rosea* crown and root have been uploaded to the EBI MetaboLights database (MTBLS566).

(145) TABLE-US-00001 TABLE 1 Kinetic parameters of characterized enzymes. Enzyme RrT4GT

RrT8GT29 Rr4HPAAS Substrate tyrosol tyrosol L-tyrosine L-DOPA
k.sub.cat (sec) 481.60 ± 3.91 576.20 ± 5.68 167.5 ± 0.82 4.92 ± 0.08 9.52 ± 0.37 K.sub.m
(mM) 4.11 ± 0.08 1.37 ± 0.05 0.53 ± 0.01 0.42 ± 0.02 1.04 ± 0.10 k.sub.cat/K.sub.m(sec.sup.-1
mM.sup.-1) 117.18 420.58 316.04 11.71 9.15
(146) TABLE-US-00002 TABLE 2 Cloning primers. Vector/ SEQ ID Gene direction NO:
Sequence Rr4HPAAS pHis8-4 97 GAAAACTTGTACTTCCAGGCCCATGGCATGGGC
Forward AGCTTGCCTTCTCCTAATG Rr4HPAAS pHis8-4 98
CTCGAATTCGGATCCGCCATGGCTAAGACACGA Reverse TGCTTTGAGCTGTTTCTTG
Rr4HPAAS pEAQ-HT 99 GTATATTCTGCCCAAATTCGCGACCGGTATGGGC Forward
AGCTTGCCTTCTCCTAATG Rr4HPAAS pEAQ-HT 100
GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG Reverse
AGCTAAGACACGATGCTTTGAGCTGTTTCTTG Rr4HPAAS p423TEF 101
GCATAGCAATCTAATCTAAGTTTTCTAGAACTAG Forward
TATGGGCAGCTTGCCTTCTCC Rr4HPAAS p423TEF 102
CAGCCCGGGGGATCCACTAGTCTAAGACACGAT Reverse GCTTTGAGCTGTTTCTTG
RrAAS pHis8-4 103 GAAAACTTGTACTTCCAGGCCCATGGCATGGAG Forward
GAGGAGTTGAAGCCG RrAAS pHis8-4 104
CTCGAATTCGGATCCGCCATGGTCATGCATTTAT Reverse
ATGCTTTTGTAGCAGTGAAGTG RrPAR1 pHis8-4 105
GAAAACTTGTACTTCCAGGCCCATGGCATGAGTT Forward TAAGCGGAGCGGGG
RrPAR1 pHis8-4 106 CTCGAATTCGGATCCGCCATGGTCAGAGTTTGGC Reverse
GAAACCCTTTTCC RrPAR1 p425TEF 107
GCATAGCAATCTAATCTAAGTTTTCTAGAACTAG Forward
TATGAGTTTAAGCGGAGCGGGG RrPAR1 p425TEF 108
CAGCCCGGGGGATCCACTAGTTCAGAGTTTGGC Reverse GAAACCCTTTTCC RrPAR2
pHis8-4 109 GAAAACTTGTACTTCCAGGCCCATGGCATGGGTT Forward
TATCTGAAGAGAAGAAGTTAG RrPAR2 pHis8-4 110
CTCGAATTCGGATCCGCCATGGTCATTTGTCTTT Reverse CAAACTTTTCGACAGTGTCTC
RrUGT1 p426TEF 111 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGT Forward
GACGAAAAAACTCACATTCTTATCC RrUGT1 p426TEF 112
CAGCCCGGGGGATCCACTAGTTCAGGTAAGACC Reverse AGACACAAACTTGAC
RrUGT2 p426TEF 113 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGG Forward
TTCTGATTCACGGCCTC RrUGT2 p426TEF 114
CAGCCCGGGGGATCCACTAGTCTAGGACAAAGT Reverse CTCTCTTCTCAACTTCAATTC
RrUGT2 pHis8-4 115 GAAAACTTGTACTTCCAGGCCCATGGCATGGGTT Forward
CTGATTCACGGCCTC RrUGT2 pHis8-4 116
CTCGAATTCGGATCCGCCATGGCTAGGACAAAG Reverse
TCTCTCTTCTCAACTTCAATTC RrUGT2 pEAQ-HT 117
GTATATTCTGCCCAAATTCGCGACCGGTATGGGT Forward TCTGATTCACGGCCTC
RrUGT2 pEAQ-HT 118 GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG Reverse AG
CTAGGACAAAGTCTCTCTTCTCAACTTC RrUGT3 p426TEF 119
CAATCTAATCTAAGTTTTCTAGAACTAGTATGTC Forward AGGCACACCACACATCG
RrUGT3 p426TEF 120 CAGCCCGGGGGATCCACTAGTTCAATGCTTCATC Reverse
GAACTCCGCC RrUGT3 pHis8-4 121 GAAAACTTGTACTTCCAGGCCCATGGCATGTCAG
Forward GCACACCACACATCG RrUGT3 pHis8-4 122
CTCGAATTCGGATCCGCCATGGTCAATGCTTCAT Reverse CGAACTCCGCC RrUGT3
pEAQ-HT 123 GTATATTCTGCCCAAATTCGCGACCGGTATGTCA Forward
GGCACACCACACATCG RrUGT3 pEAQ-HT 124
GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG Reverse
AGTCAATGCTTCATCGAACTCCGCC RrUGT4 p426TEF 125

CAATCTAATCTAAGTTTTCTAGAACTAGTATGGG Forward
TTCACAAGCCTCTCCAAAACC RrUGT4 p426TEF 126
CAGCCCGGGGGATCCACTAGTTCATTCTTGAAC Reverse
TGGAGAATATCTTTCACAAGCC RrUGT5 p426TEF 127
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGA Forward ACCGAGACCTCACGCAG
RrUGT5 p426TEF 128 CAGCCCGGGGGATCCACTAGTTTAATTAGTGTCA Reverse
CCAAGATGAGTTTTCTTTAGTAAG RrUGT6 p426TEF 129
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGA Forward
ATCTGTACAAGGTGTTCAAGAAAAGC RrUGT6 p426TEF 130
CAGCCCGGGGGATCCACTAGTTCAGTTTGAATTC Reverse CTCGACAGGAGCAC
RrUGT7 p426TEF 131 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGC Forward
TGAAAACACTCATGCTCATGC RrUGT7 p426TEF 132
CAGCCCGGGGGATCCACTAGTTCATTTCTTGAAG Reverse ATTTGTAGGTCTGTGGATG
RrUGT8 p426TEF 133 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGC Forward
TTCCTCCTCTTTAGCTTGTGATTC RrUGT8 p426TEF 134
CAGCCCGGGGGATCCACTAGTTTATTTAACTGTT Reverse
TCTTGTTTTTGCAGGACAGAATGAATG RrUGT9 p426TEF 135
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGG Forward GTCTGAGCCACTAGTCC
RrUGT9 p426TEF 136 CAGCCCGGGGGATCCACTAGTTTATGCTGAAATT Reverse
GCATCCTTAGCAACTGG RrUGT10 p426TEF 137
CAATCTAATCTAAGTTTTCTAGAACTAGTATGAC Forward GAGGCGCCACCAC RrUGT10
p426TEF 138 CAGCCCGGGGGATCCACTAGTTCATCCAAGGCC Reverse
ATTGACAAAACGAC RrUGT11 p426TEF 139
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGC Forward
AGGCGAGATTCTAATACTTCCG RrUGT11 p426TEF 140
CAGCCCGGGGGATCCACTAGTTCACTTGTGGGA Reverse GATAATGAAGTCCCTG
RrUGT12 p426TEF 141 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGA Forward
GGAGGCGGCCAG RrUGT12 p426TEF 142
CAGCCCGGGGGATCCACTAGTTTAACACAGAGT Reverse CCAAATGTCCAGCAAC
RrUGT13 p426TEF 143 CAATCTAATCTAAGTTTTCTAGAACTAGTATGCT Forward
ACCTCTCTTACATGTTACACTAAC RrUGT13 p426TEF 144
CAGCCCGGGGGATCCACTAGTTTACAAGCCAAT Reverse GTTGGTCCTGAGATCAC
RrUGT14 p426TEF 145 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGA Forward
CACCACCGCCGC RrUGT14 p426TEF 146
CAGCCCGGGGGATCCACTAGTTTATCCCCTTCCA Reverse AGTTGAGTCAACGAC
RrUGT15 p426TEF 147 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGC Forward
TGATGCTGCTCAACATGTC RrUGT15 p426TEF 148
CAGCCCGGGGGATCCACTAGTTTATTGAACTTTG Reverse
TGAAATTGAAGATGACTCAAAAGG RrUGT16 p426TEF 149
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGC Forward AGAGGAAAACAGAACCAGC
RrUGT16 p426TEF 150 CAGCCCGGGGGATCCACTAGTTCATACAGCTGA Reverse
AGATATTTTGGATATGAATTGGTC RrUGT17 p426TEF 151
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGG Forward CTCACTTCCTTCCAC
RrUGT17 p426TEF 152 CAGCCCGGGGGATCCACTAGTTCAGACGCTAAA Reverse
CTGGACCACTTTTTTCC RrUGT18 p426TEF 153
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGG Forward CTCCCGAGGAAAGCCACATG
RrUGT18 p426TEF 154 CAGCCCGGGGGATCCACTAGTTCATTTTGGGGA Reverse
ATTAGACAGCAGG RrUGT19 p426TEF 155
CAATCTAATCTAAGTTTTCTAGAACTAGTATGAC Forward GTCATCAACACCTCCTCCTC
RrUGT19 p426TEF 156 CAGCCCGGGGGATCCACTAGTCTAAAAAATGC Reverse

TTTAACATAGCTCGTCCG RrUGT20 p426TEF 157
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGG Forward TTCACTCGACGTCGTC
RrUGT20 p426TEF 158 CAGCCCGGGGGATCCACTAGTTCATTTTCATAATA Reverse
GCTTCATCAATCAACTCGG RrUGT21 p426TEF 159
CAATCTAATCTAAGTTTTCTAGAACTAGTATGAA Forward
GTCCAACACTCATCTATTCCTC RrUGT21 p426TEF 160
CAGCCCGGGGGATCCACTAGTTCATACAACCGG Reverse CTCCAGTTGAC RrUGT22
p426TEF 161 CAATCTAATCTAAGTTTTCTAGAACTAGTATGAA Forward
AACTCCTCAAAATCCACACGTAG RrUGT22 p426TEF 162
CAGCCCGGGGGATCCACTAGTTCATCCTGATA Reverse
AATCTTTGAACTCATCTTGCTC RrUGT23 p426TEF 163
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGA Forward AAGGCAGAGTGATCACCAAG
RrUGT23 p426TEF 164 CAGCCCGGGGGATCCACTAGTTCATTTGGTGGAT Reverse
ATCACATCTCTAACAACACTG RrUGT24 p426TEF 165
CAATCTAATCTAAGTTTTCTAGAACTAGTATGAG Forward CAACGCCGCCG RrUGT24
p426TEF 166 CAGCCCGGGGGATCCACTAGTTTAGTTTATGACT Reverse
TCATTCACTTGCTCCAACAAC RrUGT25 p426TEF 167
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGC Forward GCGCCACCACTTTG RrUGT25
p426TEF 168 CAGCCCGGGGGATCCACTAGTTTAGCAGGTAAC Reverse
AAGGTTATTAACCAAATCCTTGAG RrUGT26 p426TEF 169
CAATCTAATCTAAGTTTTCTAGAACTAGTATGTC Forward
ATCAGATTCCGGCCACATTATCC RrUGT26 p426TEF 170
CAGCCCGGGGGATCCACTAGTCTATATTATTTTT Reverse
CTTAATGCCATGACTTGTCGGACC RrUGT27 p426TEF 171
CAATCTAATCTAAGTTTTCTAGAACTAGTATGAG Forward
TTCAGTCAATGCTCAAAAGCC RrUGT27 p426TEF 172
CAGCCCGGGGGATCCACTAGTTCAAAAGTGCAT Reverse TAGTAGTCCTTCCACAAATC
RrUGT28 p426TEF 173 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGA Forward
CTCGGTTGATCTGAACAAG RrUGT28 p426TEF 174
CAGCCCGGGGGATCCACTAGTCTAGTTGGCACTT Reverse GGCAACACAATCG RrUGT29
p426TEF 175 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGG Forward
ATCTCTAGGAAAGAAGATTCAAC RrUGT29 p426TEF 176
CAGCCCGGGGGATCCACTAGTTTAGGTTGTAAC Reverse ACAATTTTTTTTTTTGGAC
RrUGT29 pHis8-4 177 GAAAACCTTGACTTCCAGGCCCATGGCATGGGA Forward
TCTCTAGGAAAGAAGATTCAAC RrUGT29 pHis8-4 178
CTCGAATTCGGATCCGCCATGGTTAGGTTGTAAC Reverse TACAATTTTTTTTTTTGGAC
RrUGT29 pEAQ-HT 179 GTATATTCTGCCCAAATTCGCGACCGGTATGGGA Forward
TCTCTAGGAAAGAAGATTCAAC RrUGT29 pEAQ-HT 180
GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG Reverse
AGTTAGGTTGTAAC TACAATTTTTTTTTTTGGAC RrUGT30 p426TEF 181
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGG Forward CTCCCGAGGAAAGCCACATG
RrUGT30 p426TEF 182 CAGCCCGGGGGATCCACTAGTTCATTTTGGGGA Reverse
ATTAGACAGCAGG RrUGT31 p426TEF 183
CAATCTAATCTAAGTTTTCTAGAACTAGTATGGA Forward
ATCTGTACAAGGTGTTCAAGAAAAG RrUGT31 p426TEF 184
CAGCCCGGGGGATCCACTAGTTCAGTTTGAATTC Reverse CTCGACAGGAGCAC
RrUGT32 p426TEF 185 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGA Forward
CTCGGTTGATCTGAACAAGAAACC RrUGT32 p426TEF 186
CAGCCCGGGGGATCCACTAGTCTACAATTTTTTTT Reverse
TTGGACAGAAGTACGTCATTTATAAGTC RrUGT33 p426TEF 187

CAATCTAATCTAAGTTTCTAGAACTAGTATGAG Forward
 CTTAATTGAAAAACCACTCACG RrUGT33 p426TEF 188
 CAGCCCGGGGGATCCACTAGTCTAACGGATATG Reverse TTTTGTTTTTGAGAGCAGGAC
 RrUGT33 pHis8-4 189 GAAAACTTGTACTTCCAGGCCCATGGCATGAGCT Forward
 TAATTGAAAAACCACTCACG RrUGT33 pHis8-4 190
 CTCGAATTTCGGATCCGCCATGGCTAACGGATATG Reverse
 TTTTGTTTTTGAGAGCAGGAC RrUGT33 pEAQ-HT 191
 GTATATTCTGCCCAAATTCGCGACCGGTATGAGC Forward
 TTAATTGAAAAACCACTCACG RrUGT33 pEAQ-HT 192
 GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG Reverse
 AGCTAACGGATATGTTTTGTTTTTGAGAGCAGGA C RrUGT34 p426TEF 193
 GCATAGCAATCTAATCTAAGTTTTCTAGAACTAG Forward
 TTGGACCCTGACGACAGCGTTTTG RrUGT34 p426TEF 194
 CAGCCCGGGGGATCCACTAGTTTAGTTTTGTTC Reverse
 TCGTACAAATAATGCACAACTCATC Pc4HPAAS pHis8-4 195
 GAAAACTTGTACTTCCAGGCCCATGGCATGGGCT Forward CCATCGATAATC Pc4HPAAS
 pHis8-4 196 CTCGAATTTCGGATCCGCCATGGTTAGGATAAAAT Reverse
 ATTCACGATCTTCT Pc4HPAAS pEAQ-HT 197
 GTATATTCTGCCCAAATTCGCGACCGGTATGGGC Forward TCCATCGATAATC Pc4HPAAS
 pEAQ-HT 198 GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG Reverse
 AGTTAGGATAAAATATTCACGATCTTC PsTyDC pHis8-4 199
 GAAAACTTGTACTTCCAGGCCCATGGCATGGGA Forward
 AGCCTTCCGACTAATAACCTTG PsTyDC pHis8-4 200
 CTCGAATTTCGGATCCGCCATGGCTAGGCACCAA Reverse GTATGGCATCTGTATG PsTyDC
 pEAQ-HT 201 GTATATTCTGCCCAAATTCGCGACCGGTATGGGA Forward
 AGCCTTCCGACTAATAACCTTG PsTyDC pEAQ-HT 202
 GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG Reverse
 AGCTAGGCACCAAGTATGGCATCTGTATG AAS55083 p426TEF 203
 CAATCTAATCTAAGTTTTCTAGAACTAGTATGGC Forward AGGCAGTGGGACTG
 AAS55083 p426TEF 204 CAGCCCGGGGGATCCACTAGTTCAGTGTTTAACT Reverse
 GAGGATCTCCACTTTTTCAGC EU567325 p426TEF 205
 GCATAGCAATCTAATCTAAGTTTTCTAGAACTAG Forward
 TATGGGTCTGAACTCGGCCTTTG EU567325 p426TEF 206
 CAGCCCGGGGGATCCACTAGTCTAGACTTTCTTT Reverse AACTTGAGTTCCTGAAGCAG
 Example #2: Results

(147) Enzymes of the plant aromatic amino acid decarboxylases (AAAD) family that can be used in the production of one or more of tyrosol, salidroside, and icaraside D2 were identified. These plant AAAD-family enzymes contain substitutions in one of two active site residues responsible for influencing aldehyde synthase chemistry. These activity-influencing residues are boxed in the multiple sequence alignment of biochemically characterized plant AAADs shown in FIG. 26.

(148) Plant AAAD enzymes that contain an active site histidine to asparagine or aspartic acid substitution have an aldehyde synthase activity. This active site substitution is represented by the Rr4HPAAS MF674522 histidine 198 in FIGS. 31 and 32.

(149) Plant AAAD enzymes that contain an active site tyrosine to leucine, isoleucine, phenylalanine, methionine or valine substitution have an aldehyde synthase activity. This active site substitution is represented by the Rr4HPAAS MF674522 phenylalanine 343 in FIGS. 31 and 32.

(150) These active site substitutions at positions 198 and 343 were selected from natural variation, shown in FIG. 33, found within all plant AAAD sequences available on Phytozome V12.1. We have curated 226 plant AAAD sequences from Phytozome 12. The list was queried for sequences containing substitution in one of the two activity dictating residues to make a list of 73 enzymes

that likely have some 4HPAAS activity. These 73 AAS enzymes are identified as SEQ ID NOS: 21-93. In some embodiments, any of the enzymes of SEQ ID NOS: 21-93 can provide 4HPAAS activity in a host cell or method described herein.

(151) To demonstrate the roles of these residues in aldehyde synthase chemistry, the biochemical activity of wild type and mutant *Papaver somniferum* tyrosine decarboxylase (PsTyDC) enzymes were characterized. The substitution of the active site histidine (Rr4HPAAS MF674522 histidine 198) or the active site tyrosine (Rr4 PAAS MF674522 phenylalanine 343) within a *Papaver somniferum* tyrosine decarboxylase (PsTyDC) results in aldehyde synthase chemistry. FIGS. 34A-C are chromatograms showing product formation of PsTyDC and mutants.

SEQUENCES

(152) TABLE-US-00003 TABLE 2 Summary of Sequences. SEQ ID NO.: Description 1 Rr4HPAAS DNA 2 Rr4HPAAS GenBank accession MF674522 3 Rr4HPAR1 DNA 4 Rr4HPAR1 amino acid GenBank accession MF674524 5 RrUGT2 DNA 6 RrUGT2 amino acid GenBank accession MF674527 7 RrUGT3 DNA 8 RrUGT3 amino acid GenBank accession MF674528 9 RrUGT7 DNA 10 RrUGT7 amino acid GenBank accession MF674532 11 RrUGT13 DNA 12 RrUGT13 amino acid GenBank accession MF674538 13 RrUGT17DNA 14 RrUGT17 amino acid GenBank accession MF674542 15 RrUGT29 DNA 16 RrUGT29 amino acid GenBank accession MF674554 17 RrUGT32 DNA 18 RrUGT32 amino acid GenBank accession MF674557 19 RrUGT33 DNA 20 RrUGT33 amino acid GenBank accession MF674558 21 *Arabidopsis thaliana* AT2G20340.1 22 *Brachypodium distachyon* 1g28960.3 23 *Carica papaya* 16427710 24 *Ricinus communis* 16804377 25 *Cucumis sativus* 16963476 26 *Vitis vinifera* 17835588 27 *Citrus sinensis* 18113817 28 *Capsella rubella* 20900667 29 *Malus domestica* 22636618 30 *Linum usitatissimum* 23178995 31 *Eutrema salsugineum* 20200788 32 *Populus trichocarpa* 27022899 33 *Brachypodium stacei* 06G160800.1 34 *Physcomitrella patens* Pp3c4_30790V3.1 35 *Ananas comosus* 33033299 36 *Zostera marina* 33182387 37 *Daucus carota* subsp. *sativus* 36055203 38 *Trifolium pratense* 35974269 39 *Arabidopsis lyrata* 35943929 40 *Sorghum bicolor* 002G120700.1 41 *Sphagnum fallax* 0166s0011.1 42 *Kalanchoe laxiflora* 1398s0003.1 43 *Manihot esculenta* 12G038600.1 44 *Prunus persica* 8G214500.1 45 *Eucalyptus grandis* K01418.1 46 *Amborella trichopoda* 31565185 47 *Salix purpurea* 0252s0200.1 48 *Medicago truncatula* 31080941 49 *Brassica rapa* I01156.1 50 *Brassica rapa* I04706.1 51 *Brassica rapa* G00043.1 52 *Glycine max* 03G167900.1 53 *Fragaria vesca* 27261550 54 *Kalanchoe fedtschenkoi* 0172s0035.1 55 *Capsella grandiflora* 22666s0001.1 56 *Selaginella moellendorffii* 15420188 57 *Setaria italica* 3G188200.1 58 *Kalanchoe fedtschenkoi* 0033s0078.1 59 *Daucus carota* subsp. *sativus* 36068870 60 *Daucus carota* subsp. *sativus* 36056758 61 *Solanum tuberosum* 3DMP400026166 62 *Solanum tuberosum* 3DMP400024738 63 *Solanum lycopersicum* 36137005 64 *Daucus carota* subsp. *sativus* 36065781 65 *Oropetium thomaeum* 35995617 66 *Oryza sativa* 33157740 67 *Brachypodium stacei* 01G392300.1 68 *Amaranthus hypochondriacus* 32828676 69 *Brachypodium distachyon* 5g21770.1 70 *Brachypodium distachyon* 2g02360.1 71 *Sorghum bicolor* 009G192600.1 72 *Kalanchoe laxiflora* 0994s0009.1 73 *Kalanchoe laxiflora* 0003s0173.1 74 *Panicum hallii* 32512198 75 *Prunus persica* 6G202600.1 76 *Prunus persica* 4G086700.1 77 *Prunus persica* 4G087100.1 78 *Medicago truncatula* 31073039 79 *Zea mays* GRMZM2G009400 80 *Glycine max* 07G059000.1 81 *Panicum virgatum* Ca01381.1 82 *Theobroma cacao* 27425420 83 *Fragaria vesca* 27274768 84 *Gossypium raimondii* 26786642 85 *Populus trichocarpa* 26994989 86 *Malus domestica* 22679008 87 *Citrus Clementina* 20801973 88 *Citrus Clementina* 20818150 89 *Vitis vinifera* 17834108 90 *Petunia hybrida* ABB72475.1 91 *Carica papaya* 16421889 92 *Sphagnum fallax* 0042s0024.1 93 *Eucalyptus grandis* E01788.1 94 pHs8-4 95 pEAQ-HT 96 pJKW 1410

(153) TABLE-US-00004 Rr4HPAAS DNA (SEQ ID NO: 1):
ATGGGCAGCTTGCTTCTCCTAATGATCCATCAAACACCTTCAACCCCATGGACCTC
ACCGAGTTATCCACCGAGTCGAAACTCGTCGTAGATTTCATAACTCAGTACTACCAA
ACCCTAGAGACCCGACCCGTCCAGCCACGGGTCAAGCCAGGTTTCTTAACGGGCCA

GCTTCCAGTAAACGACCTTATGTTGTAAGTGAATGTAATATTGTCTGATGT
AAATGAGAAGATTGTCCCTGGCCTCACTCATTGGCAAAGCCCTAATTTCCATGCATA
CTTTCCAGCCAGTTCCAGCAACGCAGGGCTGTTGGGAGAGTTACTATGCTCCGGACT
CAGTGTCATTGGGTTTACATGGAGCTCCTCCCCTGCCGCGACGGAGCTTGAGAATGT
CGTGGTTGACTGGATGGCCAAGATGCTTAACCTTCCATCCTCTTTCTGCTTCTCCGGC
GGAGGCGGTGGCGTTCTGCAAGCAAACACTTGCGAGGCTGTGTTGTGCACTTTAGCC
GCTGCGAGGGACAAGGCTCTTAACCGGGTGGGAGATGATCAGATCAATAAACTGGT
CCTCTACTGCTCCGACCAAACACATTTTACAATCCACAAGGGCGCAAAGTTGATAGG
AATCCGATCAAAGAACATAAAATCAATCACTACTAAGAAAGAGAACGAGTTTAAAC
TCTGTCCTAACGACCTACGCGACGCGATAAGGAGTGATCTGGAAGCAGGACTAGTT
CCGTTTTACGTATGCGGAACGATTGGAACGACCGCGTTAGGAGTTGTGGATCCGATT
AAAGAGCTGGGTAAGGTGGCAAGAGAGTTTGATTTGTGGTTACATGTTGATGGAGC
TTATGGTGGCAGTGCATGCATATGCCCTGAGTTTCAGCATTACCTTGATGGAGTTGA
CCTTGTTGACTCGATCAGCATGAATGCACATAAATGGCTTTTATCCAATCTAGATTG
CTGCTTCCTGTGGCTTCAATCTCCTAACGCCCTAATCGAATCCCTGGCCGCAGAAGC
TAACTTTCTGAAAGGTGGTAGTGAGATGGTGGATTACAAGGACTGGCAGATATCGTT
GAGTCGTCGATTTAGAGCGATCAAGATGTGGATGGTGATAAGGCGATACGGTGTGA
GTAATCTCATTGAGCATATTCGATCCGACGTGAGCATGGCGGTGAGATTCGAAGAG
ATGGTGGCGGCGGACGACCGGTTTGAAATCGTGTTTCCTAGAAAGTTTGCGCTTGTT
TGCTTCAAGCTTAGTAGCGAGAAGACACCACCGGGCCGCGACTCGGAGTTAACTCG
TGAGCTGATGGAGAGAGTCAACTCGAGTGGGAAGGCTTACTTGAGTGGAGTTCAA
TGGGTTCGGATCTTCTTCATCAGGTGTGTGATCGGGTCGAGTTTGACTGAGGAGAGAC
ACGTCGATAATCTGTGGAGGCTCATTCAAGAAACAGCTCAAAGCATCGTGTCTTAG

Rr4HPAAS GenBank accession MF674522 (SEQ ID NO: 2):

MGSLPSPNDPSNTFNPMDLSTESKLVVDFITQYYQTLETRPVQPRVKPGFLTGQLPD
KAPFHGESMEVILSDVNEKIVPGLTHWQSPNFHAYFPASSSNAGLLGELLCSGLSVIGFT
WSSSPAATELENVVVDWMAMKMLNLPSSFCFSGGGGGVLQANTCEAVLCTLAAARDKA
LNRVGDDQINKLVLYCSDQTHFTIHKGAKLIGIRSKNIKSITTKKENEFKLCPNDLRDAIR
SDLEAGLVPFYVCGTIGTTALGVVDPIKELGKVAREFDLWLHVDGAYGGSACICPEFQH
YLDGVLDLVDSISMNAHKWLLSNLDCCFLWLQSPNALIESLAAEANFLKGGSEMVDYKD
WQISLSRRFRAIKMWMVIRRYGVSNLIEHIRSDVSM AVRFEEMVAADDRFEIVFPRKFA
LVCFKLSSEKTPPGRDSELTRELMERVNSSGKAYLSGVQMGRIFIRC VIGSSLTEERHVD
NLWRLIQETAQSIVS Rr4HPAR1 DNA (SEQ ID NO: 3):

ATGAGTTTAAGCGGAGCGGGGAAGGTGGTTTGCGTTACCGGCGCGTCTGGCTACAT
AGCGTCCTGGCTCGTCAAGCTTCTTCTCCAGCGCGGTTATACCGTCAAGGCCTCCGT
TCGCGATCCTAATGATCCGAAAAAGACTCAGCACTTGACGGCACTTGATGGAGCTA
AGGAGAGGCTGCAGTTGTACAAAGCCAATTTGCTTGAACAAGGCTCGTTTGATCCCA
TAGTTGAAGGATGTGAAGGTGTTTTCCACACCGCGTCTCCCTTTTATCATGCAGTGG
ATGATCCGCAGGCCGAGTTAATTGACCCTGCTGTCAAGGGAACACTCAATGTTCTTT
CTTCATGTGCTAAAGTTGCGTCTCTTAAAAGAGTAGTCCTGACTTCTTCGATTGCTGC
TGTTGCATATAATGGGAAACCCCGTACTCCGGAGGTTGTAGTTGACGAGACTTGTT
TTCTAACCCAGATGTTTGTAAGGAGATGAAGCTTTGGTATGTCATATCCAAGACACT
CGCTGAAGAAGCAGCATGGAAGTTTGTGAAAGAGAAAGGAATAGACATGGTTACCA
TAAATCCGGCCATGGTGATTGGTCCCCTTCTGCAACCAACACTCAATACCAAGTGCTG
CTGCTATTCTGAACTTGATCAATGGATCGGAGACATACCCAAATGCTTCTTTTGGAT
GGGTCAATGTGAAAGATGTTGCAGAAGCACACGTTCTTGCATTTGAGGTTCTTTCAG
CTAATGGTAGATACTGCTTGGTGGAAAGAGTTGCCCACAGTTCTGAAGTGGTGAACA
TGCTCCATGAGCTCTACCCTGATATCAAATTTCCCGCCAAGTGTGCAGATGACAAAC
CATTTGTGCCAATTTATCAAGTTTCAAAAAGAAAAGGCACATACTTTAGGGGTAAAAT
TCATTCCTTTAGAGGTAAGCCTCAAGGAAACAGTTGAAAGCTTGAAGGAAAAGGGT

TTCCGCAAACTCTGA Rr4HPAR1 amino acid GenBank accession MF674524

(SEQ ID NO: 4):

MSLSGAGKVVCVTGASGYIASWLVKLLLQRGYTVKASVRDPNDPKKTQHLTALDGAK
ERLQLYKANLLEQGSFDPIVEGCEGVFHTASPFYHAVDDPQAELIDPAVKGTLNVLSSC
AKVASLKRVVLTSSIAAVAYNGKPRTPEVVVDETWFSNPDVCKEMKLWYVISKTLAEE
AAWKFFVKEKGIDMVTINPAMVIGPLLQPTLNTSAAAILNLINGSETYPNASFGWVNVKD
VAEAHVLAFEVPSANGRYCLVERVAHSSEVVNMLHELYPDIKLPKACADDPFVPIYQ
VSKEKAHTLGVKFIPLEVSLKETVESLKEKGFAKL RrUGT2 DNA (SEQ ID NO: 5):

ATGGGTTCTGATTCACGGCCTCTACGCGTCTTCTTCTTCCCTTCATGGCTCACGGCC
ATCTGATTCCGATGGTCGACATCGCCAGACTCTTCTCTTCTCAAGGAGTCCACTCCA
CCATCATCACCACCCCACTAAACGCCAATTACATCTCCAAAACGACGTCTCTATCCA
TCAAAACGATACCGTTTCTGCTGCGGAAGTTGGGCTTCCGGACGGCTGCGAGAATA
TCGACATGCTTCTTCGCCCCGATCTCTTCTTCAAATTTTTTCCAAGCCGCCAATTTACT
CCAAGCGCCGTTTCGAGAACCTTCTAGAACTCGAAAGGCCCGATTGCTTAATCTCCGA
CATCTTCTTCCCCTGGTCAGTCGACTCCGCCGAGAAATTCAACATCCCGAGACTCGT
TTTCCACGGCACGAGCTTCTTCGCCATGTGCGCCATGGAGAGCTTGAAGACCCACAA
GCCCTATAAATCGGTAAGCACCGACTCTGAACCGTTCTTAATCCCGAATCTCCCTGA
TGAAATCAAAATGACTAAAAGTCAGTTCACGGTTGACGCTTGGGAAGACACCGAAA
AGGGCCTTGGGAAGCTGTTGGCTGATGCGAGAGCTTCAGGGCTGAGGAGCTTCGGC
ATGATCGTAAACAGCTTCCACGAGCTCGAACC GGCTTACGCGGATTATTACAAGAAT
GTGTTGAACATGAAAGCGTGGTGTGTCGGGCCTGTTTCGTTATATAACCGAAACGAT
GACGAGAAAATTGCAAGAGGGAAGAAATCAGCAATCGATGATCATGAGTGTTTAAA
ATGGCTGGAGGGAAAGCAGCCAGACTCCGTCGTGTACGTTTGTTCGGGAGCAGCG
CGAGCTTCCCTGATGAGCAGTTGCGCGATATCGCATTGGGGCTGGAAGAATCTGGA
GTAAATTTTCATCTGGGTGATCAGGAGAAGTTCCGAGTCAGGATCAGAAGATTACTTG
CCGGAGGGGTTTGAGGACCGGGTGAAGGACAGAGGGCTCGTGATCCGAGGTTGGGC
GCCACAGGTACTGATTTTGGACCATCCGTCGGTTGGGGGATTTGTGACTCACTGCGG
ATGGAATTCGGCATTGGAGGGGATTTAGCTGGCTTGCCGATGGTGACTTGGCCACT
GTTTCGCAGAGCAGTTTTTCAACCAGAAATTGATTACGGATGTGTTGAAAGTTGGGGT
TGAGGTTGGAGTGCAGAAATGGTCTCGGAACGGGGAGGATCGCGTGACGAAGGAG
AAGGTTGAGAAGGCGGTGAGGGCTGTTATGGTTGGGGAGGACGCTGAGGAGAGGC
GTGGCAGAGCTCGTCAGCTTGGGAAATTGGCAAAGAAAGCTGTGGCGAAAGATGGG
TCTTCGTACATTGATCTCCACAATTTGCTTGATGAATTGAAGTTGAGAAGAGAGACT
TTGTCCTAG RrUGT2 amino acid GenBank accession MF674527 (SEQ ID NO:

6):

MGSDSRPLRVFFFPFMAHGHLPMDIARLFSSQGVHSTIITPLNANYISKTTSLSIKTIPF
PAAEVGLPDGCENIDMLPSPDLFFKFFQAANLLQAPFENLLELERPDCLISDIFFPWSVDS
AEKFNIPRLVFHGTSTFFAMCAMESLKTHKPYKSVSTDSEPFLLIPNLPDEIKMTKSQFTVD
AWEDTEKGLGKLLADARASGLRSFGMIVNSFHELEPAYADYYKNVNLNMKAWCVGPVS
LYNRNDDEKIDARGKKS AIDDECLKWLEGKQPDSVVYVCFGSSASFPDEQLRDIALGLE
ESGVNFIWVIRRSSESGSEDYLPEGFEDRVKDRGLVIRGWAPQVLILDHPSVGGFVTHCG
WNSALEGISAGLPMVTWPLFAEQFFNQKLITDVLKVGVEVGVQKWSRNGEDRVTKER
VEKAVRAVMVGEDAEEERRGRARQLGKLAKKAVAKDGSSYIDLHNLLDELKLRRETLS
RrUGT3 DNA (SEQ ID NO: 7):

ATGTCAGGCACACCACACATCGCCATCCTCCCCAGCCCCGGCATGGGCCACCTCATC
CCCATGGCCGAGTTCGCCAAGCGCCTAGTCCACCACCACAACCTTCAGTATCACCTTC
GTCATCCCTACCGACGGCCCCACCTTCTCCGCTACCAACAAGTCCTCACCTCCCTCC
CATCTTCCATAGATCACATCTTCTTCCACAAGTCGACTTAACCGACGTCGTATCAC
AATCACCAGCTCATCCCAGAATCGAAACCCTAATCTCCCTCACCGTCGCTCGCTCCC
TCTCCTCCCTCCGCACCACCTTATCCTCTCTCCAATCGTCTAAAAACCTCGTCTCGCT

CGTCTGTTGTTTTCCTTCCTTCCACAGCCATGACGCTCTCGCTCTTCCTATACATGCCT
GCCCTACATTTTCTTCCCTTCCACAGCCATGACGCTCTCGCTCTTCCTATACATGCCT
CAGCTTGACAAATCAGTCACGTGCGAATTTTCGTCACATGACGGATTTGGTTCGAATT
CCTGGATGCGTTCCTGTCCGTGGATCGGATTTATTCGACCCGGTTCAAGACAGGACC
GACGAGGCTTATAAATGGGTCATACATCACTCCAACAGGTACCCTATGGCGGAGGG
TGTTATAGAGAATAGCTTCATGGAGTTGGAACATGGTGCGTTAAAGTATTTGCAAAC
GGTTCAATCGGGTAAGCCGCCTGTCTACGCGGTCGGACCGTTGATTAAAATGGATTA
TGATGTTGACGATTCCGGGTCGAAGATAATCGAGTGGCTCGATGATCAACCGGTTGG
TTCGGTTTTATTTGTTTCGTTTTGGAAGCGGCGGAACGCTCTCGTATGAGCAAATGAC
CGAGCTGGCTCACGGTTTGGAAATCGAGCCAGCAACGGTTCTTATGGGTGGTTCGGAG
TCCGAATCAAATCCCCAACAGCACGTATTTTCAGTGTACAAAGCCAAAAAGACCCGT
TGGCTTACTTGCCAGAAGGATTTTTTAAACCGAACCGAGGGTAGGGGTCTGGTCGTAT
CGAATTGGGCCCCACAGGCTCAAATTTTGAGTCACGGTTCGACCGGTGGGTTCATGA
GCCACTGTGGTTGGAATTCGATTTTGGAGAGTGTGGTGCACGGCGTGCCGATCATAG
CGTGGCCGTTGTACGCCGAGCAGAAGATGAATTCGATAATCGTGGTGGAGGACGTT
AAGGTGGCGCTGAGGCCGGCGGGGGTAGGGGAGAGGGTGGTGGAGAGGTTCGGAGA
TAACCGCAGTGGTGAAGGCGTTGATGGAGGGTGAGGAGGGGAAGAAGGTAAGGAA
TAGGATGAAGGAACTCAAGGAAGCGGCGGCACGTGCGGTTAGTGATGACGGTGCGT
CGACCATAGCGATTGCGGACTTGGCGCAAAAATGGCGGAGTTTCGATGAAGCATTGA

RrUGT3 amino acid GenBank accession MF674528 (SEQ ID NO: 8):
MSGTPHIAILPSPGMGHLIPMAEFAKRLVHHNFSITFVIPTDGPSSAYQQVLTSLPSSID
HIFLPQVDLTDVVSQSPAHPRIETLISLTVARSLSSLRTTLLSSLQSSKNLVSLVVDLFGTDA
FDPAIELGISPYIFFPSTAMTLLSLFLYMPQLDKSVTCEFRHMTDLVRIPGCVPVVRGSDLFD
PVQDRTDEAYKWVIHHSNRYPMAGVVIENSFMELEHGALKYLQTVQSGKPPVYAVGPL
IKMDYDVDDSGSKIIEWLDDQPVGSVLFVSFGSGGTLSYEQMTELAHGLESSQQRFLWV
VRSPNQIPNSTYFSVQSQKDPLAYLPEGFLNRTEGRGLVVSNNWAPQAQILSHGSTGGFM
SHCGWNSILES VVHGVPIIAWPLYAEQKMNSIIVVEDVKVALRPAGVGERVVERSEITAV
VKALMEGEEGKKVRNRMKELKEAAARAVSDDGASTIAIADLAQKWRSSMKH- RrUGT7
DNA (SEQ ID NO: 9):

ATGGCTGAAAACACTCATGCTCATGCCATAGTGGTACCATTTCCAGTTCAAGGACAC
ATAAAGCCCTCGCTGAATCTAGCCCTCAAGCTAGCATCTCAAGGCTTCACCATCACT
TTTGTCACCACTCATTTACCCACCAGCAAATCTCCCAAGCTCACAAAAACAGTACA
AATACAAACCATGACATGTTTTTCCAGGCACGAACTCCAGTCTCGATATCCGCCAT
GTAACGGTGACAGACACTTTTCCTTTGGGATTCGATCGCGCAGGGAATCAGGATCAG
TTTTGGGAGGGCATGCTTCACGTATTCCCTGCACATGTTGATGAACTGGTGGATCAG
TTAATGAATTCTTCGAAGCCGAGACCAACTTGTTTGATTCTGGATACATTTTATAACT
GGGGTTCCAAAATTGCTAACAAGTTTAATTTAGTGCATATTTCATTTTGGACTCAGTC
TGCTCTTTCTTTCACTTTGTTTTACCATTGGGAACTTTAAAGAAAAATGGTCACTTT
GGCTCTCCAGATAATCGCACGGATGTCATCGATTATATTTCCCGGTGTGCAAGAGATC
AAGCCCGCAGACTTAATATCCTACCTTCAGATGAGTGATACAACACTACTGTGGCTCAC
AGGACTTGTTTCACAGCATTTGAAGATGTCAGGAAGGCAGATTTTCATCCTGGCTAAT
ACAATCCAAGAATTTGAAACTGATACAATTTCTTCTATCCGATTTACCAGCCATTTT
TCTACCCAATTGGACCTGTTTTTTTAAACAAAGTCTGAACAACAAGCTAGCTCAGCTTT
GTGGTCTGAGTCAGACTGTGAGCAGTGGCTAAGTACAAAACCAAAAGGGTCTGTTT
TCTATGCCTCATTTGGGAGCTATGCTCGTGTAAGTGGCATGATATCGCAGAGATAG
CCTACGGATTGATGCAAAGTGAGGTGAATTTTATTTGGGTGATTCGCGACGATATTG
TGGGTGCACACGAGACTGATTTTTTACCAACAGAATTCATAAATGGAATCAAACCTCA
AAGATCAGGGACTACTAGTTTCTGCTGGTCTCAAACCTGAAGTTTTGTCCAATGCGG
CGATTGGAGGATTTCTGACTCATTGTGGATGGAACCTCGATACTCGAAAGCGTATGGT
GTGAAGTTCCATTATTGTGTTTTCCAATAATGACTGATCAGCCTAGTAACAGGAAAC

TGGTGGTGGATGCCTGGAGGATCAACCTATCTGCGGCGGAGGAGGTCACT
AGAGAAGAAGTGTCAATGAAGGTCAGGAACCTTGATTTCTGGAGAATTGGGGAATGA
GTTGAGAGTGCAGATTCAAAAGTACAAAAAGTTGATGGAGAATGGTATAATGGAAG
GTGGATCATCACATTCCAATTGGAACAAGTTCATCCACGACCTACAAATCTTCAAGA
AATGA RrUGT7 amino acid GenBank accession MF674532 (SEQ ID NO: 10):
MAENTHAHAIVVPFPVQGHKPSLNLALKLASQGFTITFVTTHFTHQQISQAHKNSTNTN
HDMFFQARNSSLDIRHVTVTDTFPLGFDRAQNQDQFWEGMLHVFPAHVDELVDQLMN
SSKPRPTCLILDTFYNWGSKIANKFNLVHISFWTQSALSFTLFYHWELLKKNHFGSPDN
RTDVIDYIPGVQEIKPADLISYLQMSDTTVAHRTCFTAFEDVRKADFILANTIQEFETDI
SSIRFHQPFFYPIGPVFLTKSEQQASSALWSESDCEQWLSTKPKGSVLYASFSGSYARVTR
HDIAEIAAYGLMQSEVNFIWVIRDDIVGAHETDFLPTEFINGIKLKDQGLLVSWCSQTEVLS
NAAIGGFLTHCGWNSILES VWCEVPLL CFPIMTDQPSNRKLVVDDWRIGVNL SAAEEVS
REEVSMKVRNLISGELGNELRVQIQKYKKLMENGIMEGGSSHSNWNKFIHDLQIFKK-
RrUGT13 DNA (SEQ ID NO: 11):

ATGGCAGAAATAAGTCTCATCTTCATCCCTTTTCCCGTAATCAGCCATCTCACTCCCA
CAATCGAAATCGCCAAAATCCTCCTCAGCAGAGACCACCGCCTTTCCATCACCTTCC
TCGTCATCGACATCCCCCAACGAGACGCCTCACTCGCCTCCCTCACCACCTCCATCA
TCTCCGATCGCCTCCACTTCCTCGATGTCGTACTTCCTCCCAACCAACACTCCCAATC
ATCCAAGCCATCAGGCATCGCGGCTATCGAGTCCGCCAAACCCGCAGTCAAGAAAA
CGATCAGCGATCTTGTTGTACGATCTCAGTCCGCCGCATCTGGTCCGCGGATAGCTG
GCTTCGTGCTGGACATGTTCTGCACGGCCATGATCGACATCGCAACTGAGTTTAACC
TTCCTTCGTATATTTACTACACTTGCGGCTCTTCGTTTCTTTCAATCGTGCTCCACGTC
CAGAAGCTCTGCGATGACGACGCTCTCGATATCGCCGATTTCAAAAACCTCGAGTGTG
GAGTTTTCGTTACCTGAGTTTTCAAACCTTGATTCCGGCTAGGCTGCTTCCATCCATGG
CGCTCGATAAGGACTTCTCGGCTTCATTTCGTGCGCAAAGCTAGAGCGTTCAGGAAGA
CGAAGGGGCATTTTGGTCAACTCGCTTG TAGAGTTGGAGCCTCACGCAATCGAGTCGA
TGAAATTAGACCGGTCTGTTCCCTCCGATTTACTCGGTTCGGACCAGTGCTCAACATGA
ATAGCAACACTGCATTTATCAGACAGGAGCAGGAGAAGGAGATCATGGAGTGGCTG
GACCAACAGCCTCCAGCATCTGTAGTTTTCTTGTTGTTTTGGCAGCAGGGGAGCGTTC
AAGCCGGACCAGGTGAAGGAAATCGCACGGGGGTTGGAGTCGAGCGGCTGCCGGTT
CCTCTGGGCGCTTCGGCAGCCTTCATCAAGCAATGTGAGGTTTTTCACCTCCTACAGA
TTATGAAGATTTCTCTGAGGTTCTGCCTGAAGGGTTTTTGCAGCGGACATATGGTGTT
GGGAAAGTGATTGGTTGGGCACCCCAGACAGCTGTTTTAGACCACCCTTCGGTGGGT
GGATTTCGTATCGCATTGCGGTTGGAACCTGATACTGGAATCTCTTTGGTTTGGTGTGC
CGATTGCGACTTGGCCTCTGTATGCTGAGCAGCAGATGAATGCGTTTGAGGTTGTGA
AGGAGATGAAGATTGGAGTGGAGATAAGTTTGGATTATCGGCTTGAAATGGGCGGT
AAACAAGCAGAAGGTTCTGGGATTATAAGTGGTGAACAGATTGAGAGAGGGATTAG
AGATGTGATGCAGGAGGATAGTGAAGTGAGGAAGAAGGTGAAGCTGATGATGGAA
AAGAGTAGAGAGGCAGTTGTGGAGGGAGGCTCCTCTTATAATTATATCCAAAACCTTC
ATCAGTGATCTCAGGACCAACATTGGCTTGTA RrUGT13 amino acid GenBank
accession MF674538 (SEQ ID NO: 12):

MAEISLIFIPFPVISHLTPTIEIAKILLSRDHRLSITFLVIDIPQRDASLASLTTSIISDRLHFLD
VVLPPNQHSQSSKPSGIAAIESAKPAVKKTISDLVVRSQSAASGPRIAGFVLDMFCTAMID
IATEFNLPSYIYYTCGSSFLSIVLHVQKLCDDDALDIADFKNSSVEFSLPEFSNLIPARLLPS
MALDKDFSASFVGKARAFRKTGILVNSLVELEPHAIESMKLDRSVPPIYSVGPVLNMN
SNTAFIRQEQEKEIMEWLDQQPPASVVFLCFGSRGAFKPDQVKEIARGLESSGCRFLWAL
RQPSSSNVRFSPTDYEDFSEVLPEGFLQRTYGVGKVIGWAPQTAVLDHPSVGGFVSHC
GWNSILES LWFGVPIATWPLYAEQQMNAFEVVKEMKIGVEISLDYRLEMGGKQAEGSGI
ISGEQIERGIRDVMQEDSEVRKKVKLMMEKSREAVVEGGSSYNYIQNFISDLRTNIGL-
RrUGT17 DNA (SEQ ID NO: 13):

ATGGCTCCTTCTCCACAAATCTCCTCGCTCCCATACCCCTGCCCAA
GGCCACATCAACCCTTTTCATGCAACTTGCCAAGCTCCTACACTCAAAAGGTTTCCAC
ATAACCTTCGTCAACAATGACCACAACCATCGCCGTTTGCTCAGAACAAAAGGGCA
TGATTTTGTTCAGGGTTGGAAGGTTTAAGGTTTGAAGCTGTGCCGGATGGCCTACC
TCCATCTGACCGTGATGCCACTCAGGATGTCCCTAAGCTGACTGAATCTATTTACAA
TAAGAGCATGAACCAACCGTTCAGTGATCTGCTTCAGAGGCTAAACTCAACGCCCG
GTTCCCTCCGGTCACTTGTGTTCATATCCGATGTTGCCATGTTTTTTGCTTGGGACGT
GGCGGATGAGCTTGGCATCCCTAATGTTTCAGTTTTTGGACAGCTTCAGCTTGTGGCCT
TTTGGGATACTTACAGTATGATGAGCTCCTAAGAAGAGCCATAGTCCCATTCAAAGA
TGAAAATTTTCATGACGGATGGTTCGTTGGAGGCTTTGATTGACTGGATTCTTGGCAT
GCCTAACATGAGGCTGAAGGACTTGCCAAGCTTCATGCGGACCACAAGCCCTGACG
ACGTGTTGTTCAATTACTTGCGTACAATAACCACGAAAGCTCTAAAATCCTCGGCCT
TGTTGCTGAACACATTTGATGATTTTGAACATGAAGTAGTTGAAGAGATGAAGAAA
ATGCAACCAAACATATTCCTAGGAGGTCCACTCAACATGCTTCTCAGGCACACATCA
AAAATGAAATCACATCCTTAACAACAAGTTTATGGAAAGAGGACACTCATTGTTTA
GAATGGCTGGACAAGCAAGAACCGGAGTCAGTGGTATACATCAATTACGGATCGGT
GACGATAATGTCTGATCACCATTTAATGAGTTTGCTTGGGGTTTGGCTAACAGCAA
GCACCCTTTTTTGTGGATCGTGAGGCCGGATGTTGTGAGGGGGCGAGTCGGGGACTTT
GCCCAAGGAGTTTTATGATGAGATCAAGGACAGGGGATTGATAACGAGCTGGTGTC
CGCAACCAGAGGTGCTTAAACATCCATCCGTAGGTGTATACTTGACGCATTGTGGTT
GGAATCTATCACGGAGAGTGTGGCCGGAGGAGTGCCATTGATGTGCTGGCCGTTTT
TCGCTGAGCAACAGACGAATAGCCGATTTCGCGTGTACGGTGTGGGGCACTGGAGTG
GAGGTGAATGCGGATGTGAAGAGGGAGGAGCTAGCGGAACAAGTGATGGAGATGT
TGGAAGGAAAGAGGGGGCAAGAGTTGAGGAAAAATGCTAAGGAGTGAGAGGAGGAA
GGCGGAGGAGGCGACGGACATTGGCGGTTCTGCCTATGCTGATTTTCGATAGGTTTAT
GGAAAAAGTGGTCCAGTTTAGCGTCTGA

RrUGT17 amino acid GenBank
accession MF674542 (SEQ ID NO: 14):

MGSLPSTKSHAVLVPYPAQGHINPFMQLAKLLHSKGFHITFVNNDHNHRRLLRTKGHDF
VQGLEGLRFEAVPDGLPPSDRDATQDVPKLTESIYNKSMNQPFSDLLQRLNSTPGSPVPT
CVISDVAMFFAWDVADELGIPNVQFWTASACGLLGYLQYDELLRRAIVPFKDENFMTD
GSLEALIDWIPGMPNMRLKDLPSFMRTTSPDDVLFNYLRTITTKALKSSALLLNTFDDFE
HEVVEEMKKMQPNIFLGGPLNMLLRHTSKTEITSLTTSWKEDETHCLEWLDKQEPESVV
YINYGSVTIMSDHHLNEFAWGLANSKHPFLWIVRPDVVRGESGTLPEFYDEIKDRGLIT
SWCPQPEVLKHPSVGVYLTHCGWNSITESVAGGVPLMCWPFFAEQQTNSRFACTVWGT
GVEVNADV KREELAEQVMEMLEGKRGQELRKNKEWRRKAEEATDIGGSAYADFDRE
MEKV VQFSV- RrUGT29 DNA (SEQ ID NO: 15):

ATGGGATCTCTAGGAAAGAAGATTCAACAAAAGCCACATGCAATATGCACCCCAT
CCCAGCACAAGGCCATATTAATCCCATGCTTAAACTAGCCAAGCTCCTACACCACTC
AGGCTTCTACATAACCTTTGTTCACACAACCTACAACCTACAATCGCCTTCTCAAGAC
CCACGGGTCTGATTCCTTAAGTGGTCTACCAGATTTCCAATTTGAGACCATCCCTGAT
GGACTACCACCATCAGATGCAGCTGATGTCACACAAGACATCCCTGCCTTGTGTAAA
TCAACCACCGAAACCTGCTTAGTCCCATTCAAAGAGCTCCTGGCTAAGCTGCATAAC
AAGTCAATGGCGTCACCGGAGGAAGTTCCTCCAGTGACATGCATAGTTTCTGATGGT
TGCATGTCATTTACTGTGGATGCTGCAGAAGAGGCAGGGGTTCTTAATGTGCTTCTT
TGGACTACCAGTGCATGCGGATTTTTTAGGATATGCTAATTACCCGAAACTTATTGAC
AGAGGCATAATTCCACTCAAAGATGAGAGCTACTTTACGAATGGGTACCTAGACAA
GACAGTAGATGGAATACCTGGAATGAAAGGCATACGGCTACGAGACTTCCCAAAT
TTGTATGCACCACAAACCCAGATGAGTTTATGGTGAAATATGCAATTCAAGAGATCA
CTAGAGCTGCCAGAGCAGATGCTGTTATTTTGAACACCTTTGACGCTTTGGAACATG
ATTTCTTAGATGGCCTATCAAACATATACCCAAAGGTCCTCCCTATTGGCCCGCTCC

AGCTTCACCTCAACCAAACTCCCAAGAGACCTCTACATTCTTCTAGTCT
TCTGGAAAGATGAACCACAGTGCATTACCTGGTTAAACTCCCAAAAACCAAAATCA
GTCGTTTATGTAACTACGGAAGTATCACAGTTATGACTCCGCAACAAATGGTGGAG
TTCGCATGGGGACTGGCTAATAACAAAATACCCTTTTCTGTGGATTATTAGACCTGAT
TTGGTTGCTGGTGAGACAGCTGTCCTACCTCCAGATTTTTTTGGAAGTGACAAAAGGA
AGGAGCTGCTTGGCTAGTTGGTGCCCCACAGGAACAAGTTCTTAGTCCACATCCATA
GGAGGGTTCTTAACCCATTGTGGGTGGAAGTCAATGCTAGAAAGCGTGGTTCGAAGG
AGTTCCAATGGTATGCTGGCCGTTTTTTGCTGAGCAACAGACTAATTGCTGGGCTGC
TCGGACAAAATGGGGTATAGGTATGGAAATTGACAATGATGTTAAGAGGGATAAGG
TTCAGAAAATGGTGACAGAGCTTATGGAGGGCGAAAAGGGAAAGGAGATGAAGAG
GAAGGGCGGAGAATGGAAGAAGCTTGGGGCAGAAGCTGCCGGTCCTAATGGCTCAG
CTACCTTAACTTCAGCAGACTTATAAATGACGTACTTCTGTCCAAAAAAAAAATTG
TAGTTACAACCTAA RrUGT29 amino acid GenBank accession MF674554 (SEQ
ID NO: 16):

MGSLGKKIQKPHAICTPYPAQGHINPMLKLAKLLHHSIFYITFVHTTYNYNRLKTHG
SDSLSGLPDFQFETIPDGLPPSDAADVTQDIPALCKSTTETCLVPFKELLAKLHNKSMASP
EEVPPVTCIVSDGCMSTVDAAEEAGVPNVLLWTTTACGFLGYANYPKLIDRGIPLKDE
SYFTNGYLDKTVDGIPGMKGIRLRDFPNFVCTTNPDEFMVKYAIQEITRAARADAVILNT
FDALEHDFLDGLSNIYPKVLPIGPLQLPLNQIPESSPLHSICSSLWKDEPQCITWLNSQKPK
SVVYVNYGSITVMTPQQMVEFAWGLANTKYPFLWIIRPDLVAGETAVLPPDFLEVTKGR
SCLASWCPQEQVLSHTSIGGFLTHCGWNSMLESVVEGVPMVCWPFFAEQQTNCWAAR
TKWGIGMEIDNDVKRDKVQKMVTELMEGEKKGKEMKRKGGEWKKLGAEAAGPNGSAT
LNFSRLINDVLLSKKKIVVTT RrUGT32 DNA (SEQ ID NO: 17):

ATGGGATCTCTAGGAAAGAAGATTCAACAAAAGCCACATGCAATATGCACCCCATA
CCCAGCACAAGGCCATATTAATCCCATGCTTAACTAGCCAAGCTCCTACACCACTC
AGGCTTCTACATAACCTTTGTTCACACAACCTACAACCTACAATCGCCTTCTCAAGAC
CCACGGGTCTGATTCCTTAAGTGGTCTACCAGATTTCCAATTTGAGACCATCCCTGAT
GGACTACCACCATCAGATGCAGCTGATGTCACACAAGACATCCCTGCCTTGTGTAAA
TCAACCACCGAAACCTGCTTAGTCCCATTCAAAGAGCTCCTGGCTAAGCTGCATAAC
AAGTCAATGGCGTCACCGGAGGAAGTTCCTCCAGTGACATGCATAGTTTCTGATGGT
TGCATGTCATTTACTGTGGATGCTGCAGAAGAGGCAGGGGTTCTAATGTGCTTCTT
TGGACTACCAGTGCATGCGGATTTTTTAGGATATGCTAATTACCCGAAACTTATTGAC
AGAGGCATAATTCCACTCAAAGATGAGAGCTACTTTACGAATGGGTACCTAGACAA
GACAGTAGATGGAATACCTGGAATGAAAGGCATACGGCTACGAGACTTCCCAAAT
TTGTATGCACCACAAACCCAGATGAGTTTATGGTGAAATATGCAATTCAAGAGATCA
CTAGAGCTGCCAGAGCAGATGCTGTTATTTTGAACACCTTTGACGCTTTGGAACATG
ATTTCTTAGATGGCCTATCAAACATATACCCAAAGGTCCTCCCTATTGGCCCGCTCC
AGCTTCCGCTCAACCAAATCCCAGAGAGCTCACCTCTACATTCAATCTGTTCTAGTC
TCTGGAAAGATGAACCACAGTGCATTACCTGGTTAAACTCCCAAAAACCAAAATCA
GTCGTTTATGTAACTACGGAAGTATCACAGTTATGACTCCGCAACAAATGGTGGAG
TTCGCATGGGGACTGGCTAATAACAAAATACCCTTTTCTGTGGATTATTAGACCTGAT
TTGGTTGCTGGTGAGACAGCTGTCCTACCTCCAGATTTTTTTGGAAGTGACAAAAGGA
AGGAGCTGCTTGGCTAGTTGGTGCCCCACAGGAACAAGTTCTTAGTCCACATCCATA
GGAGGGTTCTTAACCCATTGTGGGTGGAAGTCAATGCTAGAAAGCGTGGTTCGAAGG
AGTTCCAATGGTATGCTGGCCGTTTTTTGCTGAGCAACAGACTAATTGCTGGGCTGC
TCGGACAAAATGGGGTATAGGTATGGAAATTGACAATGATGTTAAGAGGGATAAGG
TTCAGAAAATGGTGACAGAGCTTATGGAGGGCGAAAAGGGAAAGGAGATGAAGAG
GAAGGGCGGAGAATGGAAGAAGCTTGGGGCAGAAGCTGCCGGTCCTAATGGCTCAG
CTACCTTAACTTCAGCAGACTTATAAATGACGTACTTCTGTCCAAAAAAAAAATTGT AG
RrUGT32 amino acid GenBank accession MF674557 (SEQ ID NO: 18):

MSGLGKIQKQKPYPAQGHINPMLAKLLHHS GFYITFVHTT YN YNRLKLTHG
SDSLSGLPDFQFETIPDGLPPSDAADVTQDIPALCKSTTETCLVPFKELLAKLHNKSMASP
EEVPPVTCIVSDGCMSTVDAAEEAGVPNVLLWTT SAGFLGYANYPKLIDRGIPLKDE
SYFTNGYLDKTVDGIPGMKGIRLRDFPNFVCTTNPDEFMVKYAIQEITRAARADAVILNT
FDALEHDFLDGLSNIYPKVLPIGPLQLPLNQIPESSPLHSICSSLWKDEPQCITWLNSQKPK
SVVYVNYGSITVMTPQQMVEFAWGLANTKYPFLWIIRPDLVAGETAVLPPDFLEVTKGR
SCLASWCPQEQLSHTSIGGFLTHCGWNSMLESVVEGVPMVCWPFFAEQQTNCWAAR
TKWGIGMEIDNDVKRDKVQKMVTELMERGEKGKEMKRKGGEWKKLGAEAAGPNGSAT
LNFSRLINDVLLSKKKL- RrUGT33 DNA (SEQ ID NO: 19):

ATGAGCTTAATTGAAAAACCACTCACGGCCATAGAGACTCGTGAAAAACCAACACGC
TGTGTGCATCCCATAACCAGCTCAAGGCCATATCAATCCCATGATGCAACTTGCAA
GCTCCTCCACCACTCTGGTTTCCACATAACGTTTGTCCCACTGAGTATAATTATGAC
CGTCTAGTGAAGTCTCAAGGTT CAGCTTGTGTGGCTGGTTTACCGGATTTCCGCTTTG
AAGCCATCCCAGATGGCTTGCCCTCGACGAATGGTGATGTTACTCAAGACATTCCTC
TGTTGAGTAGCTCTACTTCTAAAACCTGCTTGAAGCCGTTTAAGGAGTTATTGAAGA
GGTTGCAGGACAAATGCAAAGAGTTACCTGATGATGTTCCGCCTCTGTCTGTCATCG
TGTCTGATGCAGCCATGTCGTTTACGATCGATGCATCTGAGGAGTTTGGAGTGCCCA
TAGCGCTTCTTTGGACTGCAAGTGCCTGCGGGTTCTTGGGTACACGCATTACCCAT
ATCTAATTGACAGAGGTGTCATCCCATGAAAGATGAGAGCCAATTAACAAACGGA
TACCTAGATATGAGCATAGATGGCATACTTGTATGGAAGGTATCCGCTTACGAGAC
CTCCCAAGCTTTCTACGCACA ACTGATTTAGATGATATGATGTTTAGTTATATACTGC
ACGAAATAAAACAAGTTTCAAGAGGCAGTGCTATCATTCTGAACACCTTTGAAGCTT
TGGACCATGATGTCTTGGATAGTCTCTCCAAAATTTACCAAATGTCATCCTGCCAG
TTGGCCCTCTACATGTCTCGCTCAACAAGATCCCAAACACTACCCACTTCAATCTTT
AAGCTCGAATTTATGGAAAGATGACACAGACTGCATTCCCTGGCTGAGCTCTAAGGC
TTCAAATCAGTTATATACGTTAACTTTGGGAGCATCACGACGGTATCACCAAACA
AATTGTGGAGTTTGCCTGGGGATTGGCTAACAGCAAACACCCTTTCCTTTGGATAAT
CAGACCGGACTTGGTGGCAGGTGAGGCATCCATCATTCCGCAGGACTTCATGGATG
AAACAAAAGGAAGAGGTTTGTGGCTGGTTGGTGTGACCAAGAGCTTGTTCTCAACC
ATCCATCCATTGGAGGGTTTCTTACGCACTGTGGCTGGA ACTCAATTATTGAAAGCA
TTAGCGCAGGAGTCCCTACGGTCTGCTGGCCATTTTTTGTCTGAGCAGCAAACAAATT
GTTGGTTTGTCTGCAAAAAATGGTGCATTGGGATGGAGATGCATACTGATGTAAAGA
GGGATGAGGTTGACAAGCTGTTGAGAGAGCTAATGGAAGGTGACAAAGGGGAGGA
GTTGAAGAGGAAGGCAACCAACTGGAAGAGGCTGGCAGAAGAAGCTGTTTCTCCA
CTGGCTTATCAACCTTAACTTCAGGACGTTAGTGAATCAAGTCCTGCTCTCAAAAA
CAAAACATATCCGTTAG RrUGT33 amino acid GenBank accession MF674558
(SEQ ID NO: 20):

MSLIEKPLTAIETREKPHAVCIPYPAQGHINPMMQLAKLLHHS GFHITFVHTEYN YDRLV
KSQGSACVAGLPDFRFEAIPDGLPSTNGDVTQDIPLSSSTSKTCLKPFKELLKRLQDKCK
ELPDDVPPLSCIVSDAAMSFTIDASEEFGVP IALLWTASAGFLGYTHYPYLIDRGVIPLK
DESQLTNGYLDMSIDGIPCMEGIRLRDLPSFLRTTDLDDMMF SYILHEIKQVSRGSAILN
TFEALDHDVLDLSLSKIYQNVILPVGPLHVSLNKIPKHYPQLSSLNLWKDDTDCIPWLSS
KASKSVIYVNF GSITTVSPKQIVEFAWGLANSKHPFLWIIRPDLVAGEASIIPQDFMDETK
GRGLLAGWCDQELVLNHPSIGGFLTHCGWNSIIESISAGVPTVCWPFFAEQQTNCWFAC
KKWCIGMEMHTDVKRDEVDKLLRELMEGDKGEELKRKATNWKRLAE EAVSSTGLSTL
NFRTLNVNQLLSKTKHIR *Arabidopsis thaliana* AT2G20340.1 (SEQ ID NO: 21):
MENGSGKVLKPM DSEQLREYGHLMVDFIADYYKTIEDFPVLSQVQPGYLHKLLPDSAP
DHPETLDQVLDDVR AKILPGVTHWQSPSFFAYYPSNSSVAGFLGEMLSAGLGIVGFSWV
TSPAATELEMIVLDWVAKLLNLPEQFMSKGNGGGVIQGSASEAVLVVLI AARDKVLRSV
GKNALEKLVVYSSDQTHSALQKACQIAGIHPENCRVLT TDSSTNYALRPESLQEAVSRD

LEAGLFFLCANVGTTSTAVDPLAAGVHVDAAAYAGSACICPEYRQYID
GVETADSFNMNAHKWFLT NFDCSLLWVKDQDSLTLALSTNPEFLKNKASQANLVVDY
KDWQIPLGRRFRSLKLWMVLRLYGSETLKS YIRNHIKLAK EFEQLVSQDPNFEIVTPRIF
ALVCFRLLVPVKDEEKKCNNRNRRELLDAVNSSGKLFMSHTALSGKIVLRCAIGAPLTEEK
HVKEAWKIIQEEASYLLHK *Brachypodium distachyon* 1g28960.3 (SEQ ID NO: 22):
MDGSTTSNGGGGWMRPMD EEQLRECGHRMVDFIADYYKSIETYPVLSQVQPGYLKEL
LPDSAPNQPD TLDALFDDIREKIVPGVTHWQSPNYFAYYPSNSSTAGFLGEMLSAAFNIV
GFSWITSPAATELEVIVLDWVAKMLKL PSEFLSAALGGGVIQGTASEAILVVLLSARDRT
LRKHGKKSLEKIVVYASDQTHSALKKACQIAGIFPENIRIVKADCSMNYAVTPGAVSEAI
SIDLSAGLIPFFICATVGT TSSSAVDPLHEL GQIAQAHD MWFHIDAAYAGSACICPEYRKY
LNGVEEADSFNMNAHKWFLT NFDCSLLWVKDRNYLIQALSTNPEFLKNKASQENSVID
FKDWQIPLGRRFRSLKLWMVLRLYGVENLQSYIRKHIQLAQHFEQLVISDPRFEVVTPR
NFSLVCFCCLVPPTCEVDNGHKL NLDLMDSANSSGKIFISHTVLSGKFVLR FVVGAPLTEE
QHVDAAWKLLQDEATKLLGNVVQ *Carica papaya* 16427710 (SEQ ID NO: 23):
MDAEQLRENGHKMVDFIADYYKT IENFPVLSQVEPGYL RDLIPDSAPNSPESFQQLDD
VRTKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGLNIVGFSWITSPAATELEMIVL
DWLAKLLKL PEDFHSTGNGGGVIQGTASEAILVVLLAARDKVLKRVGKNALEKLVVYT
SDQTHSAFQKACQIGGIHPENCRVLK TDSSTNYALSPDLLKEAISCDVAAGLIPFFFCATV
GTTSSSTAVDPLMALGKIATSNEIWFHVDAAYAGSACICPEYR PYIDGVEEADSFNMNAH
KWFLT NFDCSVLWVKDKYSLIQSLSTNPEFLKNKASQADMVVDYKDWQIPLGRRFRSL
KLWMVLRLYGVENLKS YIRNHIKLAKHFEELVTQDPRFEVVTPRIFSLVCFRLLPPGNDE
NHGNKLNQD LLETVNSTGKLFISHTVLSGKYILRFAVGAPLTEERHVNEAWKILQDEAS
TLENP *Ricinus communis* 16804377 (SEQ ID NO: 24):
MFREGELRPMDAEQLREHGHKMVDFIADYYKT IENFPVLSQVEPGYL RKL LPDSAPNQP
ESLQNVLDDVQAKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGINMVGFSWITSP
AATELEMIVLDWL GKMLKLPEEFLSTGQGGGVIQGTASEAVLVALVAARDKVLRRVGK
DALRKL VVYGSDQTHSALQKACQIGGIHPVNCRLLETDSSTNYALAPDLLSRAISEDISL
GLIPFFLCATVGT TSSSTAVDPLLALGKIAKSNGMWFHVDAAYAGSACVCPEYRCYMDG
VEEADSFNMNAHKWFLT NFDCSALWVKDRNALIQSLSTSPEFLQNKPSQTNTVVDYKD
WQIPLGRRFRSLKLWMVLRLYGVEKLQCYIRNHINLAKYFEGLIAEDTRFEVVSPPIFAL
VCFRLLPPDNNVDHG NKL SHDLLDAVNSTGKIFISHTVLSGKYILRFAVGAPLTEERHVT
AAWKVLQDEACALLETSRIS *Cucumis sativus* 16963476 (SEQ ID NO: 25):
MDNELKPMDAEQLREHAHKMVDFIADYYKNIEDFPVLSQVEPGYLQNLLPESAPLN PES
LQSVLDDVQKKIFPGVTHWQSPNYFAYYPSNSSIAGFLGEMLSAAFNVIGFSWVTSPAA
TELEMIVLDWLAKLLKL PDDFLSSGNGGGVIQGTASEAVLVVLLAARDRALRRFGKDY
LKKLVVYASDQTHSALQKACQIGGIHPENCRWLKADISTNYALSPDVLSEELSRDTARG
LIPFFLCATVGT TSSSTAVDPLPELG TIAKRHEMWFHVDAAYAGSACVCPEYRQYIDGVE
EADSFNMNLHKWFLT NFDCSALWIKDRHALIRSLSTNPEFLKNKASEAELVVDYKDWQI
PLGRRFRSLKVWMVLRLYGTENLQKYIRNHISLAERFEALVREDPRFEIVTPRIFSLVCFR
LLPSRK NEDGGNRLNQSLLDAVNASGNIFISHTVLSGKYILRFAVGAPLTEEKHINSAWK
LLQDVASTLLAI *Vitis vinifera* 17835588 (SEQ ID NO: 26):
MDAEQLRENGHKMVDFIADYYKSIENFPVLSQVEPGYLRELLPDSAPNQPESLQQVFDD
LQAKILPGVTHWQSPNFFAYYPSNSSTAGFLGEMLSAGLNIVGFSWITSPAATELEMIVL
DWLAKLLNLPDDFLSAGNGGGVIQGTASEAVLVVLLAARDRLRTVGKTALEKLVVY
GSDQTHSALQKACQIGGIHPENCKLLKADSSTGYALSPDLLSEAVSHDITNGLIPFFLCAN
VGTTSSSTAVDPLLELGKVTKSNGIWFHVDAAYAGSACVCPEYRHYIDGVEEADSFNMN
AHKWFLT NFDCSVLWVKDRNALVQALSTNPVFLKNKASDANMVVDYKDWQVPLGRR
FRSLKLWMVLRLYGVENLQRYIRNHIKLAKQFEELVAQDPRFEIVAPRK FALVCFRLLPP
HRNEDFSNKL NHNLLDTVNSTGKVYISHTALSGKYTLRLAVGAPLTEERHVNAAWKVI
QEKASVLLSEFGMNGLFDNINLKFILNHQIDISILLNYN *Citrus sinensis* 18113817

(SEQ ID NO: 27):

MDAEQLRENAHKMVDFIADYYKSIENFPVLSQVQPGYLHNLPDSAPHHPESLQNVLDG
YIDIQEKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGLNIVGFSWITSPAATELEMI
VLDWLAKLLKLPEDFLSSGQGGGVIQGTASEAVLVLLAARDKALKRVGKNSLEKLVV
YASDQTHSALQKACQIGGIHPQNFRVLKTDSSSTNYSLSPDSLAEAISRDLTIGLIPFFLCAT
VGTTSSSTAVDPLLALGNIAKSNGMWFHVDAAYAGSACICPEYRQYIDGVEEADSFNMN
AHKWFLTNFDCSALWVKDRNTLIQSLSTNPEFLKNKASQANMVVDYKDWQIPLGRRFR
SLKLWMVLRLYGLENLQGYIRNHIQLAKHFEGFLVAQDLRFEVVTPRIFSLVCFRLLPPHN
DEDHGNKLNHKLLDDINSTGKIFISHTVLSGKYILRFAVGAPLTEWRHVNAAWVMQD
KASALLARLSIE *Capsella rubella* 20900667 (SEQ ID NO: 28):

MGFCQIELLRHINKHNMQNGSGKNVLKPM DSEQLREYGHRMVDFIADYYKTIEDFPVL
SQVQPGYLHQLLPDSAPDHPETLDQVLDDVRakilPGVTHWQSPGFFAYYPSNSSVAGF
LGEMLSAGLGIVGFSWVTSPAATELEMIVLDWLAKLLNLPKEFLSKGNGGGVIQGSASE
AVLVVLI AARDKVLR SAGKNALGKLVVYSSDQTHSALQKACQIAGIHPENCRVLETDAS
TNYALRPELLQEAVSKDLKAGLIPFFLCANVGTTSSSTAVDPLAALGKIANSNEIWFHVDA
AYAGSACICPEYRKYIDGVETADSFNMNAHKWFLTNFDCSLLWVKEQDSLTEALSTNP
EFLKNKASQANLVVDYKDWQIPLGRRFRSLKLWMVLRLYGAETLKS YIRNHIKLAKYF
EKLVSQDPNFEIVTPRIFSLVCFRLVPKNDDEKKCNNQNRKLL EAAANSSGKLFMSHTALS
GKIVLRCAIGAPLTEEKHMKEAWKVIQDEASFL LHK *Malus domestica* 22636618

(SEQ ID NO: 29):

MSGLKPM DAEQLRENAHKMVDFIADYYKTIEDFPVLSQVQPGYL RDLLPDSAPTHPESL
QQVFDDIQAKILPGVTHWQSPNFFGYPSNSSVAGFLGEMLSAGLNIVGFSWITSPAATE
LEMIVLDWFAKMLKLPEEFLSAGQGGGVIQGTASEAVLVLLAARDRILRAEGKKALE
KLVVYASDQTHSALQKACQIGGIHPENCRVLSTDSSSTNYALSPNVLNEAISNDIASGLVP
FFLCATVGTTSSSTAVDPLLELGKITKSNGMWFHVDAAYAGSACICPEYRHHIDGVEEAD
SFNMNAHKWFLTNFDCSLLWIKDRNALVQALSTNPEFLKNKASQANLVVDYKDWQIPL
GRRFRSLKLWMVLRLYGLENLQSYIRNHIDLAKCFEDLVAQDSRFEIVTPRIFSLVCFRL
LPPHNDETYATKLNHDLLDTVNSTGKIFVSHTVLSGKYVLRFAVGAPLTEERHVLA AW
KLLQEEASALLAPL *Linum usitatissimum* 23178995 (SEQ ID NO: 30):

MGGYRSLNLIFIFISFVADIRDLGYNTKEGDDGGGALKPMDAEQLRQNAHQMVDFIADY
YKNIETYPVLSQVEPGYLRELLPDSAPNRPESLQSVLDDVQSKIMPGVTHWQSPNYFAY
YPSNSSVAGFLGEMLSAGINMVGFSWITSPAATELEMIVLDWL GKLLKLPEEFLSSGHG
GGVIQGTASEAILVLLAARDKMLRKFGKSALEKLVVYASDQTHSALQKACQIGGIYPE
NCRL LKTDSSVNYSLTPELVSEAVSQDISAGLIPFFLCGTVGTTSSATVDPLGTLGKIAKN
NDMWFHVDAAYAGSACICPEYRQYLDGVEEADSFNMNAHKWFLTNFDCSTLWVKDK
SALI QALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFRSLKLWMVLRLYGVENLQQY
LRNHIELARHFEECVNHDP RFEALSGKYTLRVAIGAPLTEKRHVAAALKVLQDEATSL L
VATSPLLENGNSS *Eutrema salsugineum* 20200788 (SEQ ID NO: 31):

MENGKNV LKPM DSEQLREYGHRMVDFIADYYKTIEDFPVLSQVQPGYLHNLLPDSAP
DQPETLEEVLDDVKGKILPGVTHWQSPSFFAYYPSNSSVAGFLGEMLSAGLGIVGFSWIT
SPAATELEMIVLDWLAKLLNLPEQFLSRGNGGGVIQGSASEAELVVLI AARDKVLR SVG
KKALEKLVVYSSDQTHSALQKACQIAGIHPENCRVLKADYSTNYALRPETLQEAVSKDL
EAGLIPFFLCANVGTTSSSTAVDPLAALGEIAKS NEMWFHVDAAYAGSACICPEYRQYID
GVETADSFNMNAHKWFLTNFDCSLLWVKDQYALTEARSTNPEFLKNKASQANLVVDY
KDWQIPLGRRFRSLKLWMVLRLYGSENLSYIRNHIKLAKDFEQLVSEDPNFEIVTPRIF
LVCFRIVPAENDEKKCNNQNRNLLDAVNSSGKLF LSH TALSGKIVLRCAIGAPLTEEKHV
KEAWKVIQEEASYLLRK *Populus trichocarpa* 27022899 (SEQ ID NO: 32):

MESKGLQPM DSEQLRENAHKMVDFIADYYKSIENFPVLSQVEPGYLRELLPDSAPNQPE
TLQNVLDDVQAKILPGVTHWQSPSYFAYYPSNSSVAGFLGEMLSAGINMVGFSWITSPA
ATELEMIVLDWL GKLLKLPEDFLSTGQGGGVIQGTASEAVLVLLAARDRVLRKLGN

ALEKLVVYASDQTHSALQKACQIGGIHPENCKLLKTGSSTNYALSPDLLGKAISDDISTG
 LVPFFLCATVGTTSSTAVDPLLSLGKIAKNNGIWFHVDAAYAGSACICPEYRCYIDGVEE
 ADSFNMNAHKWFLT NFDCSALWVKDRNALIQSLSTNPEFLKNKASQANMVVDYKDW
 QIPLGRRFRSLKLWMVLRLYGLENLQCYIRNHINLAKYFEG LVAADSRFEVVTPRIFSLV
 CFRLPPNNNEDHGNNLNHDLLDAVNSTGKIFISHTVLSGKYILRFAVGAPLTEERHVTA
 AWKVLQDEASALLGSL *Brachypodium stacei* 06G160800.1 (SEQ ID NO: 33):
 MDGSTTSNGDGGGGWMPMDDEEQLRECGHRMVDFIADYYKSIETYPVLSQVQPGYLK
 ELLPDSAPNQPD TLDALFDDIQEKIVPGVTHWQSPNYFAYYPSNSSTAGFLGEMLSAAFN
 IVGFSWITSPAATELEVIVLDWVAKMLKLPSQFLSAGLG GGVIIQGTASEAILVVLLSARD
 RTL RKH GKKSLEKLVVYASDQTHSALQKACQIAGIFSDNIRIVKADCSMNYAVTPGSVS
 E AISIDLSSGLIPFFICATLGTTSSSAVDPLHEL GQIAQA HDMWFHIDAAYAGSACICPEYQ
 QYLN GVEEADS FNMNAHKWFLT NFDCSLLWVKDRNYLIQALSTNPEFLKNKASQENSV
 IDFKDWQIPLGRRFRSLKLWMVLRLYGVENLQSYIRKHIQLAQRFEQLVISDSRFEVVT
 RNFSLVCFCLVPPTSEVDNGHKLNYDLMDSVNSSGKIFISHTVLSGKFVLRFAVGAPLTE
 EQHVNAAWKLLQDEATKLLGSVVV *Physcomitrella patens* Pp3c4_30790V3.1 (SEQ
 ID NO: 34):

MGSEAGSRSSLTKPFDPEEFRKHAHRMVDFIADYHRDIENFPVQSQVEPGYLQKLLPEN
APDEPESLDDILADVQSKIVPGVTHWQSPNFYGYYPSTAGFLGEMLSGGFNIIGFSW
ITSPAATELEIIVMDWLGLLLKLPNEFLSSGKGGGVIQGTASEAVLVVMLAARKRAVEK
LTKEQGISEFEALAKLVAYTSDQAHSCVNKASQIAGISIENLRLIPTDVSTNYAMSSKVL
NTLANDVKAGLVPFFLCGVIGSTSSAAVDPLSELGDLAQEYGMWFHVDGAYAGNACIC
PEFRPYLNGVEKADSFDMMNPHKWLLTNFDCSTLWVKNPSLLVDALSTNPVFLRNKQSD
NNLVVDYKDWQIPLGRRFRSLKLWMLRMYGSNGLRSYITNHCNLAKHFEELLRTDSR
FEVVAPRVFSLVCFRLKSPANDADNSCSLSAKLVDALNSDGNILITNTVLGGRYTIRFTV
GASRTEL RHVDA AWKVIQQLASKLLKECSS *Ananas comosus* 33033299 (SEQ ID
NO: 35):

MESELKPM DSEQ LREYAHKMVDFIADYYKMIESFPVLSQVKPGYLKELL PDSAPCKPEN
LEDVFDDIRQKIIPGITHWQSPDYFAYYPSNSSTAGFLGEMLSAGFNIGFSWIASPAATEL
EMIVLDWFAKMLKLPEQFLSTGQGGGVIQGTASEAVLVVLLAARDKILLKAGRKSLEKL
VVYCSDQTHSAMQKACQIAGIFPENFRVLKTDSSSNYALLPEVLSEAI SKDLSFGLIPFFL
CATVGT TSSAAVDPLLKLG NISKVHDMWFHVDAAHAGSACICPEYRHHIDGVEEADSF
CMNAHKWFLT NFDCSLLWVKDRSALIQSLSTNPEFLKNKASQENS VVDFKDWQIPLGR
RFRSLKLWMVLRLYGLENLQSYIREHIKLAEQFEQLISSDSRFEIVAPRTFSLVCFRLLPPL
YDQDDGYKLNYNLLDAVNRSGKIFMSHTVLSGKFVLRFAIGAPLTEERHVVA AWKVLQ
DEATILLRGS *Zostera marina* 33182387 (SEQ ID NO: 36):

MLNGNMGENEPFKPMDSEQLREYGHKMVDFIADYYKSIEKFPVLSQVQPYYLKDLLPD
AAPDQPEKFQDVLDDITKKIIPGVTHWQSPNFFGYYPGNSSIAGFLGEMICSGLNVIGFS
WITSPASTELEVIVLDWLAKLLNLPDQFLSSGHGGGVIQGTASEAILVVLLAARDKILGRI
GRNSLDKLVVYSSDQVHA AFKKACQIAGIYTENFRVLKTDASSGYGIDPKKFDQAIHDD
MEAGLIPFFLCSTVGTTSASVDPLVEIGQITEENDMWFHVDAAYAGSACICPEYRHYLD
GVEYADSFCMNAHKWLLTNFDCSALWVKDSSALVNSLSTNPEFLKNKMSEQKKVVD
KDWQIPLGRRFRSLKLWMVLRLYGAENLREYIRNHIKLANLFEQLVRSDSRFEIVCPTLF
SLVCFRFLPSNDDNDGYELNSMLLD AVNSTGQLFFTHHTIISDKYILRFAVGAAALTEERHV
RESWKVIQNQATIISRQHILSKTNMKSKEGMIANE *Daucus carota* subsp. *sativus*
36055203 (SEQ ID NO: 37):

MDGVLKPMDAEQLRENAHKMVDFIADYYKNIETFPVLSQVEPGYLRDLLPHSAPDQPE
SLQNILDDIQAKILPGVTHWQSPNYFAYFPSNSSVAGFLGEMLSAGINMVGFSWITSPAA
TELEMIVLDWLAKLLKLPDHFLSTGQGGGVIQGTASEAVLVVLLAARDKVLRLITGKDAL
GKLVVYCSDDQTHSALQKACQIAGIHPGNCRVLKTESCNDYSLSPETFEQAISTDVASGLI
PLLLCATVGTTSSTAVDPLLELGKITKMKGIWLHVDAAYAGSACVCPEFRHYIDGVEEA

DSFNMNAHIHSLSTNPEFLKNKASQENLVVDYKDWQIP
LGRRFRSLKLWMVLRLYGLENLQSYIRNHIQLAATFESFVTEDPREFVAPRKFAFALVCFR
LLPPSHKDEDCSNQLNRDLLDAVNATGKAFVSHALTSGRYVVRFAIGAPLTEESHIIEAW
KIFQEVATVLLKSLKMNHTRPLN *Trifolium pratense* 35974269 (SEQ ID NO: 38):
MVDFIADYYKTIENFPVLSQVEPGYLGKLLPDSAPTYPTTLEHVLNDVQHKILPGVTHW
QSPNYFAYFPSNSSIAGFLGEMLSAGINIVGFSWITSPAATELESIVLDWLAKALFLPQDF
LSNGKGGGVIQGTASEAVLVVLLAARDKILRTVGRSALPKLVTYASDHVHSSLLKACQI
GGLDPELCRLLKTDSSSTNFALSPDVLSEAISNDIASGLIPFFLCANVGTTSSSTAVDPLPALA
KVTKTNNIWLHVDAAYAGSACICPEYRHFIDGVEEADSFNMNAHKWFLTNFDCSLLWV
KDRSALIQLSTNPEFLKNKASEGNMVIDYKDWQIPLGRRFRSLKLWMVLRLYGLEGLR
SHIRNHIALAASFEELVVQDARFKVVTPTRTFSLVCFRLLPPPNSDNGNKLNHDLDDLNVN
STGSVFITHTVLSGEYILRLAVGAPLTEVRHVNAAWQILQEKATALLENL *Arabidopsis*
lyrata 35943929 (SEQ ID NO: 39):
MDSEQLREYGHRMVDIFIADYYKTIEDFPVLSQVQPGYLHKLLPDSAPDHPETLDQVLDD
VRAKILPGVTHWQSPSFFAYYPSNSSVAGFLGEMLSAGLGIVGFSWVTSPAATELEMIVL
DWLAKLLNLPEQFMSKGNNGGGVIQGSASEAVLVVLLAARDKVLRSVGKNALQKLVVYS
SDQTHSALQKACQIAGIHPENCRVLKTDSSSTNYALRPELLQEAVSQDL DAGLIPFFLCAN
VGTTSSSTAVDPLAALGKIANRNEMWFHVDAAYAGSACICPEYRQYIDGVETADSFNMN
AHKWFLTNFDCSLLWVKDQDSLTLALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFR
SLKLWMVLRLYGSETLKSIRNHIKLAKEFEQLVSQDPNFEIVTPRIFSLVCFRLVPVKNE
EKKCNRRNRELLDAVNSSGKLFISHTVSDFSFFLLFFLLDNVLNLRGNRLCRGKSYCVA Q
Sorghum bicolor 002G120700.1 (SEQ ID NO: 40):
MDGSGSSGGTNGGSGGDGAGWLRPMDAEQLRECGHRMVDVFADYYKSIETFPVLSQV
QPGYLKELLPDTAPNKPDTLEALFDDIREKIVPGVTHWQSPNYFAYYPSNSSTAGFLGE
MLSAAFNIVGFSWITSPAATELEVIVLDWFAKMLRLPSQFLSTALGGGVIQGTASEAVLV
VLLAARDRTL RKH GKTSLEKLVVYASDQTHSALQKACQIAGIFPENVRVLKADCNRNY
AVAPLAISDAIATDLSSGLIPFFICATVGTTSSSAVDPLPELGQIAKANDMWLHIDAAYAG
SACICPEYRHHLNGVEEADSFNMNAHKWFLTNFDCSLLWVKDRSYLIQLSTNPEFLKN
KASEANSVFDFKDWQIPLGRRFRSLKLWMVLRLYGVENLQSYIRKHIELAKEFEQLVIS
DSRFEVVTPTRTFSLVCFRLVPLASDQDNGRKLNYDLMDAANSSGKIFISHTVLSGKFVLR
FAVGAPLTEGQHIFSAWKILQDLATKQLLESS *Sphagnum fallax* 0166s0011.1 (SEQ
ID NO: 41):
MGSEAGEGSRSLSKPLDVEEFRKHAHQMVDFVADYHRDIESFPVRSQVKPGYLRPLLPS
APAEPETVEDVFADLWSKILPGLTHWQSPKFFGYPCNVSTAGMLGEMLCGGLNVNGF
SWITSPAATELETIVLDWLGLLHLPEEFLSTSGKGGGVIQGTASEAVLVVMLAARKRA
LKQVSSAAQGMSEAEALSKLVVYSSDQTHSCVIKACQVASIATENFRPLPTDASTNFALS
PAVVRKAIATDVEAGLIPFFLCGTLGTTSSAAVDPLEELGDIAKEYGMWYHIDAAYAGN
ACICPEFRHYLNGVEKADSYNMNPHKWLLTNFDCSTLWMKDSEFLLAALSNKPVFLRN
EATDNNLVVDYKDWQIPLGRRFRALKLWMVMRLYGTSGLQSFIRSHVSSAKHFESLVR
ADSRFEVMAPMTFSLVCFRLRTLPGSQDNSNSLSKLVDALNRKGNILVTHTELSGIYTV
RFAVGATHTELQHVQAAWEVIQAEASHLLNGKQ *Kalanchoe laxiflora* 1398s0003.1
(SEQ ID NO: 42):
MILSIHPFPFTLSARFSGAAAANILSKASCWLRCLRSMEGELKPMDAEQLREYGHRMVD
FVADYYKTIEDHPVLSQVEPGYLRKLLPDSAPDKPESFENVLSDVKTKIIPGVTHWQSPN
YFAYFPSNSSTAGFLGEMLSACFNIVGFSWITSPAATELEMIVLDWFAKMLKLPDFFLST
GQGGGVIQGTASEAVLVVLLAARDIFLRKLGKGFLEKLVVYASDQTHSALQKACQIAGI
HPENVRALKTDSSSTNYGLSPDLLSKEICHDIANGLVPFFACASVGTTSSSTAVDPILELANV
TKSYNIWLHVDSAYAGSACVCPEYRHHIDGVEEVDSFNMNAHKWFLTNFDCSLLWVK
DRNALIQLSTNPEFLKNKASQSNSVLDYKDWQIPLGRRFRSLKLWLVLRLYGVENLQA
YIRNHIELALNFEELVSQDMRFEIVAPRTFALVCFRLLLPCGFEDHTNDVNSDLLQAVNS

TGKIFISHTVLSGTYLRFVAGAPLTEERHIDAAWKLIQDQASSLLEKL *Manihot*
esculenta 12G038600.1 (SEQ ID NO: 43):
MEGELRPMDAEQLREYGHQMVDFIADYYKTIENFPVLSQVEPGYLHKLLPDSAPNQPE
ALQNVLDDVRVKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGINMIGFSWITSPA
ATELEMIVLDWLKGKMLKLPEEFLSSGQGGGVIQGTASEAVLVVLLAARDKVLTRVGKD
SLKKLVVYGSDQTHSALQKACQIAGVHLDNCRLKTDSSKNYALSPDILCDAISQDMSN
GLIPFFLCATVGTTSATVDPLLALGKIAKKYGMWFHVDAAYAGSACICPEYRCYIDGV
EEADSFNMNAHKWFLTNFDCSALWVKDRNALIQSLSTNPEFL1<NKASQANMVVDYKD
WQIPLGRRFRSLKLWMVLRLYGVANLQSYIRNHINLAKYFEGLVAGDSRFEVVAPRLFS
LVCFRLLPPDNDENHGKLNHDLDAANSTGKIFISHTVLSGKYILRFAVGAPLTEERHV
TAAWKVLQDEASALLGSL *Prunus persica* 8G214500.1 (SEQ ID NO: 44):
MESGLKPMDAEQLRENAHKMVDFIADYYKTIENFPVLSQVQPGYLRELLPDSAPTHPEP
LQHIFDDIQAKILPGVTHWQSPNFFGYPSNSSIAGFLGEMMSAGLNIVGFSWITSPAATE
LEMIVLDWFGKMLKLPEEFLSAGKGGGVIQGTASEAVLVVLLAARDKILRRVGKNSLE
KLVVYASDQTHSALQKACQIGGIHPENCRLLRDSSSTNYALSPNVLNEAISNDVTSGLIP
FFLCATVGTTSSTAVDPLLELGKIAKSNDMWFHVDAAYAGSACICPEYRHYIDGVEEAD
SFNTNAHKWFLTNFDCSVLWIKDRNALIQALSTNPEFLKNKASQANLVVDYKDWQIPL
GRRFRSLKLWMVLRLYGLENLQSYIRNHINLAKHFKELVAQDPRFEIVTPRLFSLVCFRL
LPPHNDETCATKLNHGLLDVAVNATGKIFISHTVLSGKYLLRLAVGAPLTEERHVNAAWK
LLQDEASALLATL *Eucalyptus grandis* K01418.1 (SEQ ID NO: 45):
MEERLKPMDAEQLRESAHRMVDFIADYYKSIESFPVLSQVEPGYLRKLLPDSAPDHPESL
QQVLEDVQAKILPGVTHWQSPNYFAYYPSNSSIAGFMGEMLSAGLNIVGFSWITSPAAT
ELEIIVLDWLAKLLNLPDDFLSTGPGGGVIQGTASEAVLVVLLAARDKFLSRIGKSSLDK
LVVYSSDQTHSALQKACQIGGIYPENCRLVKTDASTNYALSPDLLNEVISQDISTGLVPFL
LCATVGTTSSTAVDPLPALATVAKRNGMWFHIDAAYAGSACICPEYRPYIDGVEEADSF
NMNAHKWFLTNFDCSALWIKDRKALIQALSTNPEFLKNKASQANMVVDYDRDWQIPLG
RRFRSLKLWMVLRLYGVQNLQQYIRNHIELARQFEDLVIQDPRFEVVTPRIFSLVCFRLL
SPDNDGDKGNKLNDRDLLDTVNSTGKIFISHTVLSGTIYILRFAVGAPLTEERHVNEAWKV
LQDEASKLLATIQQN *Amborella trichopoda* 31565185 (SEQ ID NO: 46):
MDAEELREHGHRMVDIFISDYYKEIESYPVRSQVQPGYLRNLIPDSAPDMPESFESILEDIR
HKIIPGVTHWQSPKYFAYYPSNSSTAGFLGEMLSAGFNIVGFSWVTSPAATELEVIVLDW
LAKVLKLPEQFLSTGKGGGVIQGTASEAMLVALLAARDKALRRVGQNLLNLLVYVYSD
QTHSALIKACKIAGINPMNCRLQATFMTNYALSPEVASESISNDIAAGLLPIFLCATVGT
TSSTAVDPLAALGRLAKANDMWFHIDAAYAGSACICPEYRHYIDGVEEADSFNMNPHK
WLLTNFDCSTLWVKDSSNLIQSLSTNPEFLRNKASEEDLVVDYKDWQIPLGRRFRSLKL
WMVLRMYGVANLQNHIRTHINLAKHFEELIATDTRFEIIVPRVFALVCFALKPMPNGQD
DASKLNLKLEAVNNSGAMFLTHTVLSGRFVLRVVGAPLTEERHVNTAWKVLQDHA
NLILGTV *Salix purpurea* 0252s0200.1 (SEQ ID NO: 47):
MESKGLKPMDSQLRENAHKMVDFIADYYKSIENFPVLSQVEPGYLRELLPDSAPNQPE
TLQNVLDDVQAKILPGVTHWQSPSYFAYYPSNSSVAGFLGEMLSAGINMVGFSWITSPA
ATELEMIVLEWLKLLKLPEDFLSTGQGGGVIQGTASESVLVVLLAARDRVLTCLGKNA
LEKLVVYASDQTHSALQKACKIGGIHPENCCLKLKTDSSTNYALSPDLLSKAISDDISTGLI
PFFLCATVGTTSSTAVDPLHALGKIAKNNGIWFHVDAAYAGSACICPEYRCYIDGVEEA
DSFNMNNAHKWLLTNFDCSALWVKDRNALIQALSTNPEFLKNKASQANMVVDYKDWQI
PLGRRFRSLKLWMVLRLYGLENLQCYIRNHINLAKYFEGLVAAADSRFEVVTPRIFSLVCF
RLLPPSNNEHDHGNLNRDLLDAVNSSGKIFISHTVLSGKYILRFAVGAPLTEERHVIAAW
KVLQDESTLLGSL *Medicago truncatula* 31080941 (SEQ ID NO: 48):
MVLQIWCLTHDSKKLGGGYLLFPVIKVAYTVHTLTEWCCVTEEGGGSELKAMDAEQ
LREQGHMMVDFIADYYKTIENFPVLSQVQPGYLGKLLPDSAPTHPESLQHVLNDVQEKI
LPGVTHWQSPNYFAYFPSNSSIAGFLGEMLSAGLSIVGFSWISSPAATELETIVLDWLAK

ALLPLHFFSTGQGGVIQGTASEAVLVVLAARDKILRTVGRSALPKLVTYASDQTHS
SLQKACQIAGLNPELCRLLKTDSSTNFALSPDVLSEAISNDIASGLTPFFLCATVGTTSSTA
VDPLPALAKVTKPNNIWLHVDAAYAGSACICPEYRHFIDGVEEADSFNMNAHKWFLTN
FDCSVLWVKDRSALIQSLSTNPEFLKNKASQENTVIDYKDWQIPLGRRFRSLKLWMM
RLYGLEGLRTHIRSHIALAVYFEELVVQDTRFKVVAPRTFSLVCFRLLPPQNSDNGNKL
NHDLLDAVNSTGDVFITHTVLSGEYILRLAVGAPLTEVRHVHAAWQILQEKATALLES
Brassica rapa I01156.1 (SEQ ID NO: 49):

MQIRAKIPVFGRENGSRHVLKPMDSQLREYGHMVDIFIADYYKTIESFPVLSQVQPGY
LHNLLPDSAPDHPETVEQVLDDVKTILPGVTHWQSPNFFAYYPSNSSVAGFLGEMLSA
GVGIVGFSWVTSPAATELEMIVLDWLAKLLNLPEHFLSKGNNGGGVIQGSASEAILVMI
AARDKVLRSAGKNALGKLVVYSSDQTHSALQKACQIAGIHPENCRVLKADSSTNYALR
PELLQEAVSRDLEAGLIPFFLCGNVGTSSAAVDPLAALGKIAKSNEIWFHVDAAYAGS
ACICPEYRQYIDGVETADSFNMNAHKWFLTNFDCSLLWVKDQHALTEALSTNPEFLKN
KASQANLVVDYKDWQIPLGRRFRSLKLWMLRLYGAEALKNYIRNHIKLAKDLEQLVS
QDPNFEVITPRIFSLVCFRIVPTDNDKCKNSRNLELLEAVNSSGKLFISHTALSGKIVLRC
AIGAPLTEEKHVKETWKVIEKVSYLLRK *Brassica rapa* I04706.1 (SEQ ID NO:
50):

MDSEQLREYGHMVDIFIADYYKTITETFPVLSQVQPGYLNLLPDSAPDQPETVEQVLDD
VKTILPGITHWQSPTFYAYYPSNSSVAGFLGEMLSAGLGIVGFSWVTSPAATELEMIVL
DWLAKLLNLPEQFLSKGNNGGGVIQGSASEAILVVMIGAREKVLRRVGKNALGKLVVYS
SDQTHSALQKACQIAGIHPENCRVLKADSSTNYALRPELLQEAVSKDIEAGLIPFFLCGN
VGTTSSSTAVDPLAALGKIAKSNEIWFHVDAAYAGSACICPEYRQYIDGVETADSFNMNA
HKWFLTNFDCSLLWVKDQYVLTEALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFRS
LKLWMLRLYGAEATLKSIRNHIKLAKDLEQLVSQDPNFEVVTPRIFSLVCFRILPVDND
EKECNRNRNLLDAVNSSGKLFISHTALSGKIVLRCAIGAPLTEERHVKETWKVIEEAS
RLLGK *Brassica rapa* G00043.1 (SEQ ID NO: 51):

MDSEQLREYGHMVDIFIADYYKTITETFPVLSQVQPGYLNLLPDSAPDQPETLEQVLDD
VKEKILPGVTHWQSPSFFAYYPANSSVAGFLGEMLSAALNIVGFSWVSSPAATELEMIVL
DWFALLLNLPEQFLSRGNNGGGVIQGTASEAILVVMIAARDKVLRSGLGKKALEKLVVYS
DQTHSSLLKACQIAGIHLENCRMLKTDSSTNYALRPESLQEAVSGDLEAGLIPFFLCGT
VGTSSSTAVDPLAELGKIAKSNEIWFHVDAAYAGSACICPEYRQYIDGVETADSFNMNA
HKWFLTNFDCSLLWVKDRYALTEALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFRS
LKLWMLRLYGAEATLKSIRNHIKLAKDLEQLVSQDPNFEVVTPRIFSLVCFRIVPVDND
EKTCNNLNRSLLDVNSSGKLFISHTTSLGKFVLRCAIGAPLTEEKHVMDAWKVIEEAS
FLLASQVK *Glycine max* 03G167900.1 (SEQ ID NO: 52):

MEEESALRPMDAEQLREQAHKMVDIFIADYYKTIEDFPVLSQVQPGYLGKLLPDSAPDSP
ESLQNVLDDVQEKILPGVTHWQSPNYFAYFPSNSSIAGFLGEMLSAGLNIVGFSWITSPA
ATELETIVLDWLAKAFQLPDYFYSSGKGGGVIQGTASEAVLVLLAARDKILRRVGRNA
LPKLVMYASDQTHSALLKACQIAGINPELCRLLKTDSSTNYALSPDVLSEAISNDIAGGL
VPFFLCATVGTSSSTAVDPLPALGKIAKTNKLWFHVDAAYAGSACVCPEYRHCIDGVEE
ADSFNMNAHKWFLTNFDCSLLWVKDRSSLIQSLSTNPEFLKNKASQGNMVIDYKDWQI
PLGRRFRSLKLWMLRLYGLDGLRSHIRNHIELAANFEELVRQDTRFKVVAPRTFSLVC
FRLLPHPNSADHGKNLNSDLLDSVNSTGNAFITHTVLSGEYILRFAVGAPLTERRHVN
AWQILQDKATALLES *Fragaria vesca* 27261550 (SEQ ID NO: 53):

MDAEQLRENAHKMVDIFIADYYKTIEDFPVLSQVQPGYLRLLPDSAPTQPESLQHIFDDI
QAKILPGVTHWQSPNFFAYYPSNSSIAGFLGEMLSAGLNIVGFSWVTSPAATELEMIVLD
WLAKLIKLPDEFSLAGQGGGVIQGTASEAILVVMLAARDKILRRVGKNALEKLVVYASD
QTHSALQKACQIAGIHPENCRILSTNSTTNYALSPSVGTSSSTAVDPLGELGKIAKNEM
WFHVDAAYAGSACICPEYRHYIDGVEKADSFNMNAHKWFLTNFDCSVLWIKDRNALV
QSLSTNPEFLKNKASQANMVVDYKDWQVPLGRRFRSLKLWMLRLYGLLENLQSYIRT

HINLAKFEELVAQDPRLVCPRLYSLVCPRLPHGNEACSLKLNHDLLDVANSSTGKI
YISHTVLSGAYILRFAVGAPLTEEKHVTAAWKKLKSVIDRDLALANSFVSITFSHMYREA
NFLTDALASVGHSLSSSMCWFDGIPPQAQMALLMDSSCIGHLRGSSL *Kalanchoe*

fedtschenkoi 0172s0035.1 (SEQ ID NO: 54):

MEGELKPMDAEQLREYGHMVDVADYYKTIEDHPVLSQVEPGYLRLKLLPDSAPDKPE
SFENVLSDVKTIIIPGVTHWQSPNYFAYFPSNSSTAGFLGEMLSACFNIVGFSWITSPAAT
ELEMIVLDWFAKMLKLPDFFLSTGQGGGGVIQGTASEAVLVVLLAARDIFLRKLKGKGFLE
KLVVYASDQTHSALQKACQIAGIHPENVKALKTDSSTNYGLSPDLLSKEICHDIANGLVP
FFACASVGTTSSTAIDPILELANVTKSYNIWLHVDSAYAGSACVCPEYRHHIDGVVEEVDS
FNMNAHKWFLT NFDCSLLWVKDRNALIQSLSTNPEFLKNKASQSKSVLDYKDWQIPLG
RRFRSLKLWLVLRLYGVENLQAYIRNHIELAIHFEELVSQDMRFEIVAPRTFALVCFRLL
LPCGFEDRTNDVNGDLLQAVNSTGKIFISHTVLSGTYVMRFAVGAPLTEERHIDAAWKL

IQDQASSLLEKL *Capsella grandiflora* 22666s0001.1 (SEQ ID NO: 55):

MDSEQLREYGHMVDVFIADYYKTIEDFPVLSQVQPGYLHKLLPDSAPDQPETLDQVLDD
VRAKILPGVTHWQSPGFFAYYPSNSSVAGFLGEMLSAGLGIVGFSWVTSPAATELEMIV
LDWLAKLLNLPKEFLSKGNGGGVIQGSASEAVLVVLLAARDKVLRSAGKNALGKLVVY
SSDQTHSALQKACQIAGIHPENCRVLETDASTNYALRPELLQEAVSKDLKAGLIPFFLCA
NVGTTSSSTAVDPLAALGKIANSNEIWFHVDAAYAGSACICPEYRKYIDGVETADSFNMN
AHKWFLT NFDCSLLWVKEQDSLTEALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFR
SLKLWMVLRLYGAETLKSIRNHIKLAKYYEKLVSQDPNFEIVTPRIFSLVCFRLVPKNE
DEKKCNNQNRKLLLEANSSGKLFMSHTALSGKIVLRCAIGAPLTEEKHMKEAWKVIQD

EASFLHLK *Selaginella moellendorffii* 15420188 (SEQ ID NO: 56):

MGEANIGPKPIDAEFRKHAHEMVDVFIADYYRDIESFPVRSQVSQPGYLKTLLPPAAPED
PEALEEVFADIQSKIIPGVTHWQSPNFFGYPSNSSTAGLLGEMLSAGLNIVGFSWITSPA
ATELEIIVLDWLAKLLKLPDEFLLFGNGGGVIQGTASEAVSVVLLAARTRAISENKRKGL
SEAEILSKLAVYTSQTHSCLQKGCAGIPIENLVIVPTDSSSTNYAVSPAAMRQALEDG
VKQGLLPFFLCGTVGTTSSSAVDPLSALGDIKDFGMWFHVDAAYAGSACICPEFRHHL
DGVEKADSFNMNAHKWLLTNFDCSALWVKESHLVSALSTTPEFLRNKASDLNQVVD
YKDWQIPLGRRFRSLKLWFMVMRMNGASGLRSYIRNHVRLAKRFEGFVREDPRFQLLP
RTFGLICFRLKPESDDPDNGRTLNSTLLEAVNSSGRMFITHTVLSGVYTLRMAIGGPLTQ

DKHVDAAWKLIQEEATTLVKGPSHILANNLRLSPILANNLRLSPILANNRI *Setaria*

italica 3G188200.1 (SEQ ID NO: 57):

MDILNHADTTTANGTSPAAAAAAAVVAPATPSSLVTPPLDADEFRRQGRLVVDVFIADYY
TRINEYPVRPAVAPGFLARQLPETAPARPERDALAAALRDVRDLILPGVTHWQSPRHFA
HFAATASNVGALGEALAAGLNINPFTWAASPAATELEVVDWLGLKALHLPERLLFSG
GGGGTLLGTSCEAMLCTIVAARDRKLAEIGEERIGDLVVYFSDQTHFSFQKAARIAGIRR
GNCREIPTSRESGFTLSPKALRAAVRADEASGRVPLFLCATVGTTPTAAIDPLRELCAAV
SGHGVVHVDAAYAGAACVCPEFRHAIAGAEAVDSFSTNPHKWLLANMDCCALWVT
RPAALVAALGTDHDVILKDPASAAQDGHVVDYKDWQVALSRRFRALKLWLVLRC
HGVEGLRGFVRAHVRMAAAFEAMVRADTRFEVPVPRQFALVCFRLRPASAGEKRTRG
GEVVEPNELNRLLLEAVNATGRAYISSAVVGGVYVLRCAIGNSLTEERHVREAWSVVQ
EQANVVLAATATCPDERAVHRARCVETDAADAPASVPPVQMRFPASAQS *Kalanchoe*

fedtschenkoi 0033s0078.1 (SEQ ID NO: 58):

MGSLPSPHDPSNAFNPMMDVAELSIESRLVMDFITQYYQTLETRPVQPRVKPGFLTGQLPE
KAPFHAESMEEILSDVSEKIVPGLTHWQSPNFHAYFPASSSNAGLLGEMLCSGLSVIGFT
WNSSPAATELENVVDWLADMLNLPPSFRFSGGGGGGGVQLSNTCEAVLCTLAAARD
KVLERIGDDKINKLVAYCSDQTHFTLHKGAKLIGIRRANIKSIGTRRENGFGLCPNDLRN
AITGDLEAGLVPFYLCGTIGTTALGAVDPIKELGKVAREFDLWFHIDAAYGGSACICPEF
RHYLDGVELVDSISMNAHKWLLSNLDCCFLWLQNPCKLIQCLAAEAFLKSGSEMVDY
KDWQISLSRRFRAIKMWMVFRRYGVSNLMEHIRSDVSMAARFEEMVSADDRFEIVFPR

KFALVCFKLNTKGSVQHGEDDGLDGDSDLTREL MGRVNSSGKAYLSGVEMGRIF
IRCVIGSSLTEERHVDNLWNLIQEKTQSIMPCRA *Daucus carota* subsp. *sativus*
36068870 (SEQ ID NO: 59):

MGSLSTQKFNPLNLDFFSSES NKVIEFITAYYKNVEKYPVRSQVEPGFLLNMYPKKAPSQ
PVSLDTILQELEADIIPGMTHWQSPNFFAYFRTTTSNAAFQGEMLCNALNVAGFNWICSP
AATELEMIVMDWL GKMLS LPPQSFLFAGNGGGVLQGSTSEALICVLSAARDRAKQYGE
DSITKLVVYASDQTHFVVKAAKLVGIPTKNFRVIPTSIATCFALKPNDIKMAIERDLESG
LVPLFVCATVGATPSGSVDPVEGLGLLAKNYGLWLHIEAAYAGSAFICPELTHYLRGIEH
AHSISINLHKWLLTNMDCSCLWVKSPDVLLLESMTDEILRNEASESKKVVD FMDWQIA
TSKLFRAKLKLWFLRRYGVDNLMAHIRSDIELAKHFEALVNSDKRFEVVPVNFSLVCF
RLKPNEEGEESLKVLMNWNLM EAVNSSGRAYMTHAVLGDIFVIRCAIGTSLTEERHVNE
LWKLILEKTEVILKRDQ *Daucus carota* subsp. *sativus* 36056758 (SEQ ID NO:
60):

MNTFDTEDFRKQAHLIIDFLADYYQNI EKFPVRSQVSPGYLGEILPDSAPHDPEPIEKILED
VRSNIIPGITHWQSPNFFAYFPSCGSTAGFLGEMLANGFN VVGFNWISSPAATELETIVM
DWLGKMLQLPEAF LFSGGGGGVLQGTTC EAMLCTLVAARDRTLREQGMENFDKLLCP
VQLELEILSDVQNGLIPLFLCVTIGTTPSTAVDPLATLSEVAKKYKLWVHVDAAYAGSA
CICPEFRHFLDGL ENVNSFSMNAHKWFLTTL DCCCLWVNDPSALIKSLSTYPEFLRNHAS
ESNKVVVDYKDWQIMLSRRFRALKLWFLRSYGVEKLREFIRVHVEMAKYFEGLVAMD
QRFEVVPRLFAMVCFRVVCCGENDVNEINEKLLESVNQSGRIYVSHAVLDGVYVIRFA
IGATLTDYSHVSA AWEVVQEHADALLA *Solanum tuberosum* 3DMP400026166 (SEQ
ID NO: 61):

MGTLNINHELDDQIFNTINPLDPEEFRRQGHKIVNFLADYYQNI EQYPVCSQVNP GYLQK
IVPNSAPNNSESLEKILKDVERDIIPGLTHWQSPNFFAYFPSSGSTAGFLGEMLSVGFNVV
GFNWISSPAATELESIVMDWFGKMLNLPNCFLFASGGGGVLQGTTC EAMLCTIVAARD
QMLRKISRENFGKLVVYASDQTHFSLKKA AHIAGIDPGNFRVIPTIKANEYTLCPKSLRL
AILNDLKEGNVPLFLCATIGTTATT SVDPLRLLCEIAKEFGIWVHVDAAYAGSACICPEFQ
VFLDGVENANSFSLNAHKWFFSTLDCCCLWVKDPSALTNALSTNPECLRNKATELNQVI
DYKDWQIALSKRFRALKLWLVLRSYGVTNLRNLIRSHVNMAKHFEGLVATDKRFEIFV
PRKFAMVCFRISPLVLSQVSTKFDDEKEVNMFNTKLVESINSCGKLYLTHGVVGGTYIIR
FAIGASLTHYRHVDVAWKVIQDHANALLNQGYV *Solanum tuberosum*
3DMP400024738 (SEQ ID NO: 62):

MGMTKINPEHEFDGQFSINTSSSRLLDPEEFRRQGHMMVD FLADYFQNI EKYPVRSQVE
PGYLKKLLPDSAPYKPEPIAKILEDVERDIFPGLTHWQSPNFFAYFPCTSS TAGILGEMLS
AGLNVVGFSLIA SPAATELESIVMDWL GKMLSLPKTYLFSGGHGGGGVIQGTTC EAMLC
TIVAAREQM LEKVGREKVDKLVVYASDQTHFSFEKAVKISGIKLENFRVIPTTKDTEFAL
DPKSLSRTIEQDIKSGFIPLFM CATIGTTSTTVVDPLKLLCEITKDYGIWVHVDAAYAGGA
CICPEFQHFLDGIENANSFSFNAHKWLF SNLDCCCLWVKDPSALTNALSTRPECLRNKAT
DTKQVVDYKDWQLSRRFRALKLWLVLRSYGIDNLRNFIRSHVKMAKHFEQLVSM D
ERFEIVAPRNFSMVCFRVSPLALGNKQVNKFN MELLE SINSCGNIHMTHALVGGVYMIR
FAIAAPLTEYKHIDMAWEVICNHANAMLDVN *Solanum lycopersicum* 36137005 (SEQ
ID NO: 63):

MGTLNINHELDDQIFNTINPLDPEEFRRQGHKIVNFLADYYQNI EQYPVCSQVNP GYLQN
IVPNSAPNNPESLDKILKDVQNDIIPGLTHWQSPNFFAYFPSSGSTVGFVGEMLSVGFNV
VGFNWISSPAATELESIVMDWFGKMLNLPNCFLFASGGGGVLQGTTC EAILCTIVAARD
QMLRKISRENFGKLVVYASGQTHFSLKKS AHIAGIDPGNFRVIPTIKAKEYTLCPKSLRLA
ILNDLKEGNVPLFLCATIGTTSTTSVDPLRLLCDISKEFGIWVHVDAAYVGSACICPEFQV
FLDGVENANSFSLNDPSALTNALSTNLEFLRNKATELNQVIDYKDWQIALSRRFRALKL
WLVLRSYGVTNLRNLIRSHVNM TKHFEGLIAMDKRFEIFVPRKFAMVCFRISPLVLSQVS
IKFDDEKEVNMFNTKLLESINSCSKLYLTHGIVGGTYIIRFAIGASLTHYRHVDIA *Daucus*

carota subsp. *sativus* 36065781 (SEQ ID NO: 64):
MCKPKSSPASHINWQSPNFFAYFPSSGSTAGFLGEMLSTGFNVVGFHWMASPAATELEN
VVTDFWFGKMLQLPKSFLFSGGGGGVLQGTTCAMLCTLVAAARDKNLRQHGMENIGKL
VVYCSQDQTHSAMQKA AKIAGIDPKNFRTVETSRASNFQLCPRRLESAILTDIQNGLIPLYL
CATVGTTSSSTAVDPLPALTEVAKKYDLWVHVDAAYAGSACICPELRQYLNGVENADSF
SLNAHKWFLTTLDDCCCLWVKNPSALIKSLSTYPEFLRNNASETNKVVDYKDWQIMLSR
RFRALKLWFLRSYGVGQLREFIRGHVDMAKYFEGLVGKDKRFEVVVPRLFMSVCIRV
RPSAMTGKSCGNDVNELNRKLLLESLNESGRIYVSHTVLDGIYIIRFAIGATLTLDINHVSAA
WKVVQDHALLDLDDTNFLAKKVADIILS *Oropetium thomaeum* 35995617 (SEQ ID
NO: 65):

MAILNHADDASPANDDNPATAPAMAPATNPRPLDADEFRRQGRVLVDFIADYYARVEE
YPVRPSVTPGFLSRKLPETAPEQPEPGHGDFAFASALRDVRDLILPGITHWQSPNHFAHFA
ATASNVGALGEALAAGLNINPFTWAASSAATELEVVTDWLGKALHLPQELLFSGGGG
GTLLGTSCEAMLCTVVAARDRKLGEIGEHRIGDLVVYCSQDQTHFSFRKAARVAGIRAN
CREIPTSLESDFALSPSALLAAVRADAAAGLVPLYLCVTVGTTPTAAVDPVRELCAAVA
GRGVVWVHVDAAYAGAAARVCPELLRHAGAIVDGVDSFSTNPHKWLLANMDCCALWVQ
QPDALVAALGTDHVDILKDPA AAAAGDVVDYKDWQVALSRRFRALKLWLLLRCHG
VEGLRAHVDRDGLRMAEAFEAMVRADARFEVPVRRQLSLVCFRLRPTAVIREKQQQQRG
RRRDHDDDTAAANELNRRLLEAVNATGRTYMSCAVVGGVYMLRCAIGNSLTEDRHVE
EAWN VVQE QASAILDAAMVVRADECTVCTAAHCVQMGMVDDILAASFPTGNEVTIR
Oryza sativa 33157740 (SEQ ID NO: 66):

MAILNHSDAAFPVAATTPLLGRRPLDAGEFRRQGRQVVDIADYYAGINDYPVRPAVAP
GFLAGKLPATAPSTPEPDALTAGLRDVRELMLPGLTHWQSPRHFAHFSATASNVGALGE
ALAAGLNVNPFTWEASPAATELEVVTDWLGKALHLPERLLFAGGGGGTLLGTSCEAM
LCTIVAARDEKLAEIGEERIGDLVVYCSQDQTHFSFQKAARIAGIRRGNCREIPTCRESGFV
LTATALQA AVAADEAAAGRVPLFLCATVGTTPTAAVDPLRELCAAVEGRGVWVHVDA
YAGAACVCPEFRHAIAGAEAVDSFSTNPHKWLLANMDCCALWVARPAALVAALGTDD
DVILKDAAAARPARGDHHHHA AVDYKDWQVALSRRFRALKLWLVLRCHGVDGLRA
VVRSHVRMAAALERMVRADARFEVPVPRQFALVCFRLRGGGAAAQLVGGDELTA SNE
LNRRLLEAVNATGRAYMSSAVVGGMYVLRC AVGNSLTEEHVREAWSVVQGQAAAV
LATAGAAADTARTKDHAAGDDHGADQPHAMTTTTT MGCRSGPWEL *Brachypodium*
stacei 01G392300.1 (SEQ ID NO: 67):

MAPASSTRQVITDHKTQKENSSCTVINHLLDADEFRRQGHKVIDFIADYYSGIADYPVHP
SVTPGFLLNQLPADPPEDPDTFASALQDVRDLILPGMTHWQSPRH LAHF PASSSVTGALG
EALAAGINAVPFMWSASPAATELEMVAVDWL GKALHLPKTL LFSGAGGGTLLGTSYRK
LAETGAGRIGDLVVYGSQDQTHFALRKAARIAGIRHGRCRELRTCIADMFALSPAALSAA
MDADAGAGLVPLFLCATVGTTQTKAVDPIGALCAEAAPHGVWVHVDAAYGGSALVCP
ELARDAIDGVEAVDSFSMNAHKWLLVNTDCCALWVKRPALLVSALGTQDEDEVILRD
AAAQGH DVVDYKDWAVTLTRRFRALKLWLVLR CYGVEGLREHIRGHVRMAALFEGM
VNADPRFEVVTERRFALVCFRLRPDQLPDEGNKKKTMAAANELNRRL LQEVNAAALGP
YMSAANVGGIYVLRC AVGSTLTEKRHVRQAWEVVQE KATSILRA *Amaranthus*

hypochondriacus 32828676 (SEQ ID NO: 68):
SLHDETLQGIKYVTQYYKNVEKYPVVSKVKWGYLRQILPENAPSLPESIDQILEDVDTKI
VPGLTHWQSPNFFAYFPATASNAAMLGDIVCSGLNVVGF SWISSPAATELEAIVMDWM
AKLLMLPPTFLFSGGGGGVIHGSTCEAIVCTQAAARDVALNIHGEEKITKL VVYASDQT
HISFQKA AKLIGIPPRNFRVLPTSSATDFALSPTTLRASIEVDLSQGLVPFYICATIGATPSG
AVDPIDGLGQIARDYGAWLHVDAAFAGNACICPEYRHYLDGVELADSISMNPHKWLLT
NMECSCLWLKNPKLMVDSLSTKPEILNNKATQSGDVIDYKDWQIALSRRFRALKLWIVI
RRYGSTYLMNHVRS DIELAKYFESLIKQDERFELVVP RKFSLVCFRMKLVGREDEVETLT
NQKLEDVNSSGKAYMTHAVIGGKFVIRCAIGGTLTEKRHIDSLWKLIIEKVPLTTCEL

Brachypodium distachyon 5g21770.1 (SEQ ID NO: 69):

MSSNSCPAAAAATFTTPPGAHLPLDADAFRRQGRQVADFIADYYDRIEDYPVRPNVSP
GFLAAQLPDAAPSWPEEPDALASALRDVRDLILPGLTHWQSPRHFAHFAATASNAGAL
GEFLAAGLNVNPFMTWAASPAAAELEVVVTDWLGLQALGLPEKLLFRGGSGGGGTLLGTS
CEAMLCTIVAARDQKLLKIGEDRIGDLVVYCSAQTHFSFKKAARVAGIRRGNCRVIPTRF
EDGFALSPAALAAVRDDVARGKVPLFLCATVGTATGAVDPVRELCAAVGAGHGSG
VWVHVDAAYAGGACVCPEFRHVAAGAEADSFSSTNPHKWLLANMDCCALWIRRPGL
LVAALGAGEDEDAILNKAPPAARGMQADLMVDYKDWQVPLSRRFRALKLWLVLRCH
GVEGLRGVVRGHVRMAAAFEAMVRADPRFEVPVPPAFALVCFRLRPLAAHPGSSSGID
EVNGRLLEAVNGTGRAYMSGAVVGGAYVLRCAVGNLSTEDRHVREAWSVVQEQADA
ILAPSDDEDRCCTDQIQTEMLQRRPLGAAADVFA

Brachypodium distachyon 2g02360.1 (SEQ ID NO: 70):

MAPASSKLHAITDDKTQQQNSSCPAASNGAIEPSNAKCAASSNHLLDADEFRRQGHKVI
DFIADYYAGIADYPVHPSVTPGFLNQLPADPPSRPEDHPDGAFGPALQDVRDVILPGMT
HWQSPRHFAHFPASSSVAGVLGEALAAGINAVPFTWAASPAAAELEMVAVDWLKAL
HLPESLLFSGAGGGTLLGTSCEAILCALVAARDRKLADIGTDRIGDLVVYGSQTHFALR
KAARIAGIRHDCRELQTCLADMFAALSPAALSAAMDADAGAGLVPLFLCATVGTQT
AVDQVGALCAAAPHGVWVHVDAAYAGSALVCPALARDAIDGIEVVDSFSMNAHKW
LLANTDCCALWVKQPKLLVSLGTQNEELILRDAAAEGHDVVYKDWAITLRRFRAL
KLWLVFRCYGVGLREHIRAHVRMAALFGLVKDDPRFEVVTERRFALVCFRLRAPDQ
LMDEGNEKKKTAAANELNRLLREVNGVALGPYMSAAVVGGIYILCAVGSTLTER
HVRQAWVVQERATSILRG

Sorghum bicolor 009G192600.1 (SEQ ID NO: 71):
MGVAVTAEVVHARSCKGTPPVGAAASVMVWDGAGQGYSCQPVGTTTANGGTTAPAP
VAIAMPSPHPLLDADDEFRRQGRLLVVDIADYYARIDEYPVRPAVAPGFLARQLPETAPA
RPEPDALAAALRDVRDLILPGVTHWQSPRHFAHFAATASNAGALGEALAAAGLNVNPF
MTWAASPAATELEVVDWLGLKALHLPESLLFSGGGGGTLLGTSCEAMLCTIVAARDRKL
AEVGEERMGDLVVYCSAQTHFSFQKAARIAGIRRGNCREIPTSMEAGFTLSPKALAAV
RADEAAGRVPFLCATVGTPTAAVDPVRELCAAVAGRGVWVHVDAAYAGAAASVCPE
LRHAVAGVERVDSFSSTNPHKWLLANMDCCALWVRRPAALTAALGTDHDVILKDPSAQ
AAQEGGAVVDYKDWQVALSRRFRALKLWLVLRCHGVEGLRGLVRAHVRMAAAFEA
MVRTDARFEVPVPRQFALVCFRLRAAAVLVVGKRRARDGDDEVVTAGNELNRLL
EANATGRVYMSSAVVGGTYILCAIGNSLTEERHVREAWSVVQEQATAILAAARRPTAR
TNRRTVRRHAAL

Kalanchoe laxiflora 0994s0009.1 (SEQ ID NO: 72):
MGSLQSPHDPNAFNPMDVAELSIESRLVMDFITQYYQTLETRPVQPRVKPGFLTQQLPE
KPPFHAESMEEILSDVSEKIVPGLTHWQSPNHFAYFPASSSNAGLLGEMLCGLSVIGFT
WNSSPAATELENVVVDWLADMLNLPPSFRFSGGGGGVLQSNTCEAVLCTLAAARDKV
LERIGDDKINKLVVYCSAQTHFTLHKGAKLIGIRRANIKSISTRRENGFGLCPNDLRNAIK
SDLEAGLVPFYLCGTIGTTALGAVDPIKELGKVAREFDLWFHIDAAYGGSACICPEFRHY
LDGVELVDSISMNAHKWLLSNLDCCFLWLQNPCLIQCLAAEGEFLKSGSEMVDYKD
WQISLSRRFRAIKMWMVFRRYGVSNLMEHIRSDVSMAARFEEMVAADDRFEIVFPRKF
ALVCFKLNTKGSVQHGEVDGEDGLDGDVLTRELMGRVNSSGKAYLSGVEMGRIFIR
CVIGSSLTEERHVDNLWNLIQEKTQSIMPRRA

Kalanchoe laxiflora 0003s0173.1 (SEQ ID NO: 73):
MGSLSSPRDLTKPFNPLDPTELAVESSLVTDFAIEYYRTVEQRPVQPHVTPGFLTSQLPSA
APFASESVESILQDVYDKILPGLVQWQSPNHFAYYPATCSNAGLLGEMLCGLNVVGFT
WSAPAAAELEQVVVDWMGKMMGLPQSFLFSGGGGGVLQGSTCEAVVCTLAAARDR
ALERVGDDMFNKLVVYCSAQTHFTLKKGSKLVGIRPANVKAIKTTKNNEYGLCPTDLR
NLVASDVKAGFIPIYLCGTIGTTAFGAVDPIRELKKVAREFNMWFHVDAAYAGSAFICPE
FRHYMDGVELADSFSTNPHKWLLSNMDCCVLWLKFPKRVIKSLAAEGVFLEGGSETMV
DYKDWQIALSRRFRAIKLWMVIKRYGLKNLISHIRSDVSMAKRFEELLLSDRRFEVVFP

KFSLVCLDFVMKVPVEVDEDDGELSHDSKLTRELMASVNVVTGKAFLTGVRLGRIFFI
RCAIGSTLTEDRHIQDLWKLIQEKAHKICANHDLKFRV *Panicum hallii* 32512198
(SEQ ID NO: 74):

MAILNHGDTTANGSSPADAAVAPAMPSLVQPPLDADEFRRQGRLVVDFIADYYTRID
EHPVRPAVAPGFLARQLPDTAPARPEPGDDALAAALRDVRDLILPGVTHWQSPRHFAHF
AATASNVGALGEALTAGLNINPFTWAASPAATELEVVDWLGKALHLPESLLFSGGG
GATLLGTSCEAMLCTLVAARDRKLAEIGEERIGDLVVYCSDQTHFSFQKAARIAGIRRG
NYREIPTSRESGFTLSPKVLRAAVRADEAAGRVPLFLCATVGTTPTAAVDPLRELCATVA
GHGVVHVDAAYAGAACVCPEFRHAIAGAEAVDSFSTNPHKWLLANMDCCALWVRR
PEALTAALGTDHDVILKDPSSERDCGRGVVDYKDWQVALSRRFRALKLWLVLRCHGV
EGLRGFVRAHVRMAAAFEDMVRADARFEVPVPRQFALVCFRLRSAAAGEKRARDGDD
AEPNELNRRLLEAVNATGRAYMSSAVVGGIYVLRCAIGNSLTEERHVVREAWCVVQEQA
TVVLAACAACTEERAVHSARCADAPAAVPPVQNEGEGEPTSIKIFGTSIARCSIKSEAS
TYHSWSTLWRTLMFKLLTWIISRL *Prunus persica* 6G202600.1 (SEQ ID NO: 75):

MTSALDPVEFRRQGHMMVDFIADYYQNIDKYPVLSQVDPGYLRKRLPESAPDNPEPIETI
LQDVQEHIVPGLTHWQSPSFFAYFASNVSIAAGFLGEMLSTGFNVVGFNWVSSPAATELE
SIVMDWLGNLLSLPKSFLFSGNGGGVIHGSTCEAIVCTMAASRDQMLSRIIGDNIGKLV
VYGSDQTHSALQKASQIVGINPKNFRAIEATRSTTFALSPESLKLAISSDIEAGLVPLFLCA
TVGTTATTAVDPLGPLCDVAKHHGMWVHVDAAYAGSACICPEFRHFIDGIEGVDSFSFN
AHKWFFTGLDCCCLWVKNPGALISSLSANPEFLRNKPTDSKQVVDYKDWQIALSRRFR
AMKLWLVLSYGVVNLNRNFLRSHVKMAKLFEGLVAMDQRFEIVVPRNFSMVPPTTPTS
NSFHQNGIEINVEKCTNEVNCKLLEAINASGRVFMTHAMVGGMYVIRCAVGVTQTEEK
HIAMAWKVVQEHADVILKNNGDDGDANLKLPLLDKIA *Prunus persica* 4G086700.1
(SEQ ID NO: 76):

MGSLNFDHPQENNSAHMSGPLDLVELRRQGHMIIDFITDYYQNIKHPVLSQVQPGYLK
QRLPESAPYNPEPIETILRDVQDHIVPGLTHWQSPNHFAYFPATISTAGFLGEMLTTCFNV
VGFNWMASPAATELETIVMDWLGDMLKLKPNLSFLFSGTGGGVLHGSTHESVCTMAAA
RDQILSRIGEENIGKLVVYGSDQTHSVIQKVSQIVGIPSKNFRAIETTISSTLSPETLRLT
VCSDMEAGLVPFYLCATVGTATTAVDPLGPLCDVAKDYGMWVHVDAAYAGSACICP
EFRQYIDGIEGANSFSFNAQKWFFTALDCCCLWVKNPSALTSMSTDLEVLNRNKASESK
RVVDFKDWQIALTRRFRAIKLWLVLSYGVANLNRNFLRSHVKMAKRFEGLVRTDERFE
VVVPRIFALVCFRISPSAISKANPTPSDEKCVNEVNCKLLEAINGSGWVYMTHAVVGM
YVLRCAIGASLTKEKHVAMAWKVVQEHVDAILPLTMY *Prunus persica* 4G087100.1
(SEQ ID NO: 77):

MMGSVEFEHPQENNSAHMTTSPLDPEEFRRQGHMVIDFIADYYKTIEKYPVLSQVQPGY
LKKRLPESAPYDPEPIETILQDVQDHLVPGLTHWLSPNHFGYFPAAISTAFLGEMLTG
FNVVGFNWMAASPAATELENIVMDWLGDMLKLKPKSFLFSGNGGGVLQGTTCIAIVCTM
AARDQMLRQIGRENIGKLVVYGSDQTHSALQKASQIVGIHPKNFRAIETTTSTSFALSP
EVLKSTICSDIEAGLVPLFLCATVGTATTAVDPLGPLCEVAKEHDMWVHVDAAYAGSA
FICPEFQYFIDGVEGADSFSLNAHKWFFTLDCCCLWVKNPSALVSSLSTNPEFLRNKAT
DSKQVVDYKDWQIALSRRFKAIKLWLVLSYGVGNLNRNFLRSHVKMAKIFEGLVGMD
KRFEIVAPRHFSLVCFRVSPSAISKANPSLSDHDNGKLKAHNYELLNGVKCVVNEVNSK
LLEAINGSLVYMSHAVVGGMYVLRCAIGASLTEEKHVAMAWKVVQEHADAILGTKII
VDQT *Medicago truncatula* 31073039 (SEQ ID NO: 78):

MNTSSSNPPQSDPQKTMNPLDLEEFKRQGYMMIDFLTYYKNIENYPVLSKVEPGYLAK
ILPSSAPFQPESESILEDVQQHIIPGITHWMSPNYYAYFPSSGSIAGFIGEMLSTGFNVVGF
NWLSSPAATELETIVMNWLGLLNLPKSFIFSSNIKGGGEIKKLSQIGKDNIGKLVVYCSD
QTHSALQKATQIVGIHSENFRIKTKGSNLFALSPDSLLSTILLVDNGLIPYFLCATIGTT
STNAVDPIKLLCNVTKEYDIWVHVDAAYAGSVCICPEFRHCIDGIEELNSFSFNAHKWFL
TNLACCCLWVKDHNALTSLSTNPEFLRNKKSDSKEVIDYKDWQIPLSRKFNALKLWIV

LRSYGVENLKNVHSTHREHVEGLRFLVPSKFSLVCFRISPFASIANDSEG
YYVGKMMNDAYLVNEMNHKLLDLINSSGKAYMSHGEVEGSFVIRCAIGATLTEHHVT
MTWKL VQQIASFLLGTPLN *Zea mays* GRMZM2G009400 (SEQ ID NO: 79):
MAILNRADTSHTTTASNGSATPAAPVAIAMPSLPHPLDADEFRRQGRLVVDIFIADYYA
RIDGY PVRPAVAPGFLIRQLPEAAPARPEPDALAAALRDVRDLILPGVTHWQSPRHFAHF
AATASN VGALGEALAAGLNVNPFTWAASPAATELEV VVTDWL GKALHLPESLLFSGGG
GGTLLGTSCEAMLCTIVAARDRKLAEVGEERIGDLVVYCS DQTHFSFQKAARIAGIRRG
NCREIPTSRESGFTLSPKALAAAVRADEAAGRVPLFLCATVGTTP TAAVDPLRELCAAV
AGHDVWVHVDAAYAGAACVCPEFSHV VAGVEAAESFSTNPHKWLLANMDCCALWV
RRPAALTAALGTDHDVILKDPAAAQAQAQQQQCS DGGVVDYKDWQVALSRRFRALKL
WLVL RCHGVEGLRGLVRAHVRMAAAFEAMVRGDARFEVHVPRQFALVCFRLRAVAV
AVAGEK RAGDYDGVAAAGNELNRRLLEAVNATGRVYMSSAVVGGAYILRCAIGNSLTE
ERHVREAWSVVQE QATAILSAATATARTNGLTVRRARCD AEADVSDVPTPQQPLPLG

Glycine max 07G059000.1 (SEQ ID NO: 80):
MEMKNTMNRNPQSDAPIIKPLDPEEFKRQGYMMVDFLADYIRNVSHYPVLSKVEPGYL
KQRLPTSAPCGPEPIESILKDVQDHIIPGLTHWQSPNFYGYFPSSGSIAGFMGEMLSAGLN
VVGFNWVSSPSATELESIVMDWL GQVLNLPKSFLFCGDHGGGVVLGTTCEAILCTLVAA
REKKLSQVGKENIGKLVVYGSDQTHSALQKAAQIAGIHPANFRVIKTKRSNSFALSPDSL
LSTILLDVERGLIPCFLCATVGT TAIATIDPIGPLCNVAKDYGIVHVDAAYAGSACICPE
FRHCIDGV EEVNSFSLNAHKWFLTNLTCCLWVKDHIALTKSLTVNPQFLRNKASESKR
VIDYKDWQIPLSRKFNALKLWLVLRSYGVENIRNFLRNHVQMAKTFEGLVRLDKRFEIV
VPPKFSLVCFRIAPS AIIANGLSKGVEACYNGLVND EYMVNEVNRKLLDSVNSSGDAF
MTHGEVEGAFMIRCAIGGTLTEHHVIMAWKLVQEHANSL LGL *Panicum virgatum*
Ca01381.1 (SEQ ID NO: 81):

MAILNHGDTTAAAGTSPAAA AVNVAPPMHSLVQPVLD ADEFRRQGRLVVDIFIADYYTR
IDEYPVRPAVAPGFLARQLPEAAPARPEPGGDALAAALRDVRDLILPGVTHWQSPRHFA
HFATTGSNVGALGEALAAGL NINPFTWAASPAATELEV VVTDWL GKALHLPERLLFSG
GGGGTLLGTSCEAMLCTLVAARDRKLAEIGEERMGDLVVYCS DQTHFSFRKAARIAGIR
RGNCREIPTSRESGFALQPRTLLAAVRAD EAAAGRVPMFLCATVGTTP TAAVDPLRELCA
AVAGRGVWVHVDAAYAGAACVCPEFRGATAGAEAVDSFSTNPHKWLLANMDCCAL
WVRRPEALTAALGTDHDVILKDPSSERGGGVVDYKDWQVALSRRFRALKLWLVL RCH
GVEGLRGLVRADARFEVPVPRQFALVCFRLRAAAAAAVGEKRGRDRDND AEPNELNR
RLLEAVNATGRAYMSSAVVGGIYVLRCAIGNSLTEERHVREAWRVVQE QATAVLAAA
ACTEERAVRSAR *Theobroma cacao* 27425420 (SEQ ID NO: 82):

MSSASRKTFLEPTSF TNESKAVIDFIADYYKNIEEYPVQSGVEPGYLSAKLPDSAPYCP
ESLEDILKDVNDCIIPGLTHWQSPNFFAYFQANASTAGFLGEMLCSGFN VVGFNWISSPA
ATELESIVLDWMGKLLKLPSSFLFSGTGGGV LHGSTCEAAVCTLAAARDKALKELGGW
ENITKLMVYASDQTHFTFQKAAKLVGIPPSNFRFIETSLSTGFSMSSDQVRLAIEHDIKSG
LVPLFLCATIGTTACGAIDPIAELGQVAREYKLWLHIDAAYAGSACICPELRHFLDGVEL
ANSVSMNPHKWFLT NMDCCLWITEPRLLVDSLSTDPEILRNKASEFKAVLDYKDWQV
ALSRRFRALKLWIVIRRHGLANLVYHIRSDISMAERFEAFVAKDDRFDIVVPRKFALVCF
RLKPKQELEGLELNSRLLEAINSSGRAFMTHAVVGGIYVIRCAIGTTMTEERHVDALWK
LIQEKAQGLLME *Fragaria vesca* 27274768 (SEQ ID NO: 83):

MGSLDFHHVPEKTNSDPPMANPMDPEEFRRQGHIMIDFIADYYKNIEKYPVLSQVQPGY
LKKLLPESAPYNPEPIETILQDVQDHIVPGITHWQSPSYFAYFPSSGSIAGFLGEMLSTGFN
VVGFNWMS SPAATELERTTCEAIVCTMAAARDQMLS RIGKDNIGKLVVYGSDQTHSAL
KKASQIVGIHPNNFRAIKTTKSTEFALSPELLRSTICSDIDKGLVPLFLCATMGTTATTSDV
PLRGLCDVAKDYDLWVHVDAAYAGSICICPEFRHFIEGVDGANSFSFNAHKWFFTTLDC
CCLWVKNPTALINSLSTNPEFLRNKASDSKQVV DYKDWQVALSRRFRALKLWLVLRSY
GVANLRSFLRSHVKMAEVFEKLVRENKWFEVVVPRNFAMVCFRISPSAIRKAPTDDDGI

DVVINEVNSKLLLEAMNTSGSVYMTHAVVGGMYVLRCAIGATMTEEKHVLMAWKCGS
ALERKDVAANETLSFNFQRRFDRRARQRRGHVGFRLAITMLDLKTSERDGARRWSIGA
YANQITTISQANSSVAWTMEFHSCFIFFCGSIKLDTQVPNDDFVLSARWPPSFPVSGWSTI
NFHETIKIYVGSLSLDSWTMEFHSCFTFFCGS *Gossypium raimondii* 26786642 (SEQ
ID NO: 84):

MVSASRKTFPLDPVTFSNESKAVIDFIADYYENVEKYYPVQSTVEPGYLSAMLPEAPYC
PEPLQDILEDVSNCIIPGLTHWQSPNFFAYFHANASTAGFFGEMLCSGFNVVGFNWISSP
AATELESIVLDWMGKMLKLPSSFLFSGTGGGVVLHGSSCEAAVCVLAARDKALKELGG
WENITKLVVYASDQAHFTFQKAALVGIPPSNFRLIETSFSTGFSLSPENLRFVIEDNIRSG
LVPLFLCATIGTTSPGAVDPPIAELGKVAMEFKLWLHIDAAYAGSGCICPELRHYLDGVEL
ANSISMNPHKWFLTNDCCCLWIKPKLLVDSLSTDPEILRNNASKSKAVVDCKDWQI
ALSRRFRALKLWVIRRHGLANLMCHIRSDIAMAKRFEALVGEDERFEIVVPRKFALVC
FRLKPKVEEEDLNCKLVEAINSSGRAFM SHA VLSGIYVIRCAIGTTTLTQQHHVDALWKLI
QDKAQSLLM *Populus trichocarpa* 26994989 (SEQ ID NO: 85):

MGSLSTNTFSPLDPNGFTNDSKMVIDFIADYYKNIENNPVQSQVKPGYLLTQLPDTAPYC
EESLEDVLKDVTDSIIPGLTHWQSPNFFAYFQANASTAGFVGEMLC TGLNVVGFNWIAS
PAATELESIVMDWMGKMLKLPSTFLFSGNGGGVVLHGSTCEAIVCTLVAARDETLMIGA
ENITKLVVYASDQTHSTLLKGVKLVGIPSSNFRCLSTSFSSEFSLSPQALEDAIENDIKAGL
VPLFLCATVGT TACGAVDPVMDLGEIARKYNLWFHIDAAYAGSACICPEFRHYLDGVEL
ADSLSMNPHKWLLTNMDCCCLWVKQPRLLIESLSSDAEFLRNNA SESSDVVDYKDWQI
ALSRRFRALKLWVIRRHGLANLMCHIRSDVNLA KRFE SLVAKDSRFEVVVRRRFSLVC
FRLKHND ECQGLELNRKLLAAVNESGRAFMTHAVVGGLFIIRCAIGSTLTEERHVDDLW
KLIQEKAADLLSKKQVLLDN *Malus domestica* 22679008 (SEQ ID NO: 86):

MSLLAFYSNSGERSKRVHLSASTYGNSTPNSYISLPYALFSSATQLINIHSNSSNFQMGLSI
SQENNSPNVPTNPLDPEEFRRQGH LVIDFIADYYKSIEKHPVLSQVQPGYLLKRLPDTAP
YNPEPLETILQDVQDHIVPGITHWQSPNYFAYFPSSGSVAGFLGEMLSG FNVVGFNWM
SSPAATELESTVRDWFGNMLKLPKSFLFSGNGGDVIQGTTC EALVCAMVAARDQKLSK
FGRHNIGKLVVYGSDQTHSALQKASQIVGIHPENFRSIETTRSTSFALSPESLKVIIYSDIEA
GLVPLFLCATVGT TAIATVDPLGPLCGVAGDYGMWVHVDAAYAGSACICPSFDISLMA
SRVQIHSVSTRTNGSSPLSTVVAFGLRIPTRWNKATELKQVVDYKDWQIALSRRFRSMK
LWLVLRSYG VANLRNFLRSHVKMAKIFEGLVAMDKRFEIVAPRNFSLVCFRVSPSSISN
KASSDQNGKTDYCCDANGDENS VIINEVNRKLLESINVSGHVYMT HGVVGGLYMLRFA
VGATLTEEHIALAWKV VQEHADQILTKY *Citrus Clementina* 20801973 (SEQ ID
NO: 87):

MRAGEASIIKMG SFGLSANNITHGSSFSADLEPKSFSDESKAVIDFIADYYK NIEKYYPVQS
KVEPGYLSARLPDTAPHSPESLDDILKDVTDCILPGLTHWQSPNFFGYFQANASTAGFLG
EMLCSGFNVVGFNWL ASPVATELESIVMDWMGKMLKLPSSFLFSGTGGGVVLHGSTCES
LVCTLAAARDKALEKLGGGFDNITKLAVYASDQTHFALQKS AKLIGIPPANFRPLRTSFS
TEFSLSPDTVRAAIEDDIKSGHVPLYLCATVGT TGAGAVDPIEELGKIANEYKLWLHIDA
AYAGSACICPEYRHYLNGVELADSISLNPHKWFLTNDCCCLWVKHPSFLVDSLSTESD
IMRNRSPASNTSTNAAPVIDYKDWQIALSRRFKALKLWTVIRKHGYSGLMYHIRSDVSM
AKRFAAMVAKDERFEIVVPRKFALVCFRLKPKRESEGSELNRELVDALNGSGRAFLTQA
MLGGVYVIRCSIGTTTLTQDRHVDDLWKLIQEKA DRLLSLQEPEHASR *Citrus*
Clementina 20818150 (SEQ ID NO: 88):

MGSLNSDHELKTNSASFNNPMDSEEFRRQGHMIIDFIADYYRDVEKYYPVLSQVEPGYLQ
KRLPESAPYNPEPIETILQDVQQHIVPGITHWQSPYFAYFPSSGSIAGFLGEMLSG FNV
VGFNWMSSPAATELENIVMDWLGEMLKLPKSFLFSGTGGGVVIQGTTC EAILCTLAAARD
QILNEIGRENISRLVVYGS DQTHSALQKAAQIAGIDPKNFRAIKTTKSSSFTLTPESLQAAI
DLDIQSGLIPLFLCATVGT TAITTV DPLGPLCDIAKRYSIWIHVDAAYAGSACICPEFRHFI
DGIESADS FSLNAHKWFFTTLDCCCMWVKNPNALIKALSTNPEFLRNKASDSKQVVDY

KDWQISLSRRFRALKLWLVLRSFGVANLRNFLRSVGM AQLFQELVGGDNRF EIVAPR
NFAVVCFRVLPSASGLGNGKANEGANELNRKLLSINASGQLVSHGMVAGIYFIRFAV
GATLTEDRHVIAAWKV VQEKL DGILATS *Vitis vinifera* 17834108 (SEQ ID NO: 89):
MGSLSFNTFSPLDPQS FSEESKMVVD FDIADYYKNVEKYPVQSQVDPGYLMHHCPDTAPY
CPEPLETILKDVSDGIIPGLTHWQSPNFFGYFQANASTAGFLGEM LCTGLNVVGFNWIAS
PAATELESIAIICSLAAARDKVLKKGHHKITKL VVYGSDQTHSTLQKASKLVGIPASNFR
SLPTSFSNYFALCPDDVRTAMEEDIGAGLVPLFLCATVGT TSSGAVDPLEALGHVAKDF
KVHHLNGVELAHSISMNPHKWLLTNMDCCCLWIK EPKLFVDSLSTAPEFLRN NASESK
KVIDYKDWQIALSRRFR AIKVWAVVPRRFALVCFRLRPREEGESTELNSRLLMAVNGSG
AAFMTHAVVGGIYIIRCAIGSTLTETRHVDSLWKL IQEKAQLVLQEPGLALEEDYIDPCIG
VSATSLHAVVRWYCNYSS EINAHLVFIAFFVVCKENRENYVLGVNGPPN *Petunia*
hybrida ABB72475.1 (SEQ ID NO: 90):

MDTIKINPEFDGQFCKT TSLLDPEEFRRNGHMMVDFLADYFHNIEKYPVRSQVEPGYLE
RLLPDSAPIQPEPIEKILKDVRSDIFPGLTHWQSPNFFAYFPCSSTAGILGEMLSAGLNVV
GFSWIASPAATELESIVMDWL GKLINLPKTYLFSGGGGGVMQGT TCEVMLCTIVAARDK
MLEKFGRENIDKL VVYASDQTHFSFQKAVKISGIKPENFRAIPTTKATEFSLNPESLRRAI
QEDKKAGLIPLFLCTSIGTTSTTAVDPLKPLCEIAEEYGIWVHVDAAYAGSACICPEFQHF
LDGVEHANSFSFNAHKWLFTTLDCCCLWLKDPSSLTKALSTNPEVLRNDATDSEQVVD
YKDWQITLSRRFRSLKLWLVLKSYGVANLRNFIRSHIEMAKHFEELVAMDERFEIMAPR
NFSLVCFRVSLLALEKKFN FVDETQVNEFNAKLLESIISSGNVYMTHTVVEGVY MIRFAV
GAPLTDYPHIDMAWNVVRNHATMMLNA *Carica papaya* 16421889 (SEQ ID NO:
91):

MSSLSRDLNASPLEPENFRVESKRVIDFIADYYKNIETY PVQSRVKPGYLAGRLPSSAPFS
PESLETILQDIAENISPGLTHWQSPNFFGYFQANASTAGFHGEMLC SGLNVVGFNWISSP
AATELESIVMDWMGNMLKL PSSF LFSGSGGGVLHGSTCEAVVCTLAAARDKTLNQLG
GNYQNITKFVVYASDQTHFTLQKAAKLIGIPPSNFRSLTTSFPSGFSLSPEKLQSAIKDDIK
SGYVPLYVCATVGTTAAGAVDPILELGKVAQEYNLWFHIDAAYAGSACICTEFRHYLN
GVELADSISTNPHKWLLTNMECSCLWVKSPSSLVDSLSTKSEIMRNAATDSNQVIDYKD
WQIALSRRFRALKLWIVIRRHGLSGLTSHIHKDIKMAELFESLVAKDKRFEIVVPRKFAL
VCFRFRKPEKENQDLSELNSKLLNAVNSSGCAFMTHAVLEGVYTIRCAIGTTLTEEHV V
NLWKL IQEKAQSLINEY *Sphagnum fallax* 0042s0024.1 (SEQ ID NO: 92):

MSSKVAPWSRLSKPLDVEEFRTHAHRMVD FDIADYHHNIQSFPVHSQLKPGYLRPLL PDT
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ITSPVVTELEIIVLDWL GKLHLPEEFLSSGKGGGVIQGT SSEAVVCTS QHMSEAEALTKL
VVYTSDQAQSCVLRACQIAGIATANFRPLPTDASSHFSLSPAVLIKAAATDVAAGLFPFF
LCGKVGTTSSSAVDPLLELGDI AKRYGMWYHIDAAYAGSACICPEFRHYLNGVEKADS
YNMNP HDWMLTNFDCSTLWVKNS ELLVAALSNKPVYLQNEATDNNLVDCSHIRNHISI
AKHFESLVRADFRFEMIVPTNFSLVCFRLRTPAGSKDNSRTLNSKLVEALNRKGDILVTH
TELSGRYTLRFAVGGTHMELHHVQA AAWNLR LQRQVF *Eucalyptus grandis* E01788.1
(SEQ ID NO: 93):

MNPLDPGEFRRQGHMVVD FLAKYYENIEKYPVLSQVEPGYLSKRLPSSAPQDEEPMEAI
LDDVHQHIFPGLTHWQSPNFFAYYQTNTSTAAILGEMLCAGFN VAGFNWVSSPAATELE
SLVMDWL GKMLDLPRPFLPFGNGGGVIEGNTSEAIIC TLTAARDRVLRKLGHNSIAKLV
VYGSDQTNC SFQKAARVVGIDPRNFRALKMTRSTL FGLSPDSLEKAIRLDINAGLIPLYL
CATVGT TSCAAVDPLEPLCKVASKFSMWIHVDAAYAGASCICPEYRK FINGVEFADSFS
FNAHKWLLTPLDCCCLWVKDPNALVKS LSTDPEYLKNEATESKQVIDYADWQLSLSRR
FRALKLWLVL RSHGVQNL RSHIKNHCRLAKLFEELVEEDPQFEVVFP RN FALVCFRIHPS
GVAGMLNAQLLHAINASGRVFM SHTTVGGVYVLRFAVGATLVTEKHVIMAWKV VQE
HANSLLSMPASEQHSA pHis8-4 (SEQ ID NO: 94):
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GAT pEAQ-HT (SEQ ID NO: 95):

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[illegible]

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1410 (SEQ ID NO: 96):

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 GGAAAAACGCCAGCAACGCGGCCTTTTTACGGTTCCTGGCCTTTTGCTGGCCTTTTG
 CTCACATGTTCTTTCCTGCGTTATCCCCTGATTCTGTGGATAACCGTGCGGCCGC

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INCORPORATION BY REFERENCE AND EQUIVALENTS

(155) The teachings of all patents, published applications and references cited herein are incorporated by reference in their entirety.

(156) While example embodiments have been particularly shown and described, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the embodiments encompassed by the appended claims.

Claims

1. A host cell comprising a transgene encoding a heterologous tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) operably linked to a promoter, wherein the T8GT comprises an amino acid sequence having at least 95% identity to one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20.
2. The host cell of claim 1, wherein the host cell is a fungi cell, a bacterial cell, or a plant cell.
3. The host cell of claim 2, wherein the fungi cell is a yeast cell.
4. The host cell of claim 3, wherein the yeast cell is a *Saccharomyces cerevisiae* cell.
5. The host cell of claim 2, wherein the bacterial cell is a cell from a genus selected from *Agrobacterium*, *Escherichia*, *Bacillus*, *Pseudomonas* and *Streptomyces*.
6. The host cell of claim 5, wherein the bacterial cell is an *Agrobacterium tumefaciens* cell, an *Escherichia coli* cell, a *Bacillus subtilis* cell, a *Pseudomonas aeruginosa* cell, or a *Streptomyces griseus* cell.
7. The host cell of claim 2, wherein the plant cell is a *Nicotiana benthamiana* cell, or a cell from a genus selected from the group consisting of *Arabidopsis*, *Beta*, *Glycine*, *Helianthus*, *Solanum*, *Triticum*, *Oryza*, *Brassica*, *Medicago*, *Prunus*, *Malus*, *Hordeum*, *Musa*, *Phaseolus*, *Citrus*, *Piper*, *Sorghum*, *Daucus*, *Manihot*, *Capsicum*, and *Zea*.

8. The host cell of claim 1, wherein the host cell further comprises a transgene encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR), wherein the 4HPAR comprises SEQ ID NO: 4.
 9. The host cell of claim 8, wherein the host cell further comprises a transgene encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS), wherein the 4HPAAS comprises SEQ ID NO: 2.
 10. The host cell of claim 1, wherein the amino acid sequence of the T8GT has at least 99% identity to the one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20.
 11. The host cell of claim 1, wherein the amino acid sequence of the T8GT comprises the amino acid sequence of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, or SEQ ID NO: 20.
 12. A method of producing the host cell of claim 1, wherein the method comprises introducing into the host cell a vector comprising the transgene.
 13. A method of producing tyrosol 8-O-glucoside (salidroside), wherein the method comprises inserting into a host cell a transgene that encodes a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT), wherein the transgene is operably linked to a promoter, and wherein the T8GT comprises an amino acid sequence having at least 95% identity to one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20; and culturing the host cell in a culture media comprising tyrosol.
 14. The method of claim 13, wherein the host cell is a fungi cell, a bacterial cell, or a plant cell.
 15. The method of claim 14, wherein the host cell is the fungi cell, and wherein the fungi cell is a yeast cell.
 16. The method of claim 15, wherein the yeast cell is a *Saccharomyces cerevisiae* cell.
 17. The method of claim 14, wherein the host cell is the bacterial cell, and wherein the bacterial cell is a cell from a genus selected from the group consisting of *Agrobacterium*, *Escherichia*, *Bacillus*, *Pseudomonas* and *Streptomyces*.
 18. The method of claim 17, wherein the bacterial cell is an *Agrobacterium tumefaciens* cell, an *Escherichia coli* cell, a *Bacillus subtilis* cell, a *Pseudomonas aeruginosa* cell, or a *Streptomyces griseus* cell.
 19. The method of claim 14, wherein the host cell is the plant yeast cell, and wherein the plant cell is a *Nicotiana benthamiana* cell, or a cell from a genus selected from the group consisting of *Arabidopsis*, *Beta*, *Glycine*, *Helianthus*, *Solanum*, *Triticum*, *Oryza*, *Brassica*, *Medicago*, *Prunus*, *Malus*, *Hordeum*, *Musa*, *Phaseolus*, *Citrus*, *Piper*, *Sorghum*, *Daucus*, *Manihot*, *Capsicum*, and *Zea*.
 20. The method of claim 13, wherein the amino acid sequence of the T8GT has at least 99% identity to the one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20.
 21. The method of claim 13, wherein the amino acid sequence of the T8GT comprises the amino acid sequence of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, or SEQ ID NO: 20.
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